

C System Chromatic Button Accordion

2026 Revision

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Peter Wilkens

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Website: silver-space-couscous-69rg597j4rxhrjr5.github.dev

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About This Document

The material in this document is written for the C System Chromatic Button Accordion with both Stradella and Free Bass.

It is not an Accordion Tutor but covers material that I have found useful in learning about the Accordion after many years of playing the Guitar.

Where I have covered a topic I have attempted to include as much information as is needed to assist someone new to this type of Accordion to understand the topic.

The manual will focus primarily on chord and scale construction on both treble and free-bass keyboards.

I was drawn to the button accordion, rather than the piano accordion, noting that chord and scale construction on this type of keyboard allow the use of moveable shapes specific to different chord and scale types. Each shape may be moved anywhere on the keyboard, giving a chord or scale identified by the tonic in the shape.

Thus knowledge of any single shape enables the player to generate the chord type represented by that shape all over the keyboard in all keys.

This use of keyboard shapes is similar to the way in which different chords and scales are represented in many resources used by guitar players, removing the need to read music whilst still being able to learn about scale and chord construction.

The shapes shown for the treble keyboard are mirrored on the free-bass keyboard so it is easy to generate the chord shapes for the bass once you have grasped how the treble side works.

As a matter of interest, on the button accordion, chord shapes are consistent across the keyboard, unlike on the guitar where the interval between strings alters between the third and second string, which necessitates some alteration of chord shapes in that area of the fingerboard.

The main scales and chord types, in various inversions, will be considered.

Contents

Chapter 1

The Chromatic Button Accordion

This type of accordion (see picture in Section 1.1) is characterised by the Buttons on the Right hand (Treble Side). The Treble buttons are arranged chromatically in a number of columns. In the accordion I am considering in this manual, there are 5 columns, but the last two, nearest the bellows, duplicate notes already found in the first three columns. The additional columns allow more options in fingering. There are some smaller accordions that have only the first three columns without duplication. Section 2.2 shows a detailed arrangement of the keys on the treble side.

In the most common type of chromatic accordion, with a Stradella bass, the bass and chord buttons are arranged in six columns parallel to the bellows. The columns are: Counter Bass, Bass, Major Chord, Minor Chord, 7th Chord and Diminished Chord. The basses are arranged in intervals of a fifth and the chords are built on the second row of basses. See Section 1.2 for detailed arrangement of the keys on Stradella Bass instrument.

In the Free Bass accordion the outer-most 4 columns are arranged chromatically. In the C System the left hand keyboard mirrors the treble key arrangement allowing much more freedom of Chord choice and construction than on a Stradella bass instrument at the expenditure, initially, of more mental effort by the player. See Section 1.3 for the Free-Bass layout.

The Chromatic Button Accordion is isomorphic, meaning a chord or scale "shape" retains the exact same shape regardless of the key you are in or where you are on the keyboard. Knowledge of one type of scale or chord means that you have knowledge of all chords or scales of that identity since the shape can be moved anywhere on the keyboard taking the name of the "Root" of the chord or scale.

1.1 A Chromatic Button Accordion

The chromatic button accordion is a versatile instrument, distinguished by its button-based keyboard layout and rich tonal possibilities. This section provides an overview of a C system instrument with Stradella and Free Bass systems.

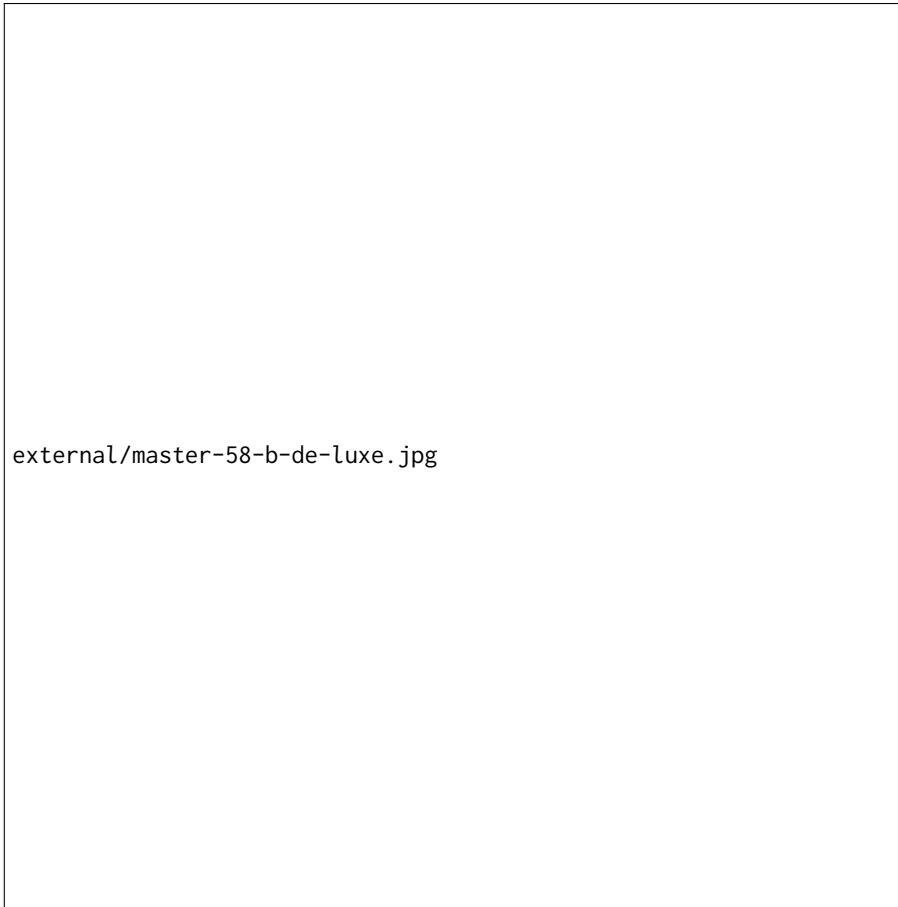
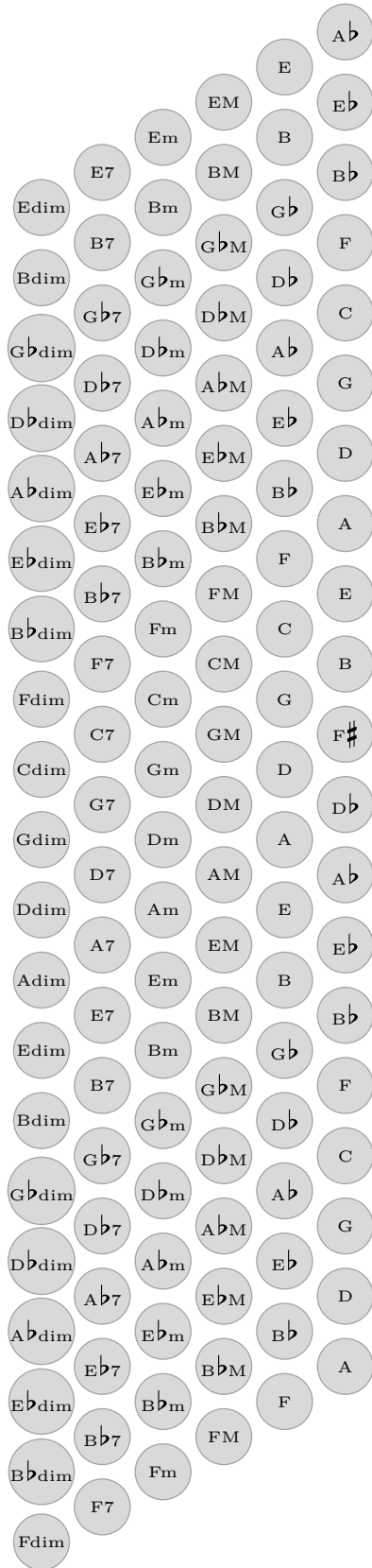


Figure 1.1.1 High quality Italian Chromatic Button Accordion with Stradella and Free Bass.

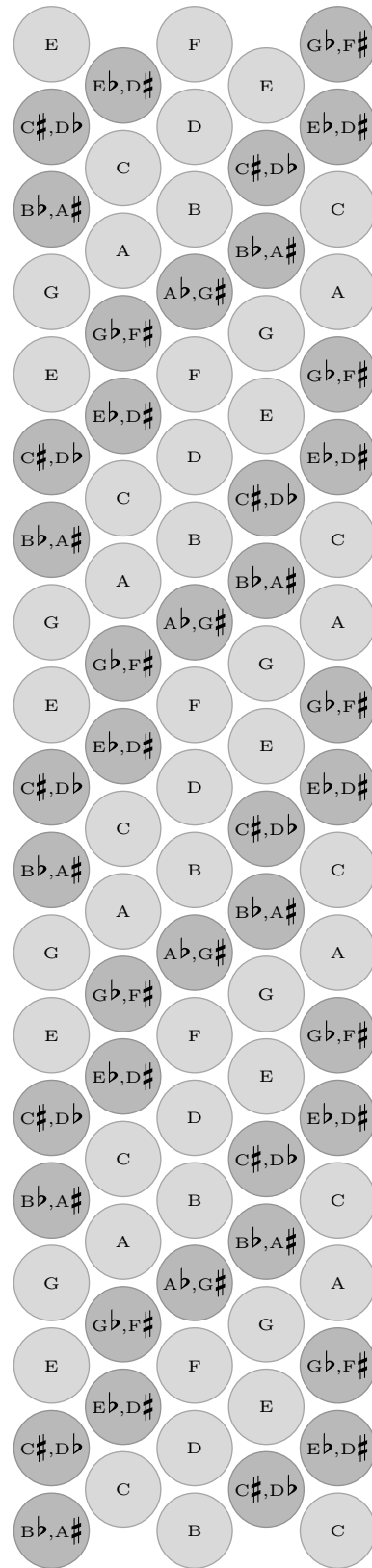
1.2 C System Accordion Keyboards with Stradella Bass

This section shows the complete accordion keyboard in Stradella Bass configuration. The left (bass) side shows the Counter Bass column closest to the bellows then the Standard Bass, Major Chord, Minor Chord, 7th Chord and Diminished Chord columns.

The right (treble) side is the same C system chromatic button layout as Section 1.2.1. Three faint vertical lines represent the bellows.



Stradella Bass

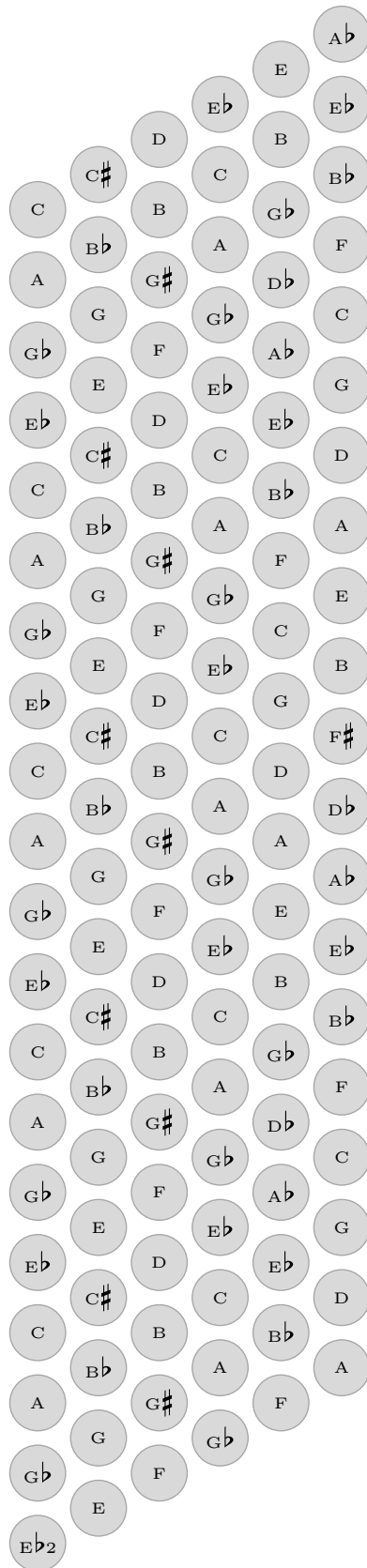


Treble

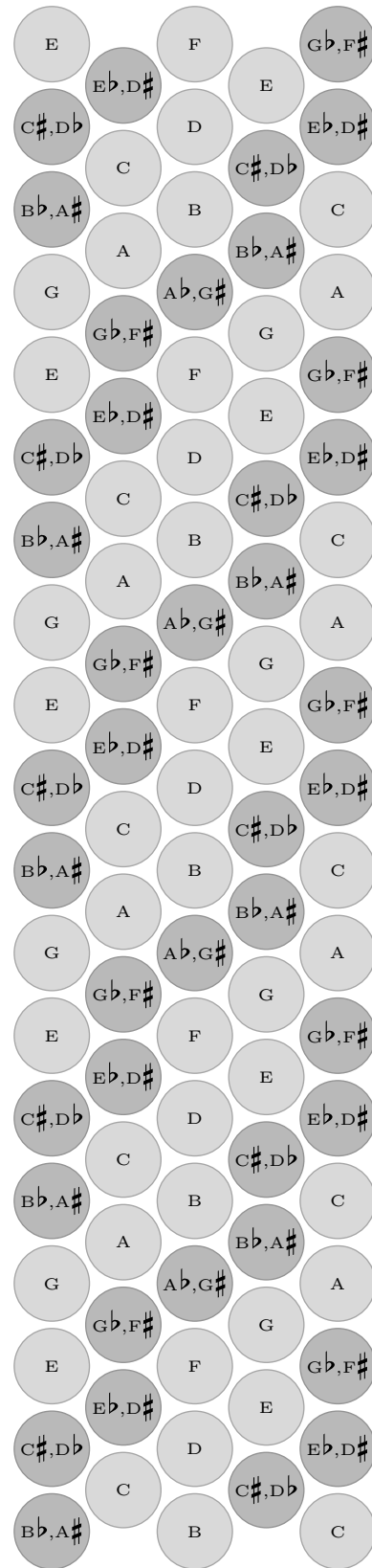
Figure 1.2.1 Accordion Keyboards Stradella Bass.

1.3 C System Accordion Keyboards with Free Bass

This section shows the complete accordion keyboard in Free Bass configuration. The left (bass) side shows the four chromatic free-bass columns alongside the standard Bass and Counter Bass columns. The right (treble) side is the same C system chromatic button layout as Section 1.3.1 Three faint vertical lines represent the bellows.



Free Bass



Treble

Figure 1.3.1 Accordion Keyboards Free Bass.

Chapter 2

Scales and Intervals

Before discussing scales it is useful to understand how the distance between notes is described. The musical distance between notes on your instrument keyboard or on a musical staff is known as an **interval**. This interval is dependant on the number of semitones between notes and the Interval is described as a minor 2nd, a major 2nd, a minor 3rd, a major 3rd, a 5th, a 7th etc;

The interval between two notes can be obtained by counting between notes on the keyboard or staff but, when counting, the first note, should be counted as one and not zero. Using this type of count the actual number of semitones is 1 less than the count. Considering the C Major scale in Figure 2.1.1 the interval C to D is a Major second because there are 2 semitones between C and D. The Count is 3. That is, C to C#:1 semitone. C# to D:1 semitone. Count C is 1, C# is 2 and D is 3. Similarly E to F is a Minor Second. One semitone between E and F and a count of 2. Count 1 at E and 2 at F.

There are a number of different scale types eg; Major, Minor, Diminished etc;. Different Scale Types are characterised by the interval between each successive note. A Major Scale has a sequence of whole tone, whole tone, half tone, whole tone, whole tone, whole tone, half tone between the tonic notes.

The interval structure of each of the main scale types will be discussed in the following sections of Chapter 2.

2.1 C Major Scale Construction

In C major, the interval pattern remains the same ascending and descending.

All major scales have the same pattern of intervals, so any major scale can be built by using this same pattern.

The pattern of intervals in a major scale is: whole, whole, half, whole, whole, whole, half. This means that starting from the root note (C in this case), you would move up a whole step to D, then another whole step to E, then a half step to F, then a whole step to G, then a whole step to A, then a whole step to B, and finally a half step back to C.

A whole step (also called a whole tone) consists of two semitones, while a half step (also called a semitone) consists of one semitone. So the interval pattern for the C major scale can also be expressed in terms of semitones as: 2, 2, 1, 2, 2, 2, 1.

The C major scale notes are C, D, E, F, G, A, B, and C.

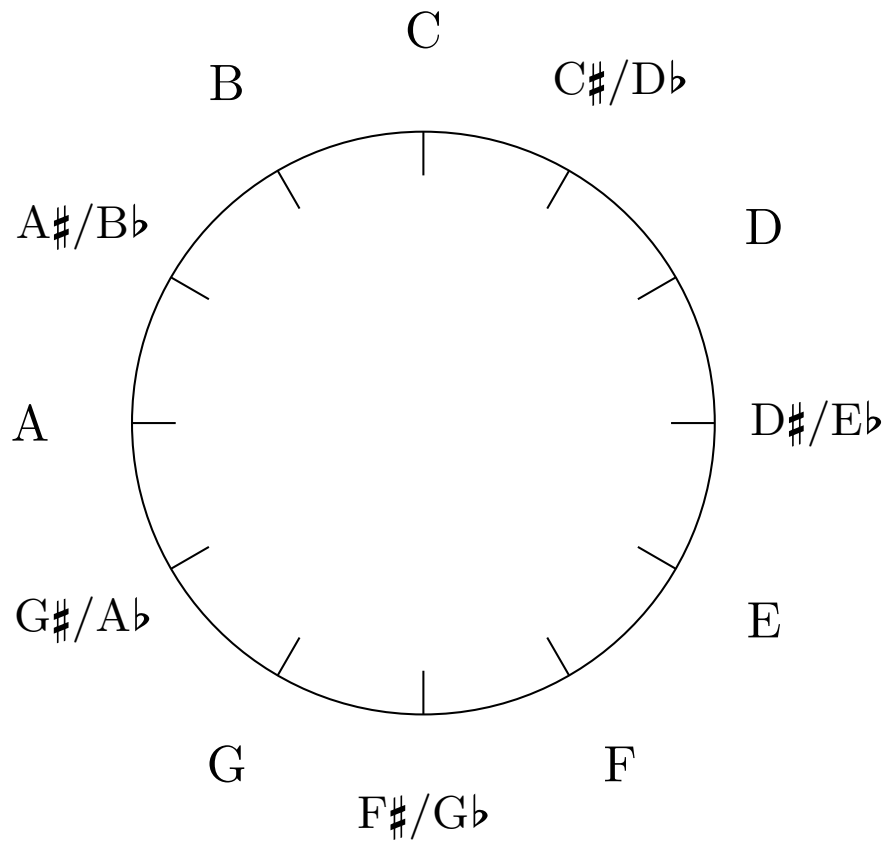


Figure 2.1.1 The 12 notes of the chromatic scale arranged clockwise around a circle, provide a useful means of visualising how scales are constructed by stacking 2nd intervals above the Scale Root note.



Figure 2.1.2 C major scale notation on a musical staff (ascending and descending).

Interval	Quality	Notes	On Staff	Number of Semitones	Count
Second	Major	C - D		2	3
Second	Major	D - E		2	3
Second	Minor	E - F		1	2
Second	Major	F - G		2	3
Second	Major	G - A		2	3
Second	Major	A - B		2	3
Second	Minor	B - C		1	2

Figure 2.1.3 Examples of intervals with notation and semitone counts (C major scale).

2.2 C Natural Minor Scale Construction

In C natural minor, the interval pattern differs from the major scale in that the third, sixth, and seventh scale degrees are lowered.

All natural minor scales follow the same interval pattern, so any natural minor scale can be built by using this pattern of major and minor seconds.

The C natural minor scale notes are C, D, E $\flat$, F, G, A $\flat$, B $\flat$, and C.

A natural minor scale is the relative minor of a major scale. The A minor natural scale uses these same notes, the sixth note of the major scale becoming the root note of the relative minor. ie.) the notes of the A Minor Natural Scale are A, B, C, D, E, F, G and A.

C minor natural is the relative minor of E $\flat$ major.



Figure 2.2.1 C natural minor scale notation on a musical staff (ascending and descending).

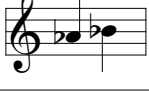
Interval	Quality	Notes	On Staff	Number of Semitones	Count
Second	Major	C - D		2	3
Second	Minor	D - E flat		1	2
Second	Major	E flat - F		2	3
Second	Major	F - G		2	3
Second	Minor	G - A flat		1	2
Second	Major	A flat - B flat		2	3
Second	Major	B flat - C		2	3

Figure 2.2.2 Examples of intervals with notation and semitone counts (C natural minor scale ascending).

2.3 C Harmonic Minor Scale Construction

In C harmonic minor, the interval pattern remains the same ascending and descending.

All harmonic minor scales have the same pattern of intervals, so any harmonic minor scale can be built by using this same pattern.

The C harmonic minor scale notes are C, D, E $\flat$, F, G, A $\flat$, B, and C. In this scale, the interval from A $\flat$ to B is an augmented second, spanning three semitones while still being spelled as a second.



Figure 2.3.1 C harmonic minor scale notation on a musical staff (ascending and descending).

Interval	Quality	Notes	On Staff	Number of Semitones	Count
Second	Major	C - D		2	3
Second	Minor	D - E flat		1	2
Second	Major	E flat - F		2	3
Second	Major	F - G		2	3
Second	Minor	G - A flat		1	2
Second	Augmented	A flat - B		3	4
Second	Minor	B - C		1	2

Figure 2.3.2 Examples of intervals with notation and semitone counts (C harmonic minor scale).

2.4 C Melodic Minor Scale Construction (Ascending)

In C melodic minor ascending, the sixth and seventh scale degrees are raised to A and B.

All melodic minor scales use this same ascending interval pattern, so any melodic minor scale can be written ascending by applying this pattern from tonic to tonic.

The C melodic minor ascending scale notes are C, D, E $\flat$, F, G, A, B, and C.



Figure 2.4.1 C melodic minor scale notation on a musical staff (ascending and descending).

Interval	Quality	Notes	On Staff	Number of Semitones	Count
Second	Major	C - D		2	3
Second	Minor	D - E flat		1	2
Second	Major	E flat - F		2	3
Second	Major	F - G		2	3
Second	Major	G - A		2	3
Second	Major	A - B		2	3
Second	Minor	B - C		1	2

Figure 2.4.2 Examples of intervals with notation and semitone counts (C melodic minor scale ascending).

2.5 C Melodic Minor Scale Construction (Descending)

In C melodic minor descending, the sixth and seventh scale degrees are lowered to A $\flat$ and B $\flat$, following the C natural minor form.

All melodic minor scales use this same descending interval pattern, so any melodic minor scale can be written descending by applying this pattern from tonic to tonic.

The C melodic minor descending scale notes are C, B $\flat$, A $\flat$, G, F, E $\flat$, D, and C.



Figure 2.5.1 C melodic minor scale notation on a musical staff (ascending and descending).

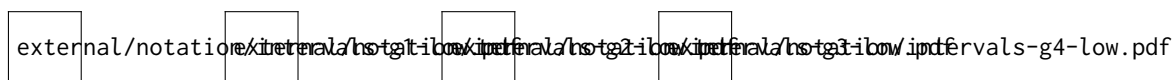
Interval	Quality	Notes	On Staff	Number of Semitones	Count
Second	Major	C - B flat		2	3
Second	Major	B flat - A flat		2	3
Second	Minor	A flat - G		1	2
Second	Major	G - F		2	3
Second	Major	F - E flat		2	3
Second	Major	E flat - D		2	3
Second	Major	D - C		2	3

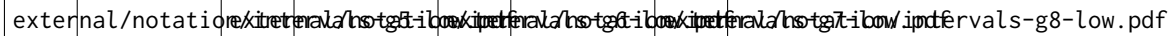
Figure 2.5.2 Examples of intervals with notation and semitone counts (C melodic minor scale descending).

Intervals and the Button Keyboard

The next part of this chapter will provide diagrams showing the main intervals as configured on the accordion keyboard.

The following grid diagrams shows examples of interval pairs on the chromatic button accordion keyboard. Each grid highlights two notes; the interval is identified by the number of semitones between them.





The following diagrams show the Free Bass keyboard (left) and the Treble keyboard (right) side by side as they appear on the accordion. The bellows are indicated by the vertical lines between the two keyboards.



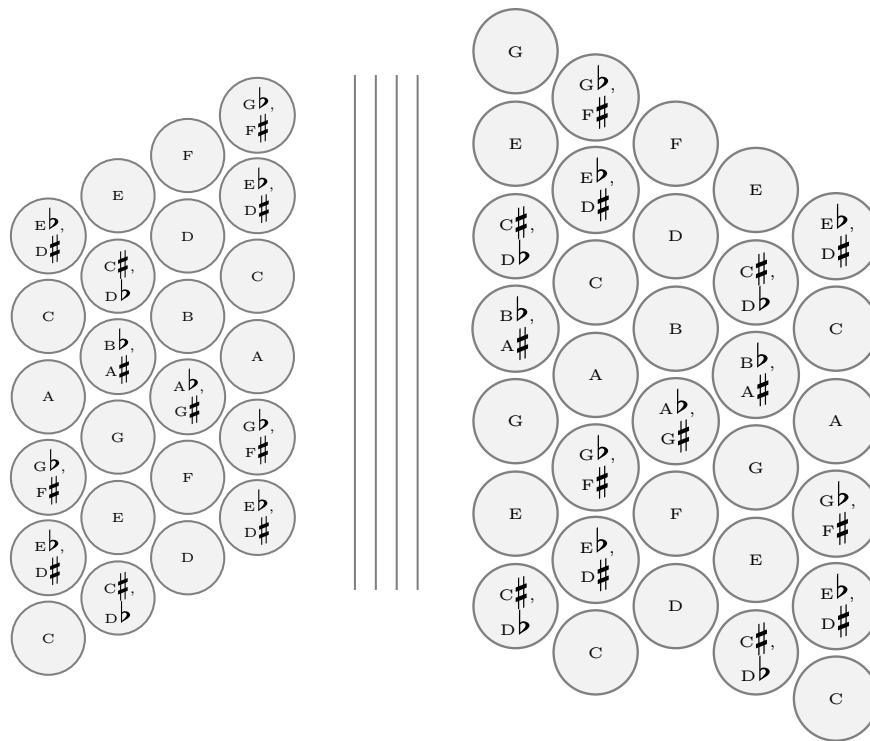


Figure 3.2.2

3.3 Using Intervals to Navigate the C System Keyboard

Because the C System Free Bass mirrors the arrangement of the Treble keyboard, every interval on the freebass keyboard is a mirror image of the shape on the treble keyboard. The diagrams below show the Free Bass keyboard (left) and the Treble keyboard (right) side by side, as they appear on the accordion. The bellows are indicated by the four vertical lines between the two keyboards. Recognising these physical relationships is important as the player is unable to view the keyboards directly and must rely on tactile feedback and muscle memory to navigate the buttons.

It is also important to recognise the four directions that are used for navigating the buttons of the keyboard. The playing directions are vertical, horizontal and diagonal as shown below. This aspect of the accordion will be discussed further in this chapter.

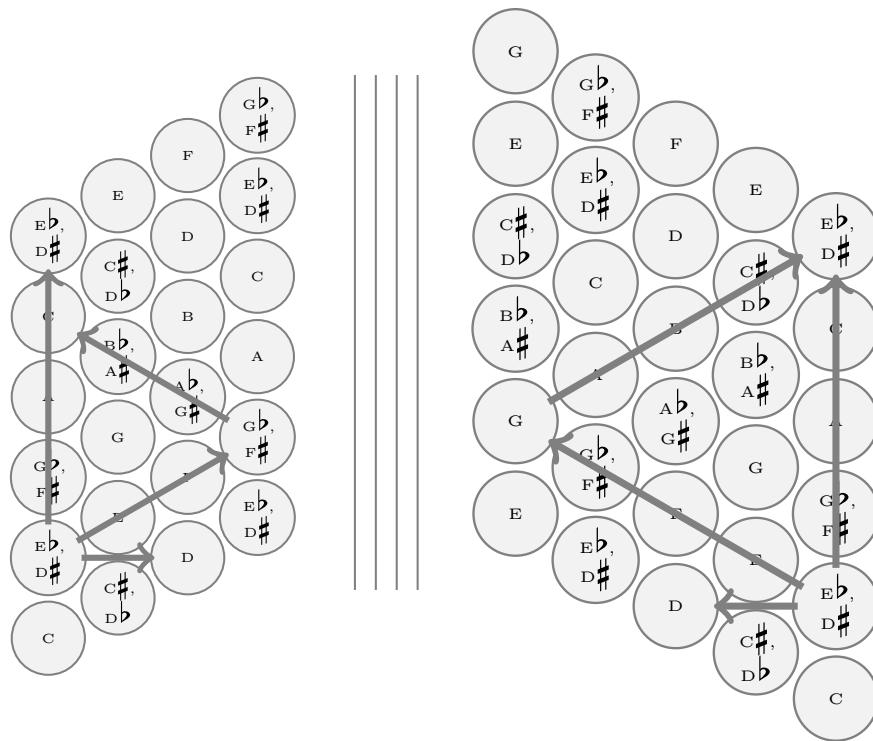


Figure 3.3.1 Playing Directions on each keyboard

Now considering first the vertical direction. The treble keyboard has 5 vertical columns and the freebass keyboard has 4 useable vertical columns. Any 2 immediately adjacent vertical buttons on either keyboard, played together, form a minor third eg; C and E♭, D and F, E and G, F and A♭, G and B♭ etc.

If a 2nd minor third is stacked on top of the first this forms a tritone (an interval consisting of 3 whole tones). For example C to E $\flat$ to G $\flat$, D to F to A $\flat$. The tritone being the interval between C and G $\flat$, D and A $\flat$ etc.

Continuing in a vertical direction adding a third minor third gives a minor 6th interval eg: C to A, D to B etc; Adding a fourth minor third forms a perfect octave: C to C, C# to C# etc.

Chapter 4

Scale Shapes

Scale Shapes with fingering on Free Bass and Treble Keyboard

4.1 C Major Scale Shape

Use this shape to form other major scales where the shape fits, for example E D, G, A.

4.1.1 Basic C Shape Scale

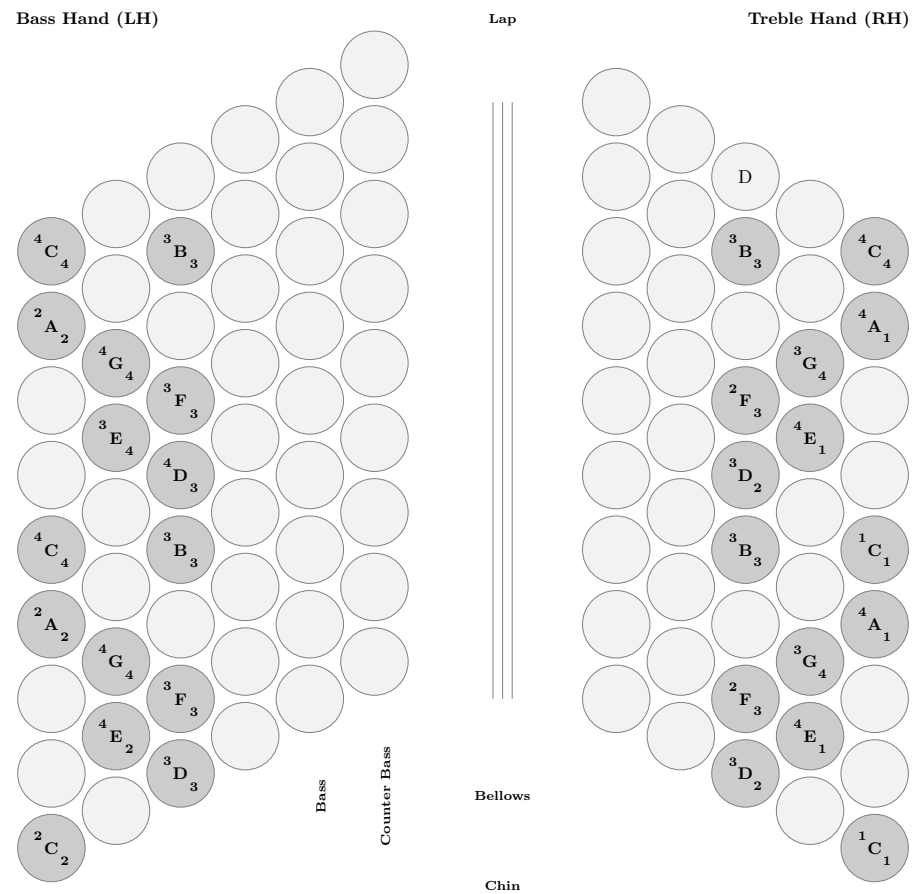


Figure 4.1.1 Basic C shape scale. Ascending fingering numbers appear at the top of each button and descending fingering numbers at the bottom.

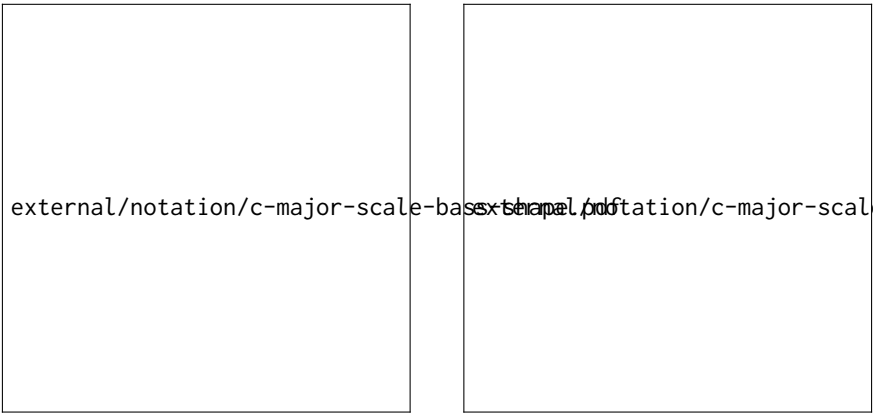


Figure 4.1.2

4.2 D Major Scale Shape

Use this shape to form other major scales where the shape fits, for example F, G A, B.

4.2.1 Basic D Shape Scale

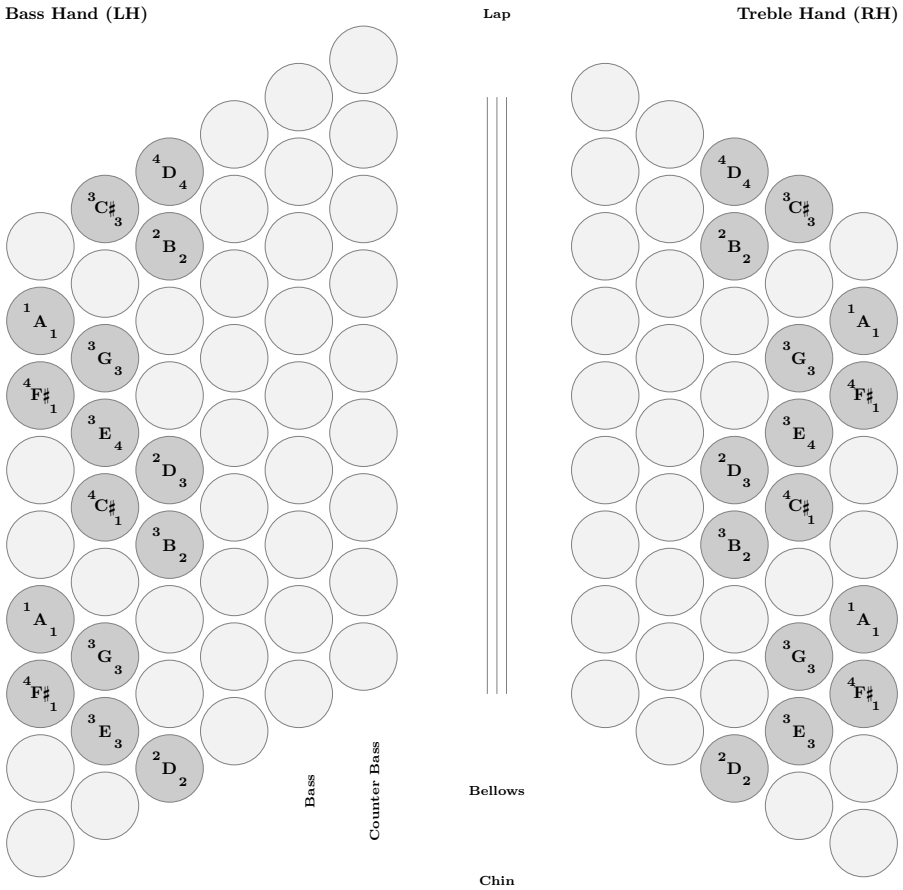


Figure 4.2.1 Basic D shape scale. Ascending fingering numbers appear at the top of each button and descending fingering numbers at the bottom.

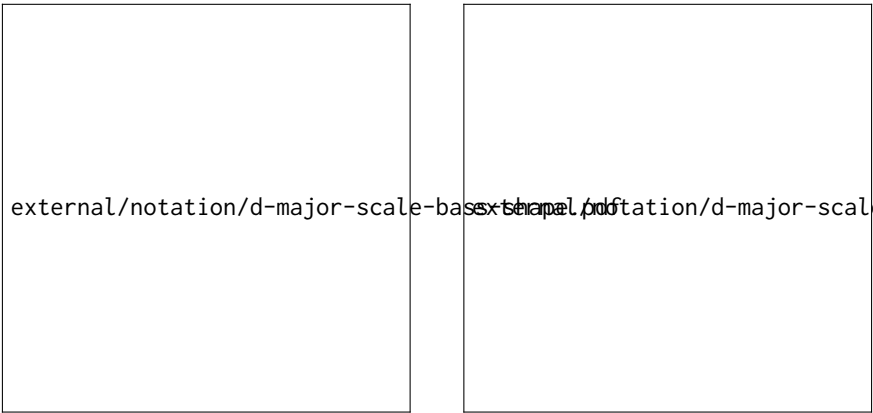


Figure 4.2.2

4.3 G Major Scale Shape

Use this shape to form other major scales where the shape fits, for example A B, C D, E.

4.3.1 Basic G Shape Scale

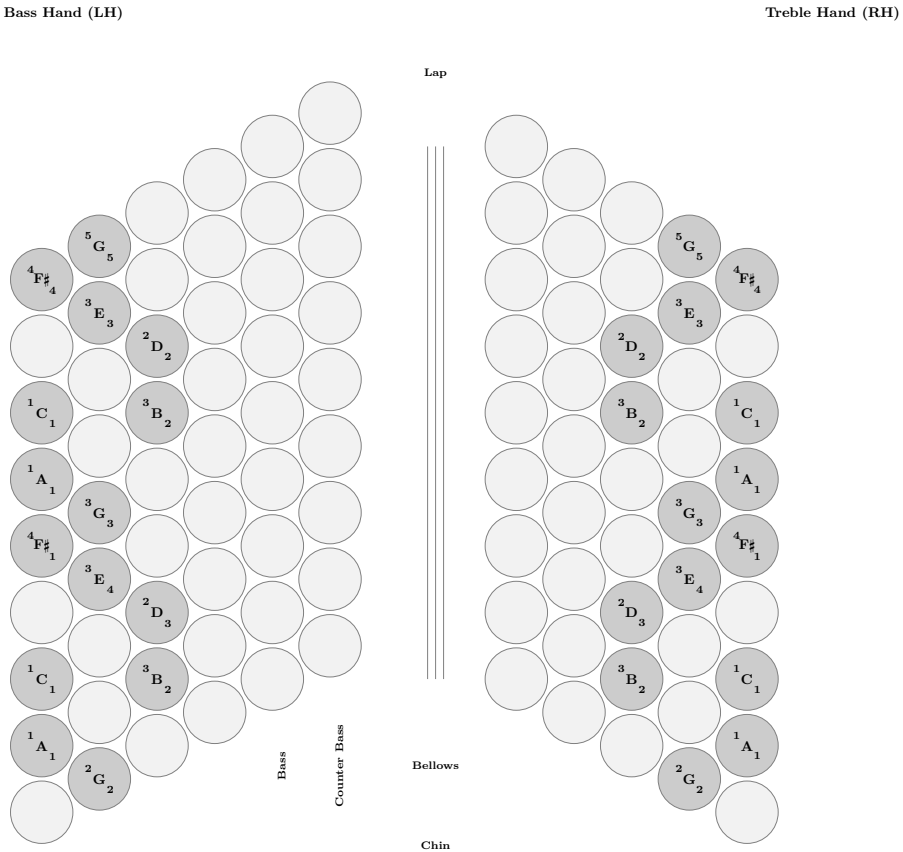


Figure 4.3.1 Basic G shape scale. Ascending fingering numbers appear at the top of each button and descending fingering numbers at the bottom.

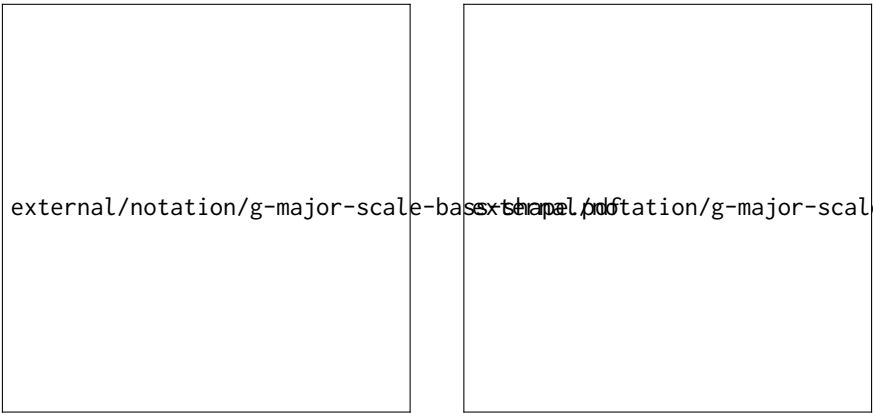


Figure 4.3.2

4.4 C Harmonic Minor Scale Shape

Use this shape to form other harmonic minor scales where the shape fits.

4.4.1 Basic C Harmonic Minor Shape

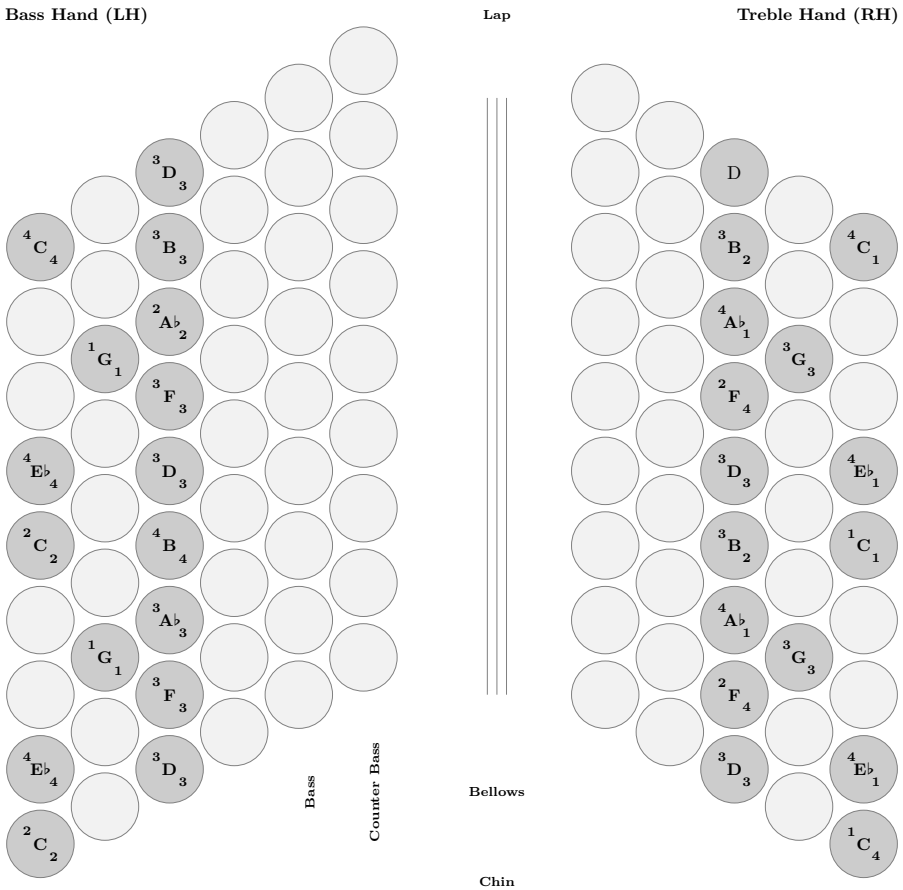


Figure 4.4.1 Basic C harmonic minor shape. Ascending fingering numbers appear at the top of each button and descending fingering numbers at the bottom.

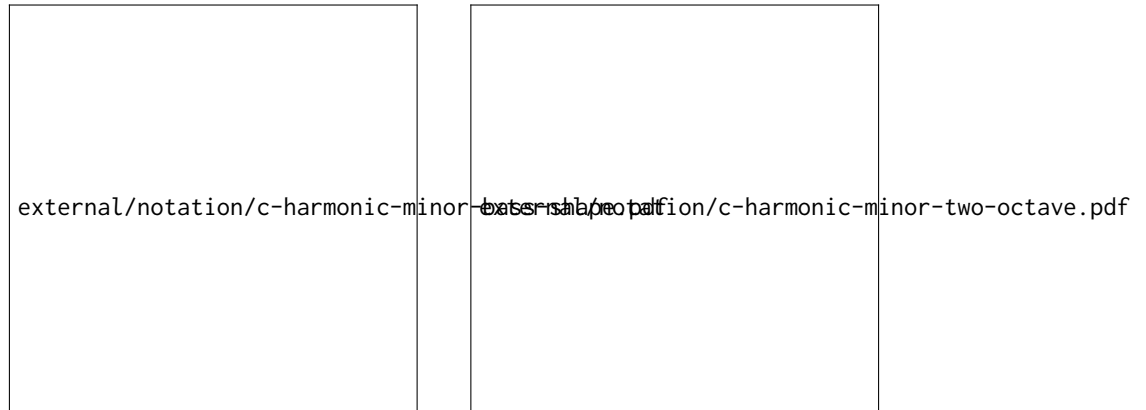


Figure 4.4.2

4.5 C Melodic Minor Scale Shape Ascending

Use this shape to form other melodic minor scales where the shape fits.

4.5.1 Basic C Melodic Minor Shape

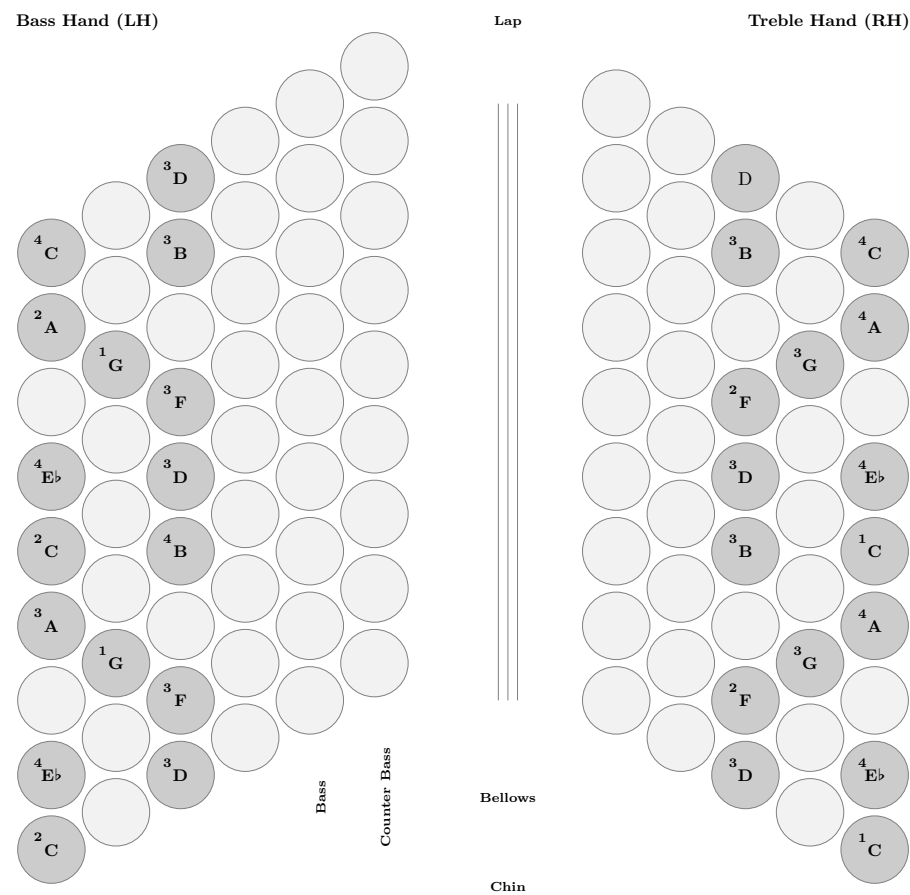


Figure 4.5.1 Basic C melodic minor shape. Ascending fingering numbers appear at the top of each button.



Figure 4.5.2

4.6 C Melodic Minor Descending Scale Shape

The descending form of the melodic minor scale is the same as the natural minor scale. Use this shape to form other descending melodic minor scales where the shape fits.

4.6.1 Basic C Melodic Minor Descending Shape

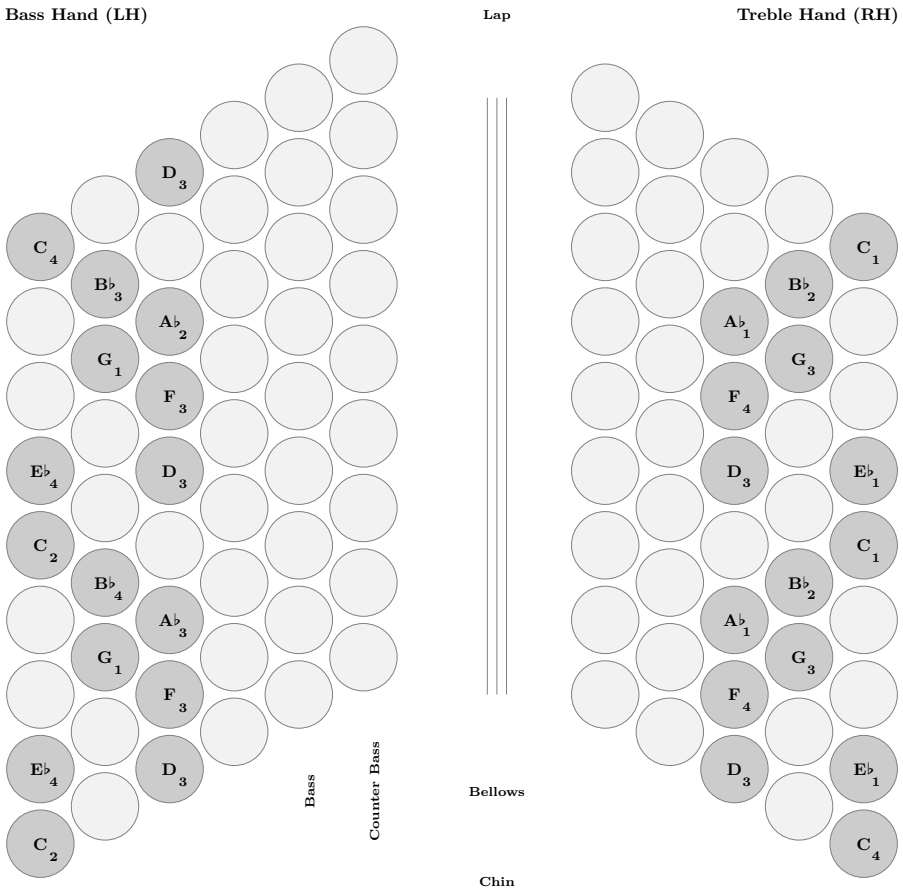


Figure 4.6.1 Basic C melodic minor descending shape (natural minor). Descending fingering numbers appear at the bottom of each button.

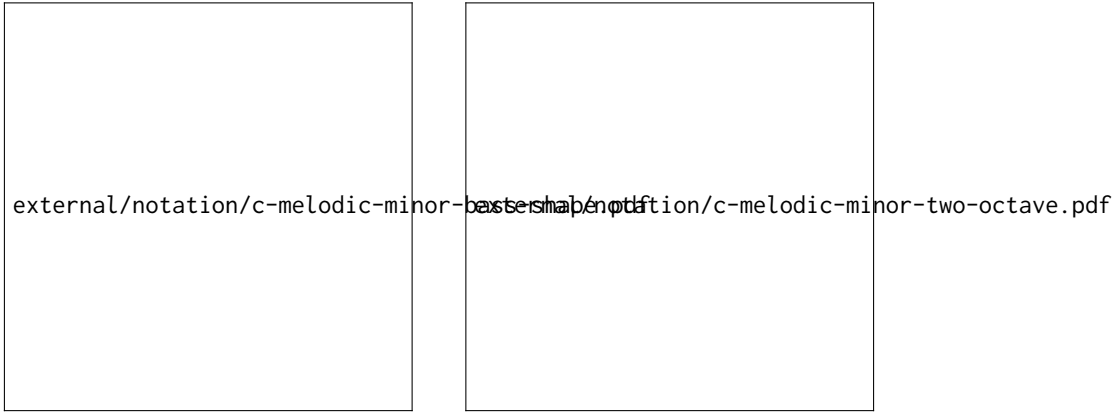


Figure 4.6.2

4.7 D Harmonic Minor Scale Shape

Use this shape to form other harmonic minor scales where the shape fits.

4.7.1 Basic D Harmonic Minor Shape

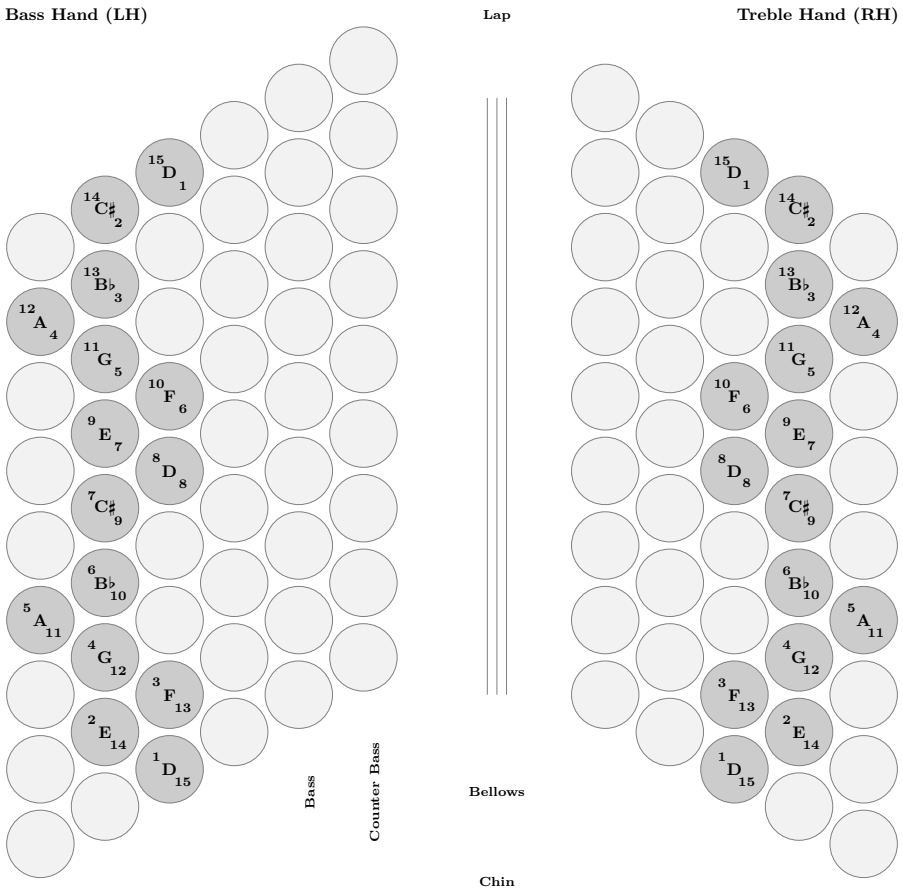


Figure 4.7.1 Basic D harmonic minor shape. Ascending fingering numbers appear at the top of each button and descending fingering numbers at the bottom.



Figure 4.7.2

4.8 D Melodic Minor Scale Shape Ascending

Use this shape to form other melodic minor scales where the shape fits.

4.8.1 Basic D Melodic Minor Shape

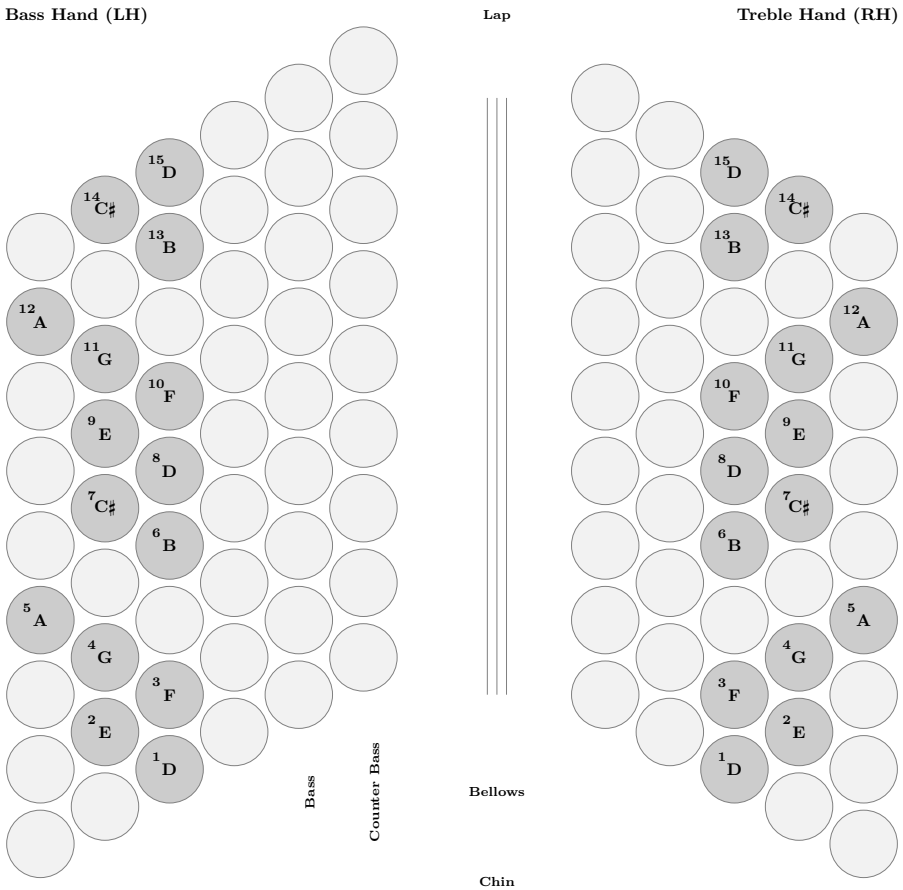


Figure 4.8.1 Basic D melodic minor shape. Ascending fingering numbers appear at the top of each button.



Figure 4.8.2

4.9 D Melodic Minor Descending Scale Shape

The descending form of the melodic minor scale is the same as the natural minor scale. Use this shape to form other descending melodic minor scales where the shape fits.

4.9.1 Basic D Melodic Minor Descending Shape

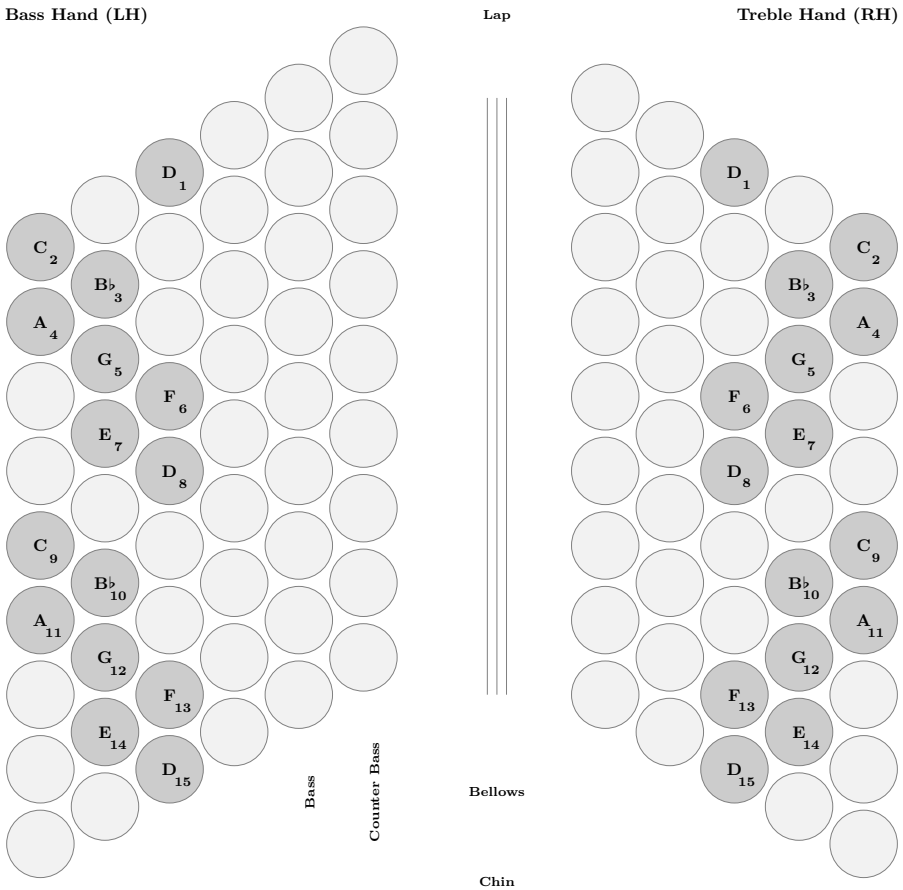


Figure 4.9.1 Basic D melodic minor descending shape (natural minor). Descending fingering numbers appear at the bottom of each button.



Figure 4.9.2

4.10 G Harmonic Minor Scale Shape

Use this shape to form other harmonic minor scales where the shape fits.

4.10.1 Basic G Harmonic Minor Shape

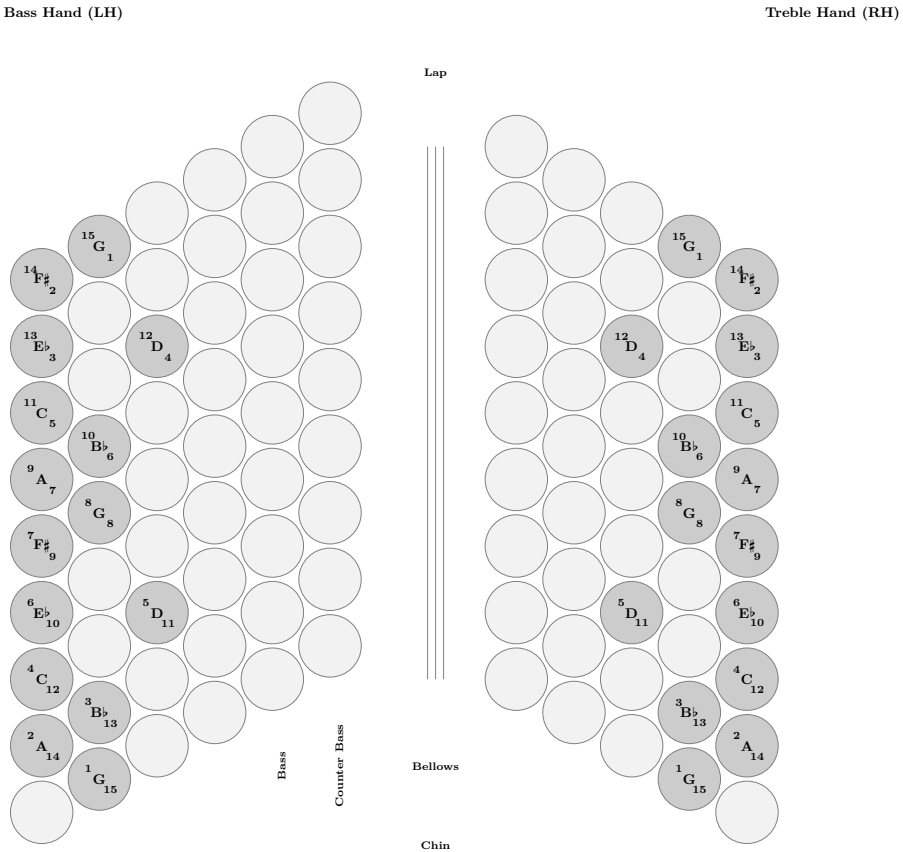


Figure 4.10.1 Basic G harmonic minor shape. Ascending fingering numbers appear at the top of each button and descending fingering numbers at the bottom.

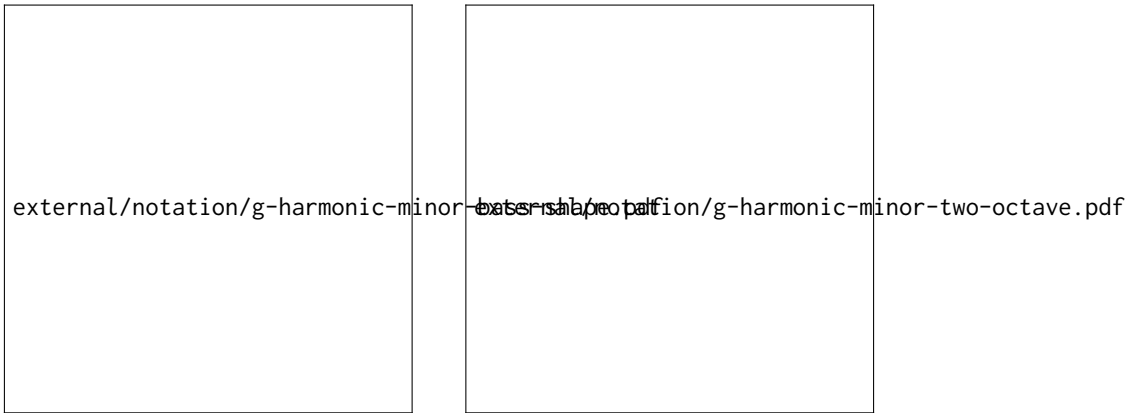


Figure 4.10.2

4.11 G Melodic Minor Scale Shape Ascending

Use this shape to form other melodic minor scales where the shape fits.

4.11.1 Basic G Melodic Minor Shape

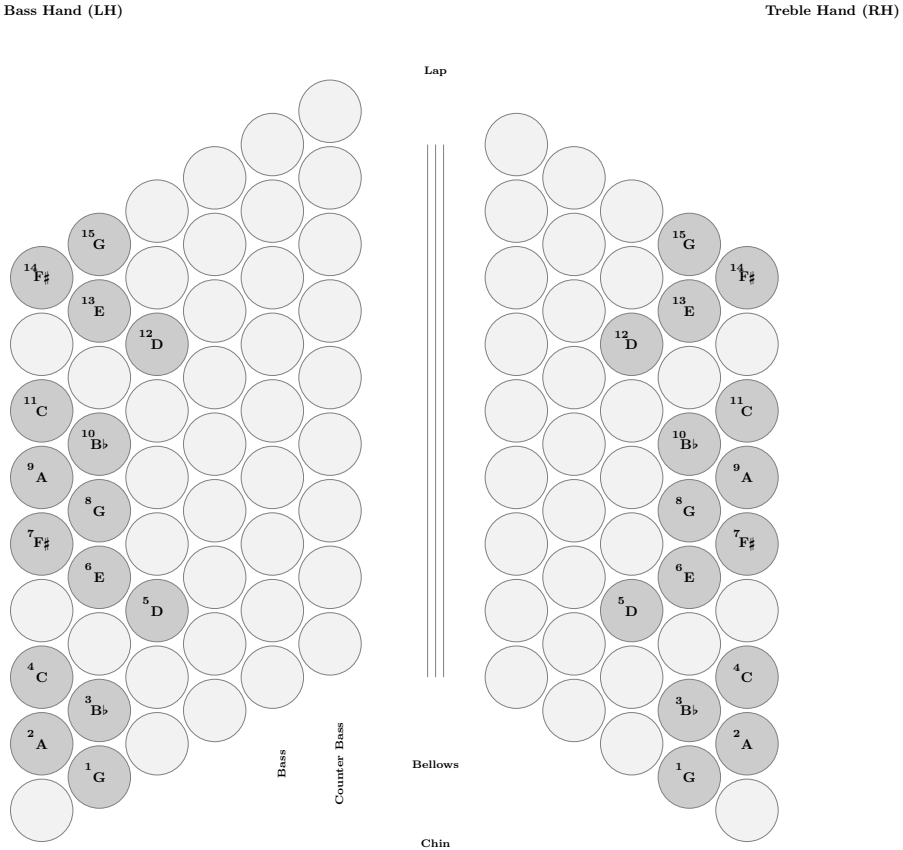


Figure 4.11.1 Basic G melodic minor shape. Ascending fingering numbers appear at the top of each button.



Figure 4.11.2

4.12 G Melodic Minor Descending Scale Shape

The descending form of the melodic minor scale is the same as the natural minor scale. Use this shape to form other descending melodic minor scales where the shape fits.

4.12.1 Basic G Melodic Minor Descending Shape

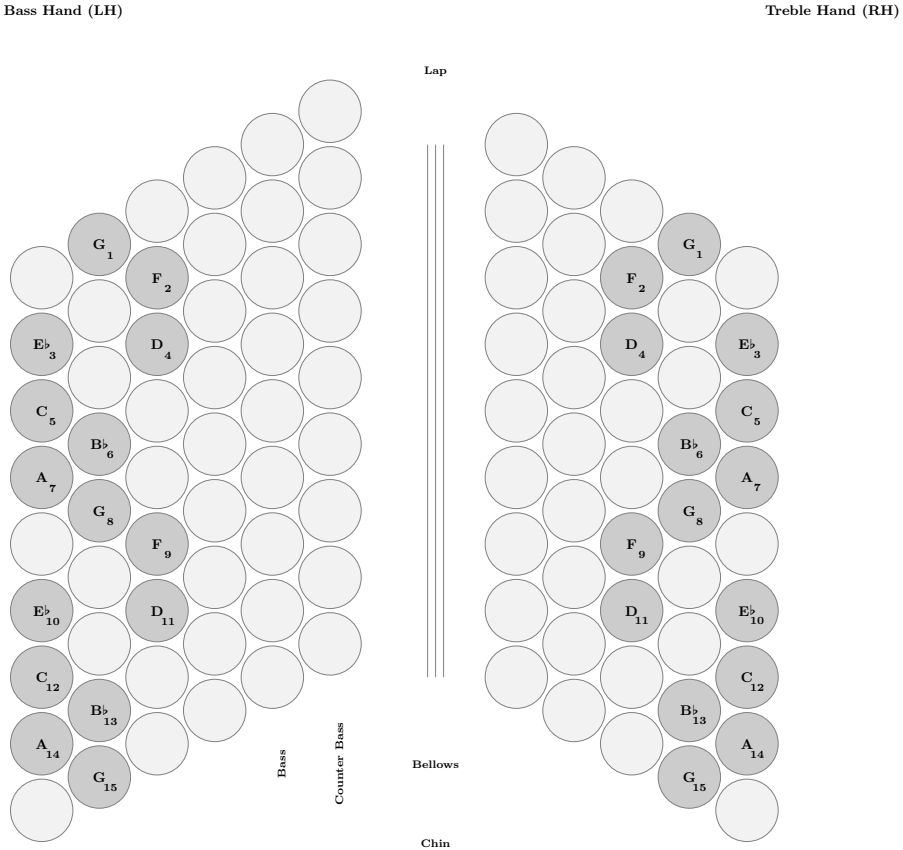


Figure 4.12.1 Basic G melodic minor descending shape (natural minor). Descending fingering numbers appear at the bottom of each button.



Figure 4.12.2

Chapter 5

Chords and Chord Construction

A **chord** is a combination of three or more notes played simultaneously. Chords are the harmonic building blocks of music — they define the tonal colour and emotional feel of a piece. Understanding how chords are constructed from intervals gives the accordion player the ability to build any chord from any root note, taking full advantage of the isomorphic layout of the chromatic button accordion.

This chapter explains the theory behind chord construction, starting with the intervals that make up a chord and progressing through triads and seventh chords. Because the chromatic button accordion keyboard is isomorphic, a chord shape learned in one position works identically in every key — learning the construction once gives you the chord everywhere on the keyboard.

5.1 Building Chords from Intervals

Chords are built by stacking intervals above a **root note**. The root is the note that names the chord — for example, the root of a C major chord is C. The other notes of the chord are determined by the specific intervals measured from the root.

5.1.1 Stacking Thirds

The most common way to construct a chord is by stacking intervals of a **third** above the root. There are two types of third:

- A **major third** spans 4 semitones (e.g. C up to E).
- A **minor third** spans 3 semitones (e.g. C up to E♭).

By choosing which type of third to place at each position in the stack, we produce chords with different qualities. A basic three-note chord built by stacking two thirds is called a **triad**.

Just as the twelve notes of the chromatic scale can be arranged around a clock face to visualise scale construction by stacking 2nds (one step clockwise each time), the seven notes of any scale can be arranged around a smaller circle to visualise chord construction by stacking 3rds. Moving *two steps clockwise* at each turn skips one scale note and lands on the next chord tone — the interval

of a third. The dashed triangle in the figure below connects scale degrees 1, 3, and 5: these are the three notes of a triad. Continuing the same two-step pattern extends the chord: degree 7 gives a seventh chord, degree 9 (= 2) gives a ninth chord, degree 11 (= 4) an eleventh chord, and degree 13 (= 6) a thirteenth chord. The note names shown are those of the C major scale, but the degree numbers apply to any major scale — replace the note names with those of the chosen scale and the pattern is identical.

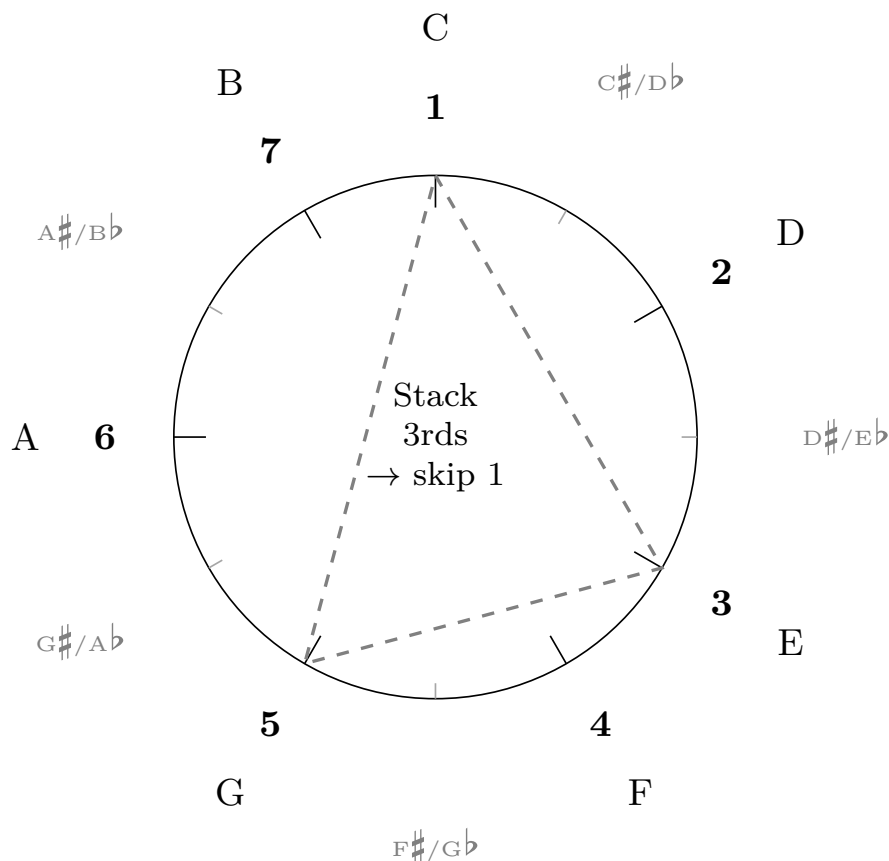


Figure 5.1.1 The 7 notes of the C major scale placed at their correct semitone positions around a 12-note circle (Root = C at 12 o'clock, one step = one semitone). Small grey ticks mark the 5 chromatic passing notes that fall between whole-tone scale steps. Moving two scale-degree steps clockwise at a time stacks thirds and builds chord tones: the dashed triangle connects the triad (1–3–5).

5.1.2 The Triad Formula

Every triad consists of three notes:

- The **root** — the note the chord is named after.
- The **third** — a third above the root (major or minor).
- The **fifth** — a third above the third (which lands a fifth above the root).

The interval from the root to the fifth may be a **perfect fifth** (7 semitones), an **augmented fifth** (8 semitones), or a **diminished fifth** (6 semitones), depending on the combination of thirds used.

5.2 Types of Triad

The four main triad types differ in the intervals between their notes. Using C as the root, the notes and interval structure for each type are as follows.

5.2.1 Major Triad

A **major triad** is built from a *major third* followed by a *minor third*:

- Root to third: major third (4 semitones) — e.g. C to E
- Third to fifth: minor third (3 semitones) — e.g. E to G
- Root to fifth: perfect fifth (7 semitones) — e.g. C to G

The major triad has a bright, stable sound. The C major triad consists of the notes C, E, and G.

5.2.2 Minor Triad

A **minor triad** is built from a *minor third* followed by a *major third*:

- Root to third: minor third (3 semitones) — e.g. C to E $\flat$
- Third to fifth: major third (4 semitones) — e.g. E $\flat$ to G
- Root to fifth: perfect fifth (7 semitones) — e.g. C to G

The minor triad has a darker, more melancholic sound. The C minor triad consists of the notes C, E $\flat$, and G.

5.2.3 Augmented Triad

An **augmented triad** is built from two consecutive *major thirds*:

- Root to third: major third (4 semitones) — e.g. C to E
- Third to fifth: major third (4 semitones) — e.g. E to G $\sharp$
- Root to fifth: augmented fifth (8 semitones) — e.g. C to G $\sharp$

The raised fifth gives the augmented triad a tense, unresolved quality. The C augmented triad consists of the notes C, E, and G $\sharp$.

5.2.4 Diminished Triad

A **diminished triad** is built from two consecutive *minor thirds*:

- Root to third: minor third (3 semitones) — e.g. C to E $\flat$
- Third to fifth: minor third (3 semitones) — e.g. E $\flat$ to G $\flat$
- Root to fifth: diminished fifth (6 semitones) — e.g. C to G $\flat$

The diminished fifth (also called a tritone) gives the diminished triad a dissonant, unstable sound that strongly seeks resolution. The C diminished triad consists of the notes C, E $\flat$, and G $\flat$.

The table below summarises the four triad types using C as the root. The number in brackets in this case is the "Interval Count" See the 2nd paragraph in Chapter 3 Scales and Intervals for an explanation of Interval Count which is always 1 more than the number of semitones:

Table 5.2.1 Summary of Triad Types (Root: C)

Triad Type	Root–3rd	3rd–5th	Root–5th	Notes (C root)
Major	Major 3rd (5)	Minor 3rd (4)	Perfect 5th (8)	C – E – G
Minor	Minor 3rd (4)	Major 3rd (5)	Perfect 5th (8)	C – E $\flat$ – G
Augmented	Major 3rd (5)	Major 3rd (5)	Aug. 5th (9)	C – E – G $\sharp$
Diminished	Minor 3rd (4)	Minor 3rd (4)	Dim. 5th (7)	C – E $\flat$ – G $\flat$

5.3 Extending Chords: Seventh Chords

A chord can be extended by adding a further third above the fifth, producing a **seventh chord**. The added note is a seventh above the root. The quality of the seventh (major, minor, or diminished) combined with the quality of the underlying triad determines the type of seventh chord.

The most commonly used seventh chords are:

- **Major seventh (maj7)** — major triad + major seventh (11 semitones above root). Example: C maj7 = C, E, G, B.
- **Dominant seventh (7)** — major triad + minor seventh (10 semitones above root). Example: C7 = C, E, G, B $\flat$.
- **Minor seventh (m7)** — minor triad + minor seventh (10 semitones above root). Example: Cm7 = C, E $\flat$, G, B $\flat$.
- **Half-diminished seventh (m7 $\flat$ 5)** — diminished triad + minor seventh (10 semitones above root). Example: Cm7 $\flat$ 5 = C, E $\flat$, G $\flat$, B $\flat$.
- **Diminished seventh (dim7)** — diminished triad + diminished seventh (9 semitones above root). Example: Cdim7 = C, E $\flat$, G $\flat$, B $\flat\flat$ (A).

The physical shapes of these chords on the chromatic button accordion keyboard are explored in detail in the following chapter.

5.4 Extended and Suspended Chords

Beyond seventh chords, chords can be extended still further by continuing to stack thirds above the seventh. Each new note is named for its interval above the root when counted through the full scale: the **ninth**, **eleventh**, and **thirteenth**. In practice these are the same pitches as the second, fourth, and sixth — but placed an octave higher and heard as colour tones layered above the harmonic foundation. All extended chords include the seventh; omitting it collapses the chord back to a triad or simple added-note chord.

Suspended chords (sus chords) work differently: the third is *replaced* by either the second or the fourth, removing the major/minor quality and creating an open, unresolved sound.

5.4.1 Ninth Chords

A **ninth chord** extends a seventh chord by adding the note a major ninth (14 semitones) above the root — the same pitch as the major second, an octave higher. The three most common ninth chords, using C as root, are:

- **Major ninth (maj9)** — major seventh chord + major 9th. Formula: 1 – maj3 – per5 – maj7 – maj9. Notes: C – E – G – B – D.

- **Dominant ninth (9)** — dominant seventh chord + major 9th. Formula: 1 – maj3 – per5 – min7 – maj9. Notes: C – E – G – B♭ – D.
- **Minor ninth (m9)** — minor seventh chord + major 9th. Formula: 1 – min3 – per5 – min7 – maj9. Notes: C – E♭ – G – B♭ – D.

A **flat ninth (♭9)** uses a minor ninth (13 semitones) instead, most commonly over a dominant chord (C7♭9 = C – E – G – B♭ – D♭), creating a strongly dissonant, tension-building sound widely used in jazz and flamenco.

5.4.2 Eleventh Chords

An **eleventh chord** adds the perfect eleventh (17 semitones above the root — the same pitch as the perfect fourth, an octave up) to a ninth chord. In major and dominant contexts the third is typically *omitted* in practice, because it sits only a semitone below the eleventh and causes a harsh clash. The third is retained in minor eleventh chords because the minor third sits a whole tone below the eleventh.

- **Major eleventh (maj11)** — 1 – (3) – 5 – maj7 – 9 – 11. Notes: C – (E) – G – B – D – F. The third E is commonly omitted.
- **Dominant eleventh (11)** — 1 – (3) – 5 – min7 – 9 – 11. Notes: C – (E) – G – B♭ – D – F. The third E is commonly omitted.
- **Minor eleventh (m11)** — 1 – ♭3 – 5 – min7 – 9 – 11. Notes: C – E♭ – G – B♭ – D – F.

A **sharp eleventh (#11)** raises the eleventh by a semitone (18 semitones from the root = F♯ for C root), associated with the Lydian mode. Example: Cmaj7♯11 = C – E – G – B – D – F♯.

5.4.3 Thirteenth Chords

A **thirteenth chord** completes the stack of thirds by adding the major thirteenth (21 semitones above the root — the same pitch as the major sixth, an octave up). A full thirteenth chord theoretically contains all seven notes of the major scale; in practice the eleventh (and sometimes the ninth) is omitted to keep the voicing playable and avoid excessive dissonance.

- **Major thirteenth (maj13)** — 1 – 3 – 5 – maj7 – 9 – (11) – 13. Notes: C – E – G – B – D – (F) – A.
- **Dominant thirteenth (13)** — 1 – 3 – 5 – min7 – 9 – (11) – 13. Notes: C – E – G – B♭ – D – (F) – A.
- **Minor thirteenth (m13)** — 1 – ♭3 – 5 – min7 – 9 – (11) – 13. Notes: C – E♭ – G – B♭ – D – (F) – A.

5.4.4 Added Sixth Chords

Added sixth chords are built by adding the major sixth (9 semitones above the root) to a triad — *without* a seventh. This distinguishes them from thirteenth chords, which always include the seventh. Added sixth chords have a bright, open character widely heard in jazz, pop, and bossa nova.

- **Major sixth (6)** — major triad + major 6th. Formula: 1 – maj3 – per5 – maj6. Notes: C – E – G – A.

- **Minor sixth (m6)** — minor triad + major 6th. Formula: 1 – min3 – per5 – maj6. Notes: C – E♭ – G – A.

The **6/9 chord** extends the added sixth chord by also including the major ninth (same pitch class as the major second, an octave higher), still with no seventh. This produces a lush, full sound popular in jazz and neo-soul.

- **Major 6/9 (6/9)** — major triad + major 6th + major 9th. Formula: 1 – maj3 – per5 – maj6 – maj9. Notes: C – E – G – A – D.
- **Minor 6/9 (m6/9)** — minor triad + major 6th + major 9th. Formula: 1 – min3 – per5 – maj6 – maj9. Notes: C – E♭ – G – A – D.

5.4.5 Suspended Chords

In a **suspended chord** the third is replaced by either the major second or the perfect fourth. This removes the chord's major or minor quality, producing an open, floating sound that traditionally resolves by moving the suspended note back to the third.

- **Sus2 (suspended second)** — the third is replaced by the major second (2 semitones above root). Notes: C – D – G.
- **Sus4 (suspended fourth)** — the third is replaced by the perfect fourth (5 semitones above root). Notes: C – F – G.
- **Dominant seventh suspended fourth (7sus4)** — sus4 triad with a minor seventh added. Extremely common in pop and jazz as an unresolved dominant colour. Notes: C – F – G – B♭.
- **Major seventh suspended fourth (maj7sus4)** — sus4 triad with a major seventh added. Notes: C – F – G – B.

The matrix below gives a complete overview of seventh, extended, and suspended chord construction using C as the root. Notes shown in parentheses are present in the full theoretical voicing but are *commonly omitted* in practice. A dash (—) indicates the note is not part of that chord type. The column “3/sus” shows the third, or the note that replaces it in suspended chords (marked *).

Table 5.4.1 Chord Construction Matrix (Root: C)

Chord Type	Symbol	1	3/sus	5	7	9	11	13	Notes (C root)
Major 7th	Cmaj7	C	E	G	B	—	—	—	C E G B
Dominant 7th	C7	C	E	G	B $\flat$	—	—	—	C E G B $\flat$
Minor 7th	Cm7	C	E $\flat$	G	B $\flat$	—	—	—	C E $\flat$ G B $\flat$
Half-diminished	Cm7 $\flat$ 5	C	E $\flat$	G $\flat$	B $\flat$	—	—	—	C E $\flat$ G $\flat$ B $\flat$
Diminished 7th	Cdim7	C	E $\flat$	G $\flat$	A $\flat$	—	—	—	C E $\flat$ G $\flat$ A $\flat$
Major 9th	Cmaj9	C	E	G	B	D	—	—	C E G B D
Dominant 9th	C9	C	E	G	B $\flat$	D	—	—	C E G B $\flat$ D
Minor 9th	Cm9	C	E $\flat$	G	B $\flat$	D	—	—	C E $\flat$ G B $\flat$ D
Dom. flat 9th	C7 $\flat$ 9	C	E	G	B $\flat$	D $\flat$	—	—	C E G B $\flat$ D $\flat$
Major 11th	Cmaj11	C	(E)	G	B	D	F	—	C (E) G B D F
Dominant 11th	C11	C	(E)	G	B $\flat$	D	F	—	C (E) G B $\flat$ D F
Minor 11th	Cm11	C	E $\flat$	G	B $\flat$	D	F	—	C E $\flat$ G B $\flat$ D F
Dom. sharp 11th	C7 $\sharp$ 11	C	E	G	B $\flat$	D	F $\sharp$	—	C E G B $\flat$ D F $\sharp$
Major 13th	Cmaj13	C	E	G	B	D	(F)	A	C E G B D (F) A
Dominant 13th	C13	C	E	G	B $\flat$	D	(F)	A	C E G B $\flat$ D (F) A
Minor 13th	Cm13	C	E $\flat$	G	B $\flat$	D	(F)	A	C E $\flat$ G B $\flat$ D (F) A
Major 6th	C6	C	E	G	—	—	—	A $\dagger$	C E G A
Minor 6th	Cm6	C	E $\flat$	G	—	—	—	A $\dagger$	C E $\flat$ G A
Major 6/9	C6/9	C	E	G	—	D	—	A $\dagger$	C E G A D
Minor 6/9	Cm6/9	C	E $\flat$	G	—	D	—	A $\dagger$	C E $\flat$ G A D
Suspended 2nd	Csus2	C	D*	G	—	—	—	—	C D G
Suspended 4th	Csus4	C	F*	G	—	—	—	—	C F G
Dom. 7th sus4	C7sus4	C	F*	G	B $\flat$	—	—	—	C F G B $\flat$
Maj. 7th sus4	Cmaj7sus4	C	F*	G	B	—	—	—	C F G B

Parentheses () indicate a note that is theoretically present but commonly omitted in practice. An asterisk * in the “3/sus” column indicates a suspended note replacing the third: D (2nd degree) for sus2, F (4th degree) for sus4. A dagger $\dagger$ in the “13” column indicates an added major sixth (same pitch class as the 13th, but no seventh is present in those chords).

Chapter 6

Building Chords from Scales

Any major scale can be used as a source for building chords. By stacking thirds at each scale degree — **selecting every other note of the scale above a starting note** — we produce a **diatonic** chord: a chord whose notes all belong to that key.

Adding a second stacked third (a fourth note) above the initial triad produces a **seventh chord**. Seventh chords add harmonic richness and are fundamental to jazz, classical, and popular music. There are exactly seven diatonic seventh chords in every major key — one built on each scale degree.

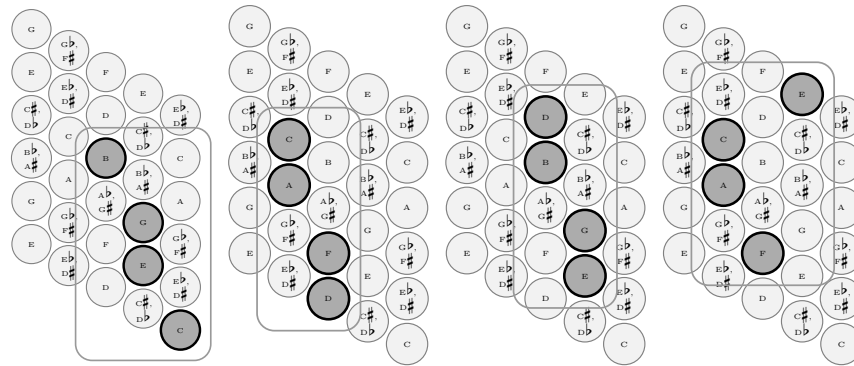
The table below shows the seven diatonic seventh chords built from the C major scale, together with the quality (type) of seventh chord produced at each scale degree.

Scale Degree	Roman Numeral	Chord	Notes	Quality
1st — C	I	Cmaj7	C, E, G, B	Major 7th
2nd — D	II	Dm7	D, F, A, C	Minor 7th
3rd — E	III	Em7	E, G, B, D	Minor 7th
4th — F	IV	Fmaj7	F, A, C, E	Major 7th
5th — G	V	G7	G, B, D, F	Dominant 7th
6th — A	VI	Am7	A, C, E, G	Minor 7th
7th — B	VII	Bm7♭5	B, D, F, A	Half-Diminished 7th

Because the chromatic button accordion keyboard is isomorphic, each chord shape is *moveable*: learn the shape in C and you can play the same chord type from any root button on the keyboard.

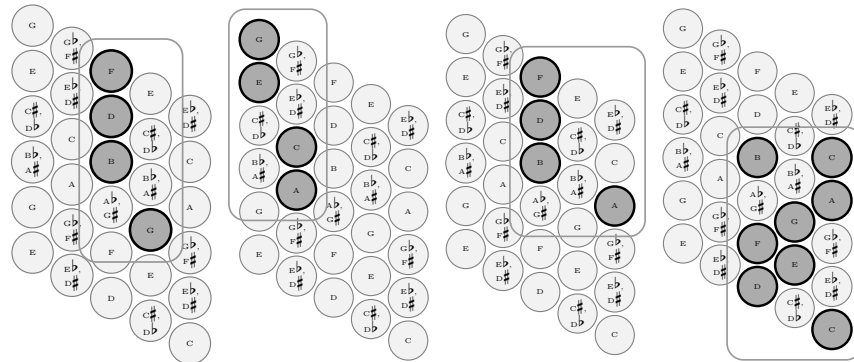
6.1 Diatonic Seventh Chords of C Major

The grids below show each of the seven diatonic seventh chords of C major in root position. Highlighted buttons are the four chord tones. The final grid shows the full C major scale for reference, making it easy to see how the chord tones are drawn from the scale.

**Figure 6.1.1** Diatonic 7th Chords — Scale Degrees I to IV

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I — Cmaj7 (C, E, G, B) II — Dm7 (D, F, A, C) III — Em7 (E, G, B, D) IV — Fmaj7 (F, A, C, E)

**Figure 6.1.2** Diatonic 7th Chords — Scale Degrees V to VII and C Major Scale

external/notation/diat-7maj5.pdf external/notation/diat-7maj6.pdf external/notation/diat-7maj7.pdf external/notation/diat-bm7b5.pdf

V — G7 (G, B, D, F) VI — Am7 (A, C, E, G) VII — Bm7b5 (B, D, F, A) C Major Scale (C–D–E–F–G–A–B–C)

Chapter 7

Chord Shapes

7.1 Major and Minor Triads

These triad shapes are moveable: the shape stays the same regardless of which button you start on, taking the name of the root note. The first row shows the four root-position triad types; the second row shows their common inversions.

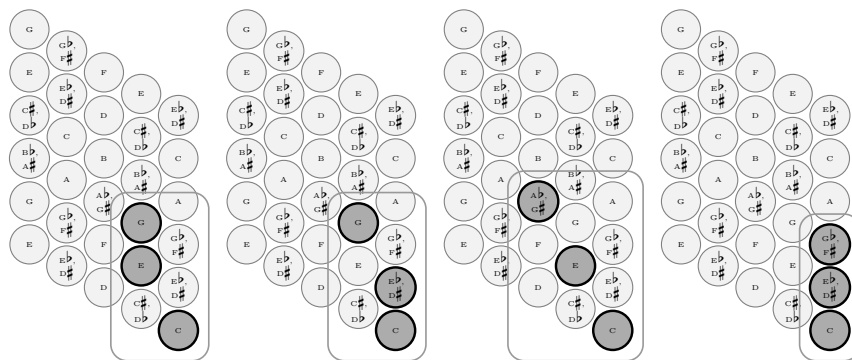


Figure 7.1.1

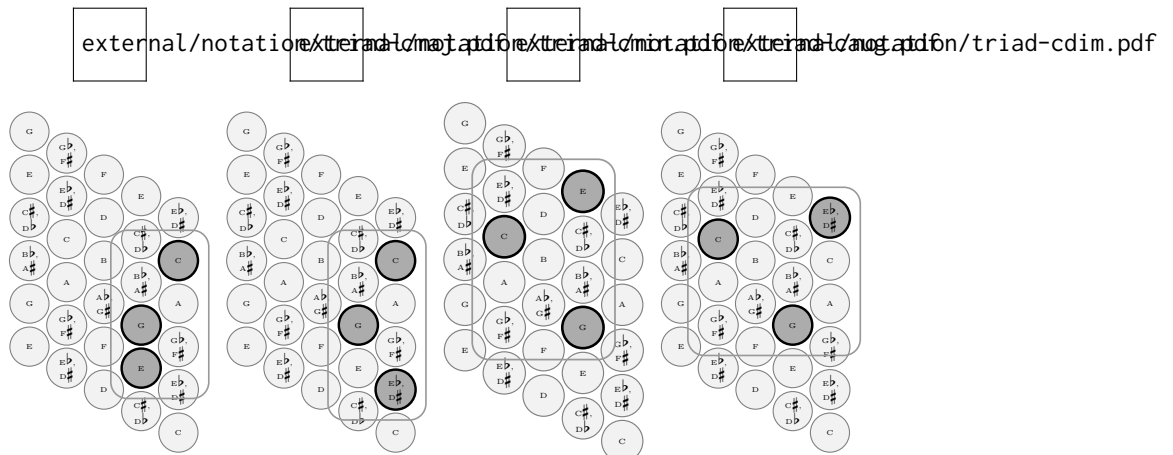


Figure 7.1.2

7.2 Seventh Chords

The Chord Shapes shown can be moved to different positions on the keyboard and take the name of the Root Chord. Thus, if you know the shape of one chord type, you know the shape of all chords of that type in that shape or inversion.

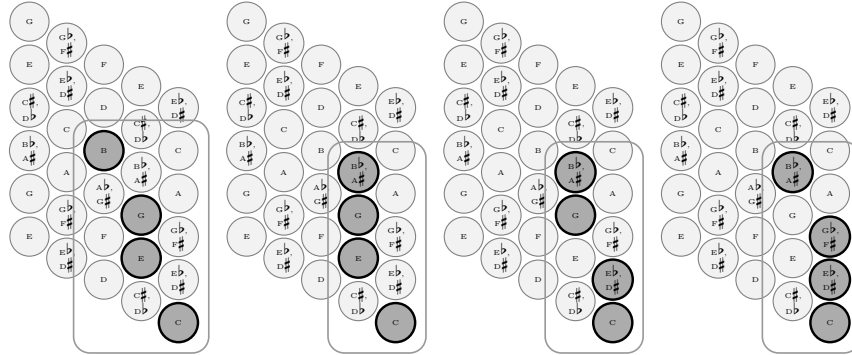


Figure 7.2.1 7th Chords Root Position

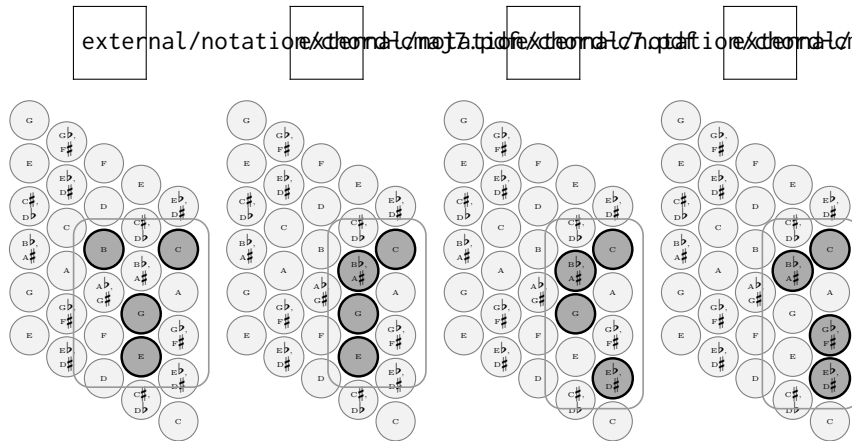


Figure 7.2.2 7th Chords 1st Inversion

7.3 Seventh Chords (2nd Inversion)

The Chord Shapes shown can be moved to different positions on the keyboard and take the name of the Root Chord. Thus, if you know the shape of one chord type, you know the shape of all chords of that type in that shape or inversion.

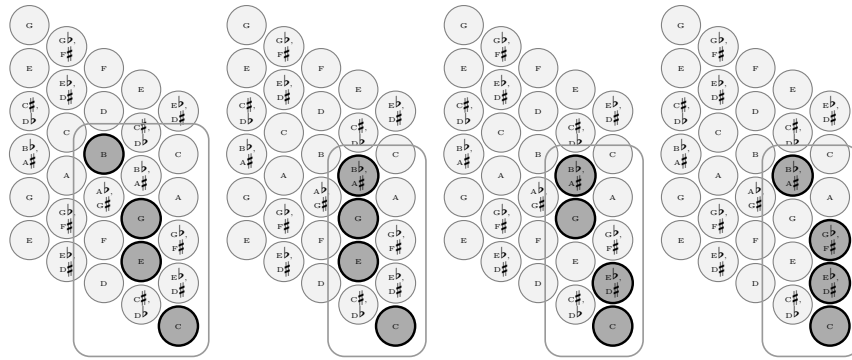


Figure 7.3.1 7th Chords Root Position

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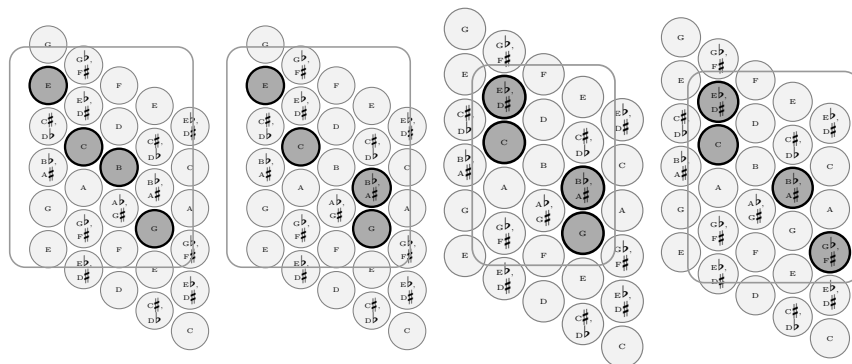


Figure 7.3.2 7th Chords 2nd Inversion

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7.4 Seventh Chords (3rd Inversion and Cdim7)

The Chord Shapes shown can be moved to different positions on the keyboard and take the name of the Root Chord. Thus, if you know the shape of one chord type, you know the shape of all chords of that type in that shape or inversion.

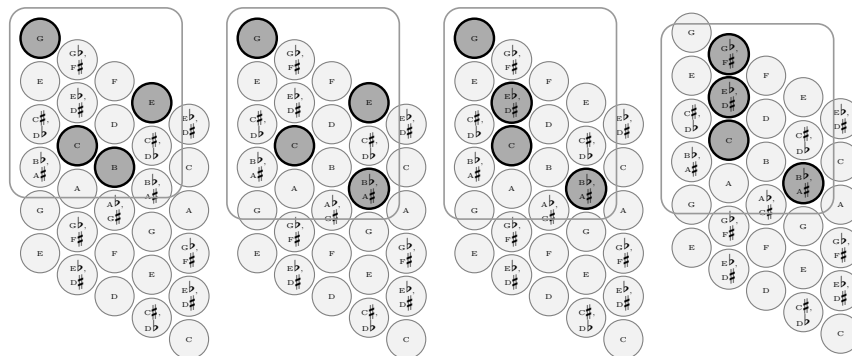


Figure 7.4.1 7th Chords 3rd Inversion

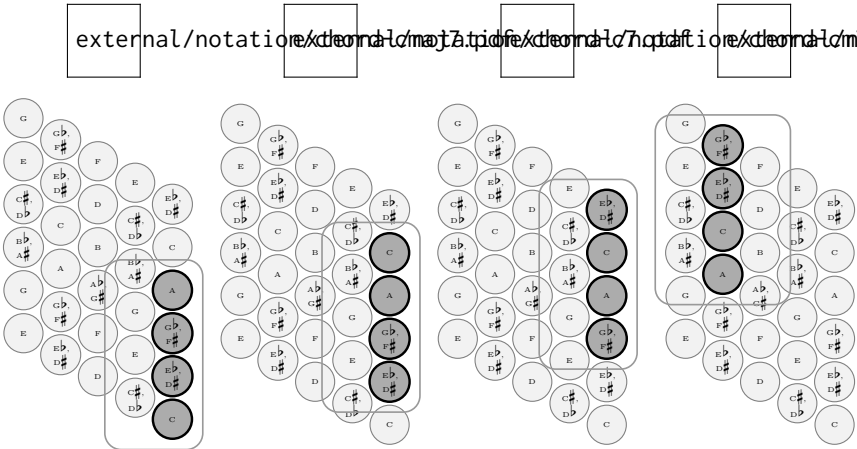
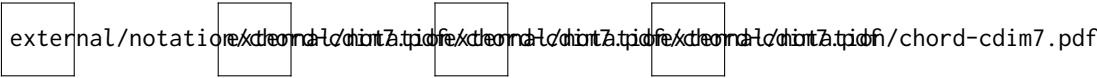


Figure 7.4.2 Cdim7 Root Position and Inversions



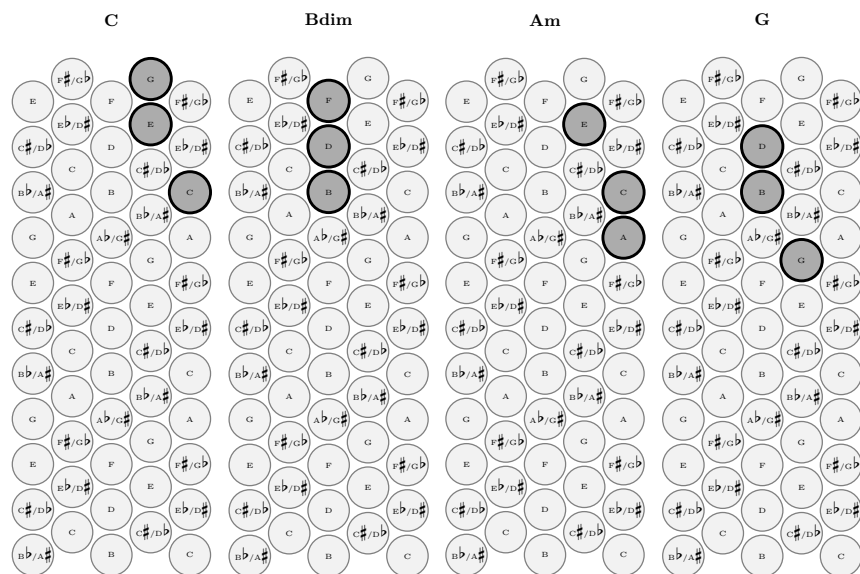
Chapter 8

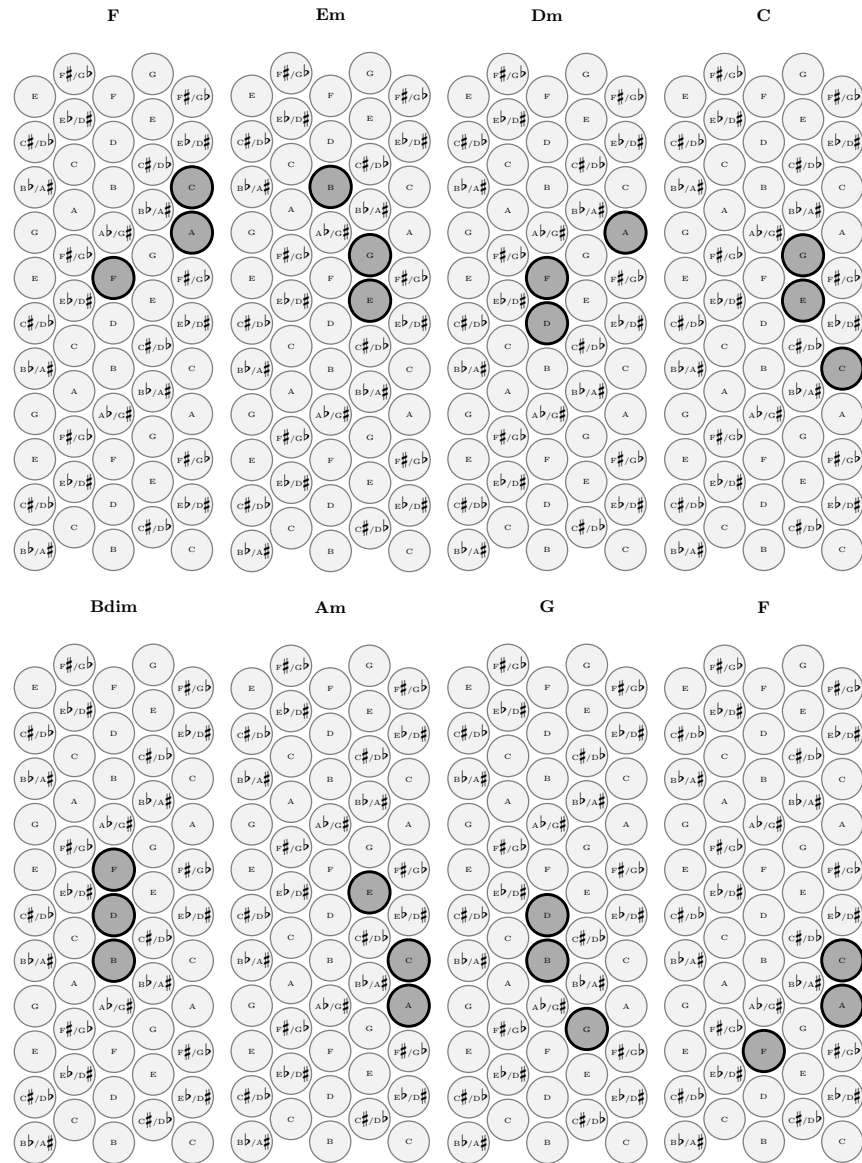
Chord Scale Cyclic Movement

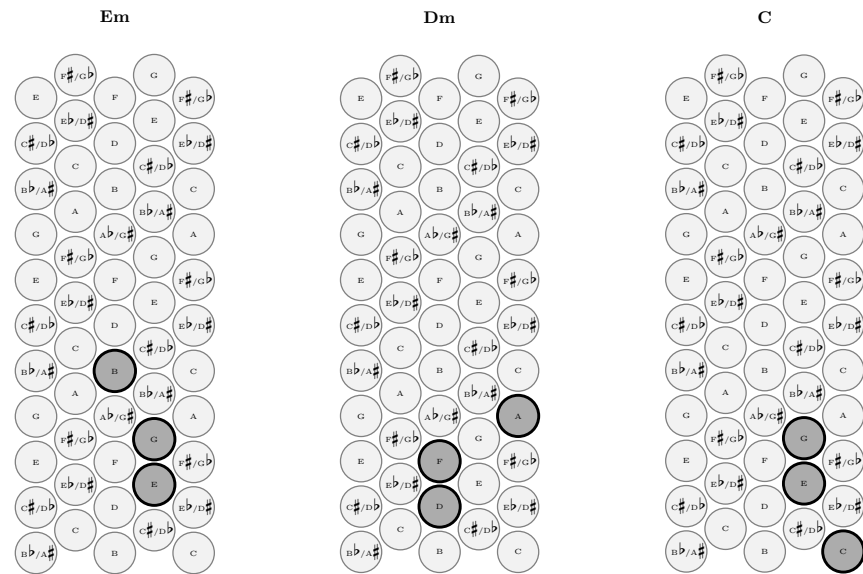
Use these exercises to practice chord scale movement providing useful ear training as well as familiarity with the keyboard when moving different voices independently as each chord moves to the next. Each voicing is arranged for minimal movement and smoothest voice leading.

New chord shapes will be introduced in these exercises and exposure to them, as well as looking at how each chord is constructed, will help you understand their function. Each cycle will use different inversions.

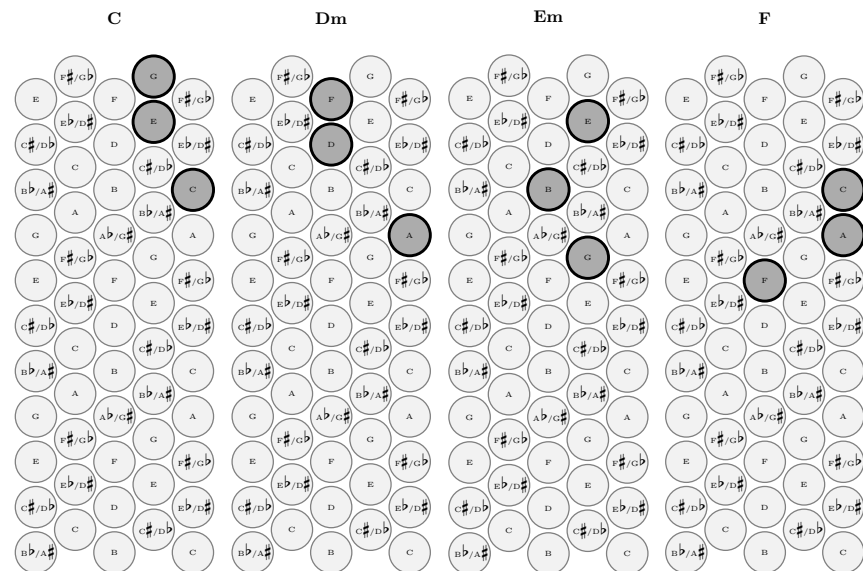
8.1 Closed Voice Triads from C Major Scale, Progressing Down in Intervals of a Second

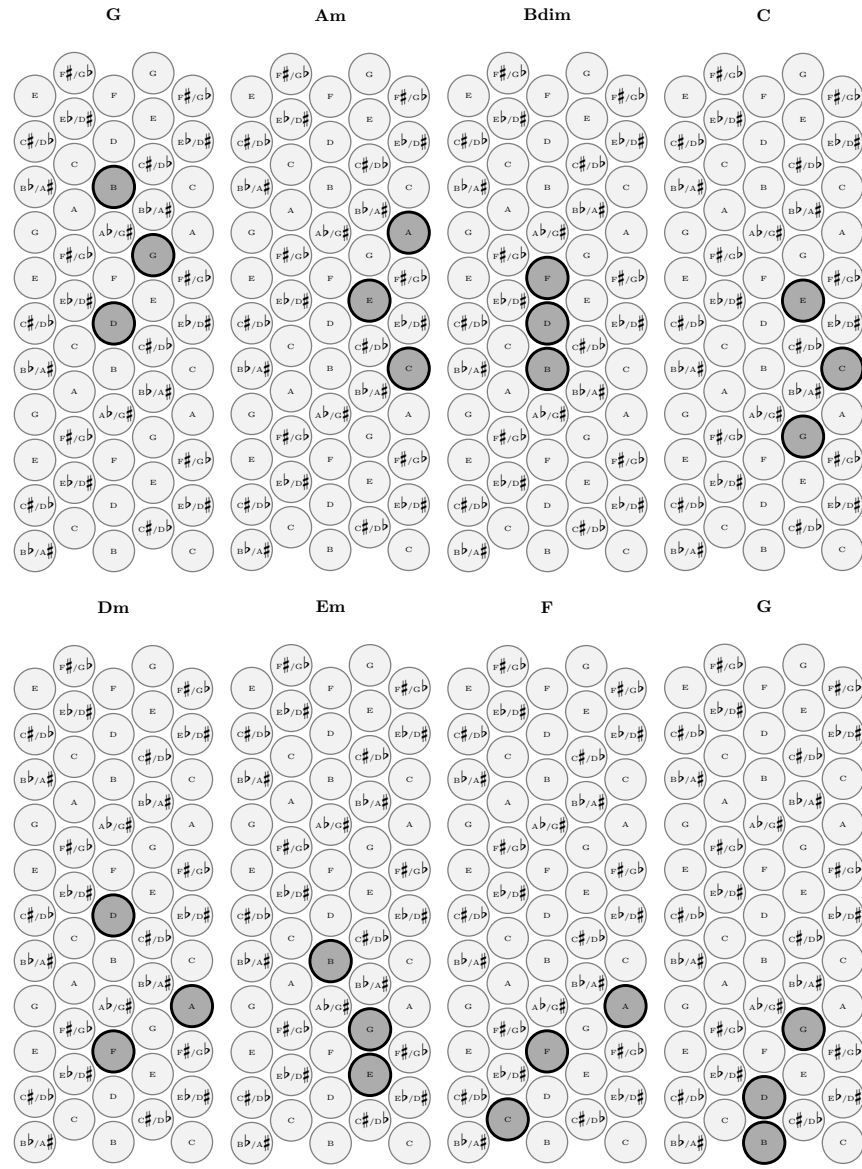


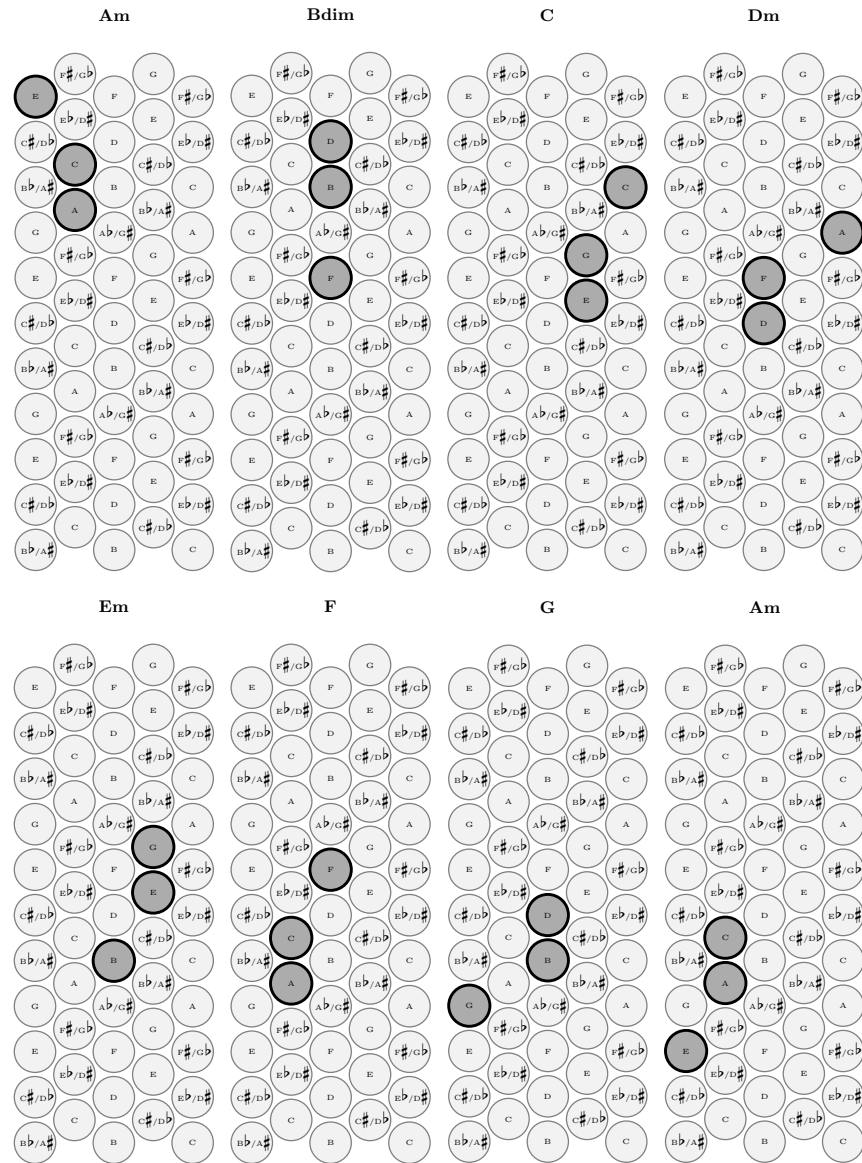


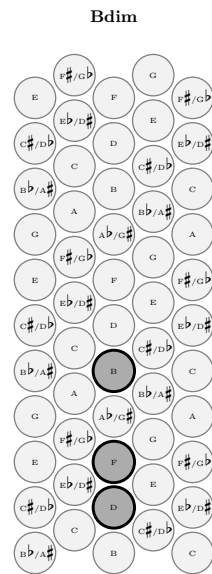


8.2 Closed Voice Triads from C Major Scale, Roots Ascending in 2nds (Cycle 2)

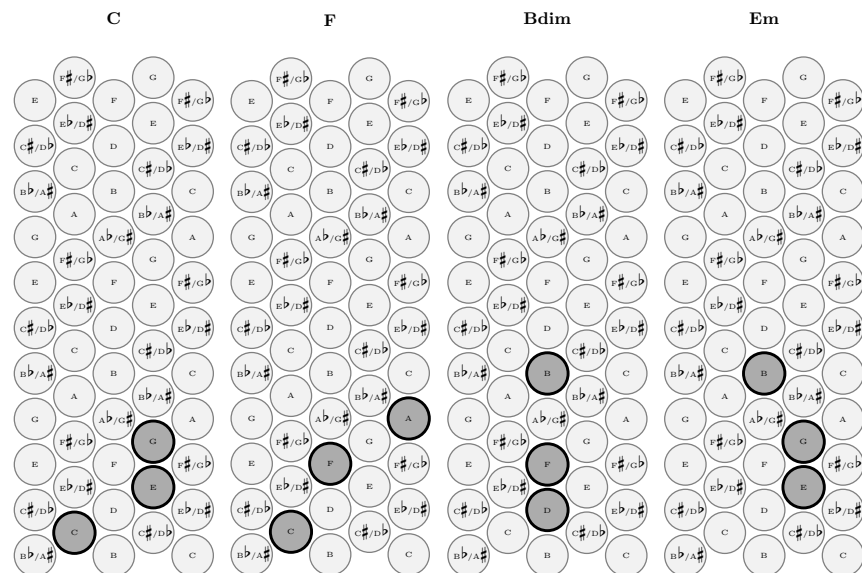


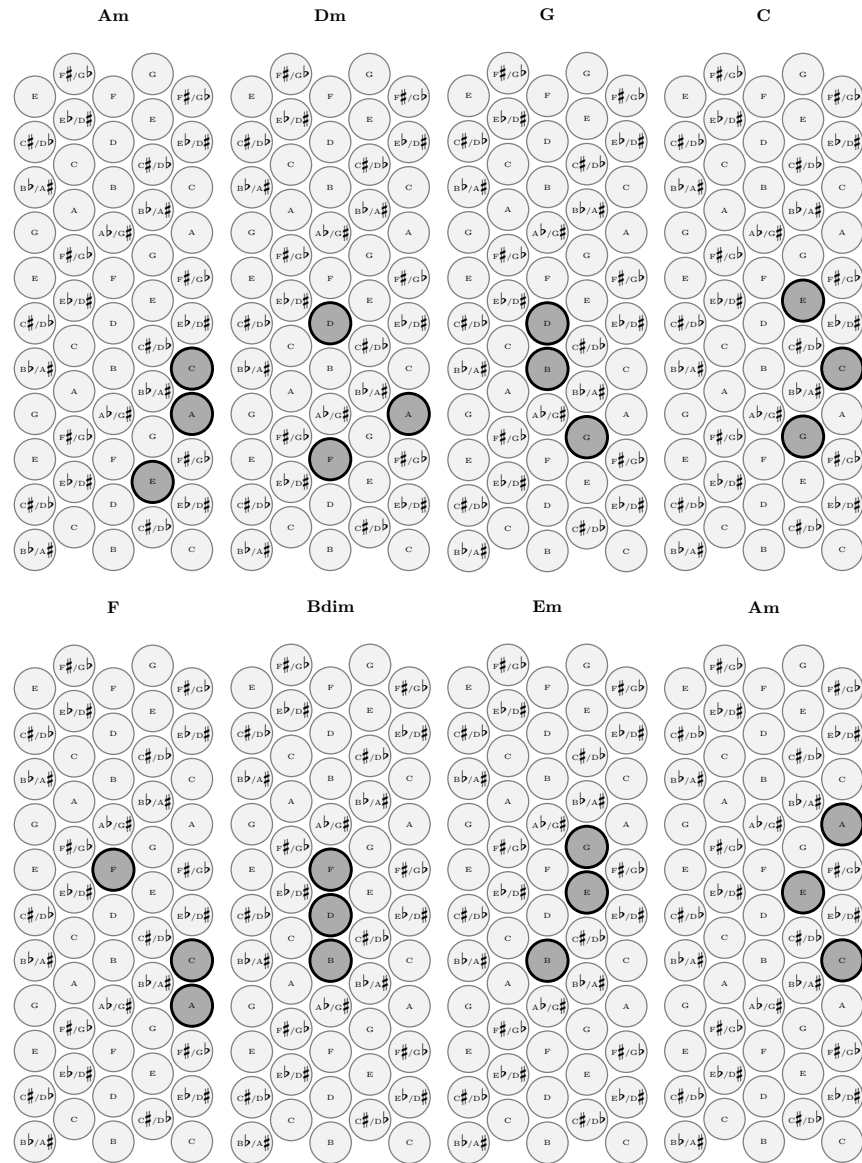


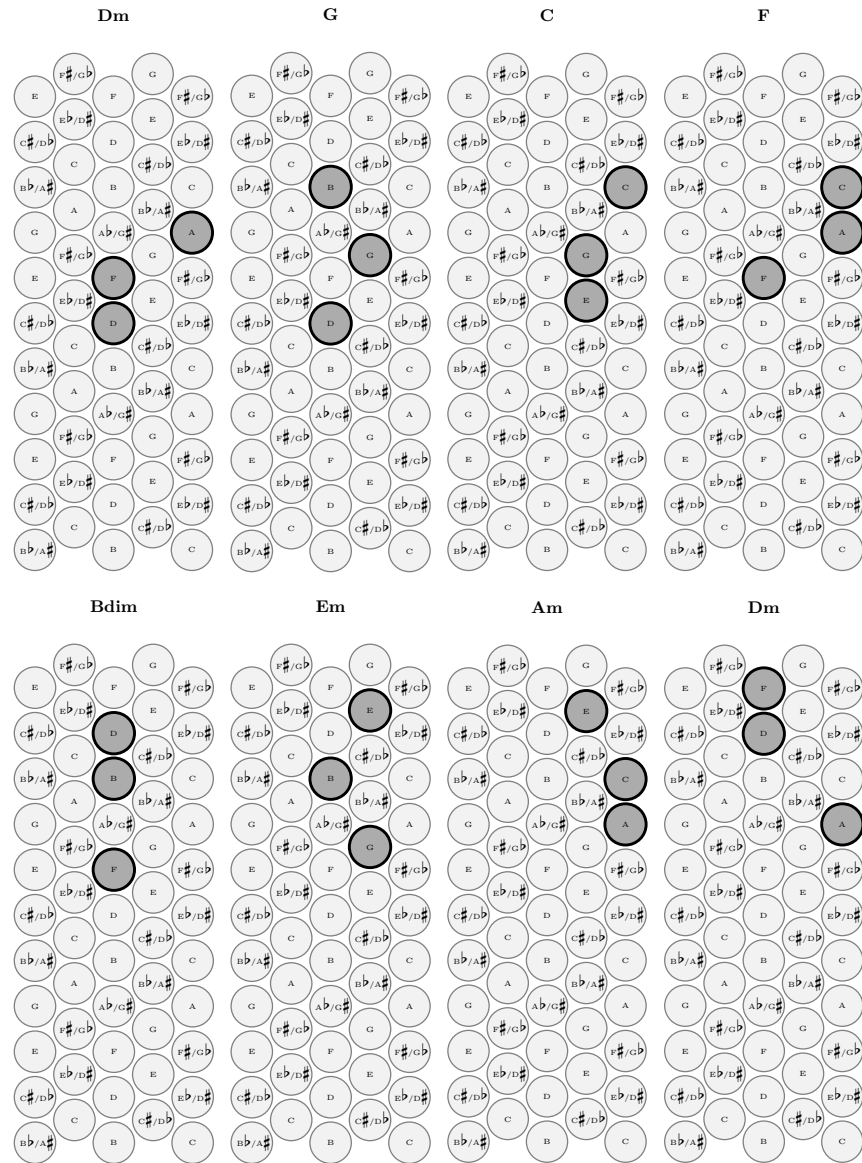


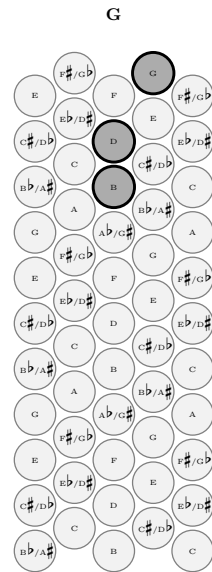


8.3 Closed Voice Triads from C Major Scale, Roots rising by diatonic 4th (Cycle 4)

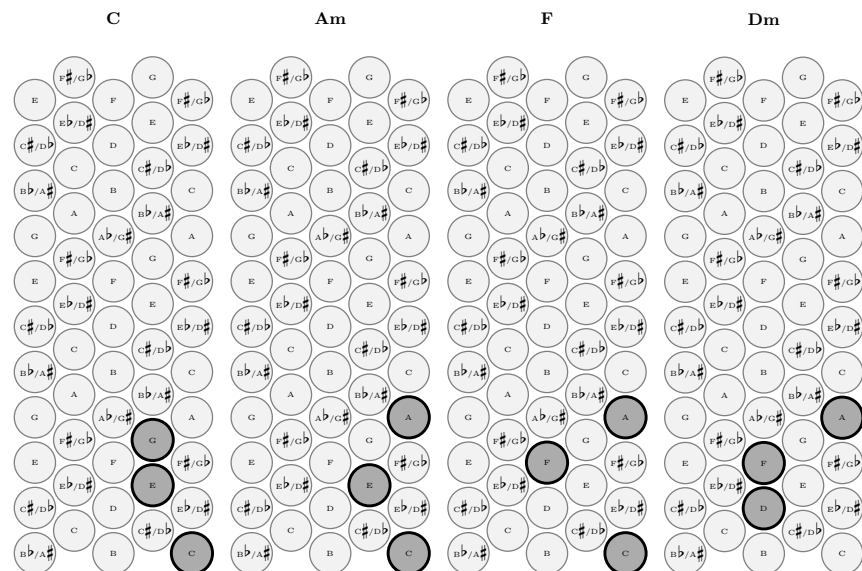


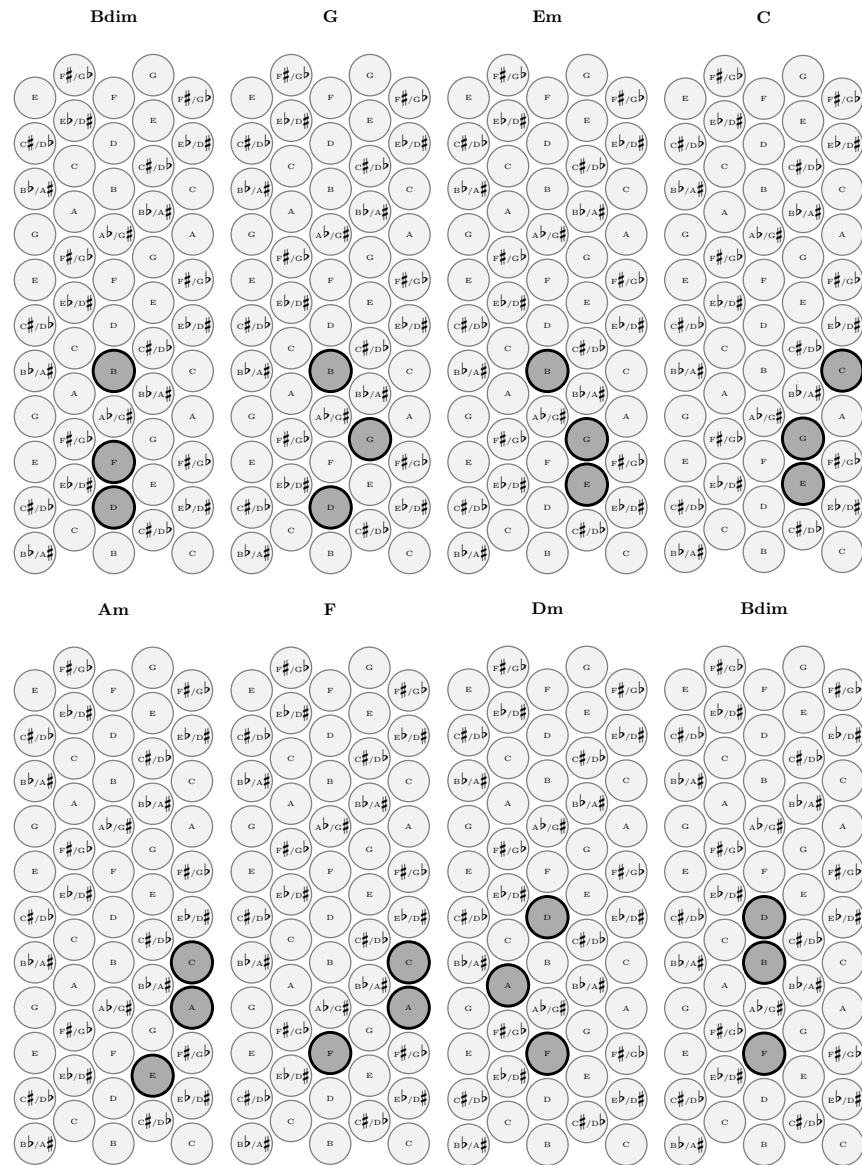


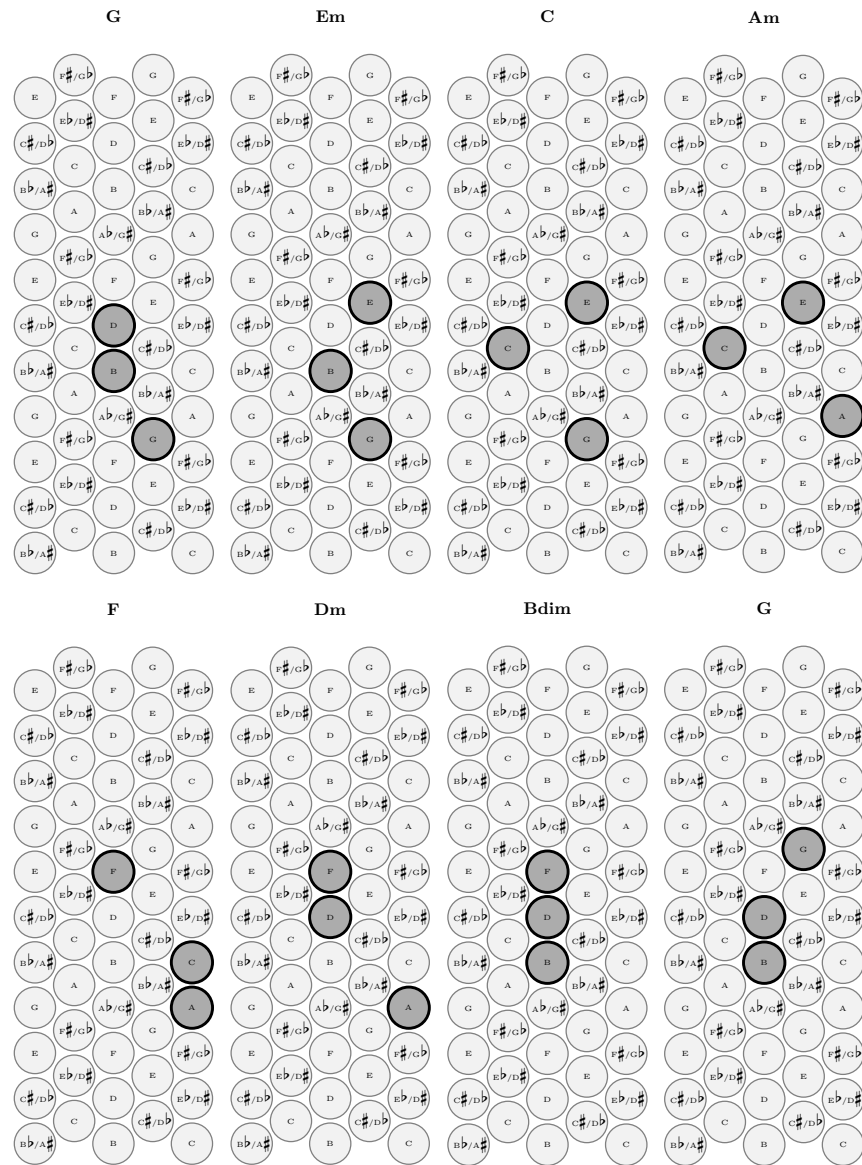




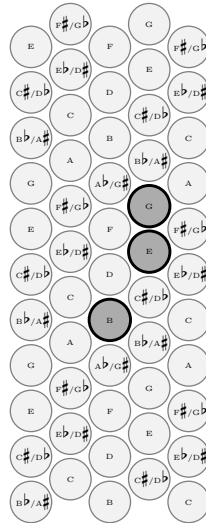
8.4 Closed Voice Triads from C Major Scale, Roots Ascending by a Diatonic 6th (Cycle 6)





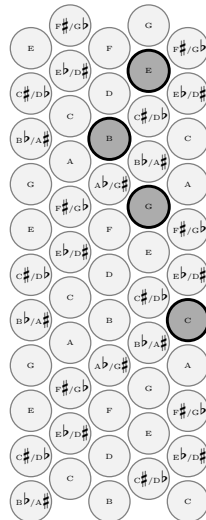


Em

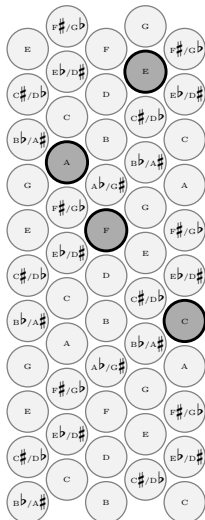


8.5 Progression of Drop 2 Diatonic Chords from C Major Scale (Cycle 4)

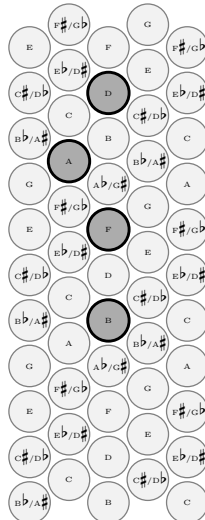
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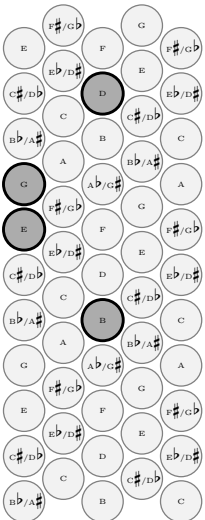
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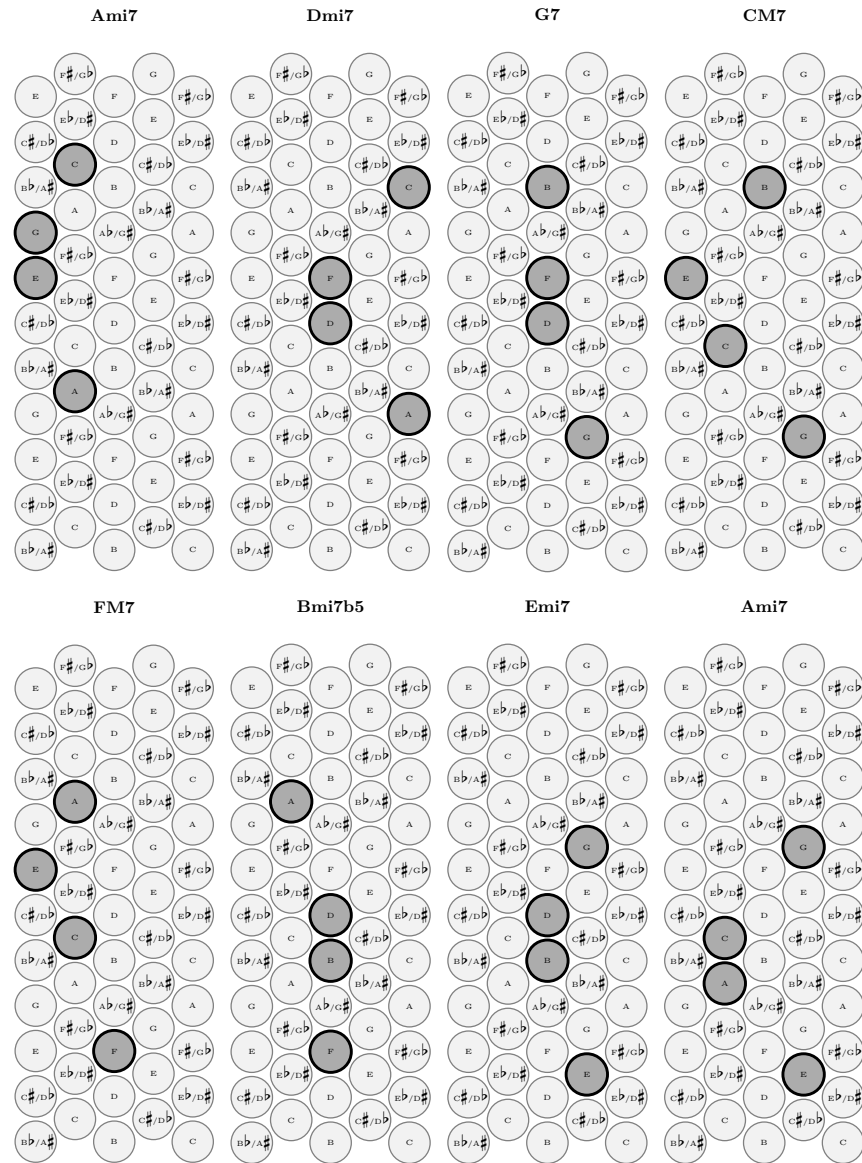


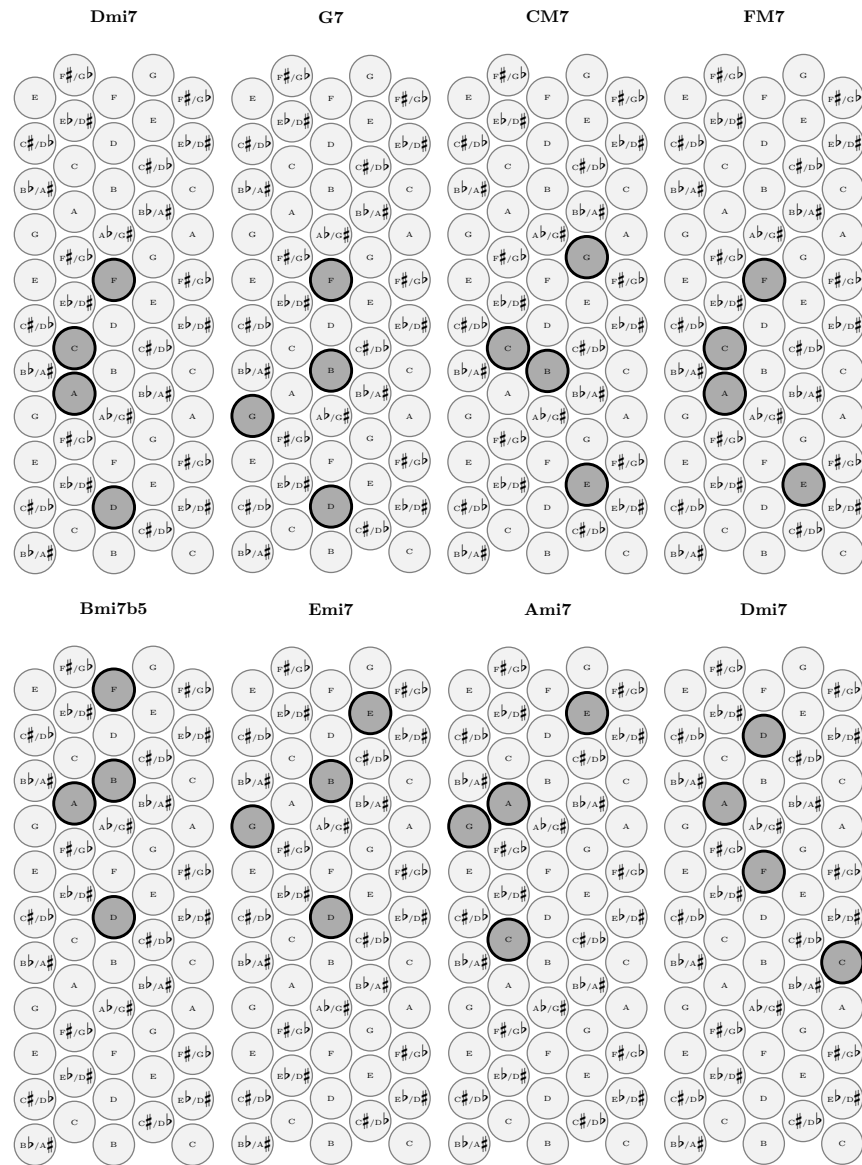
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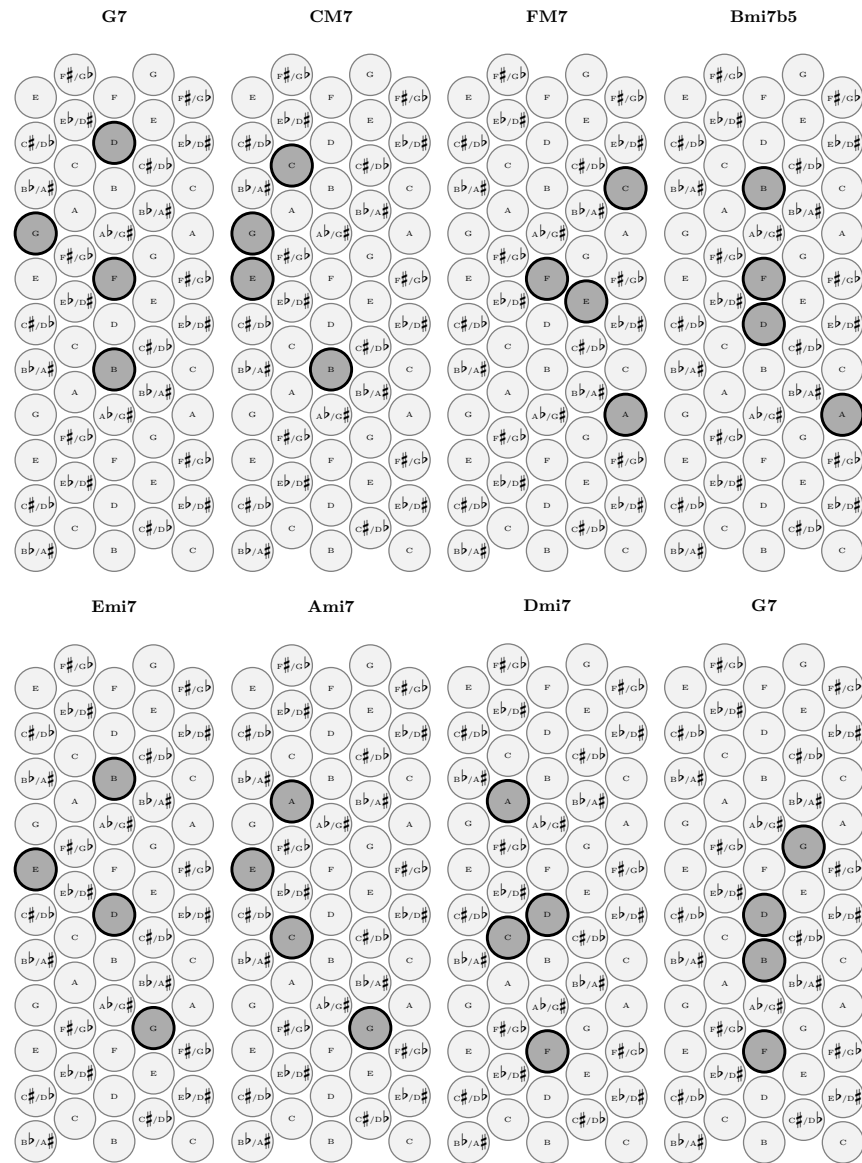


Emi7

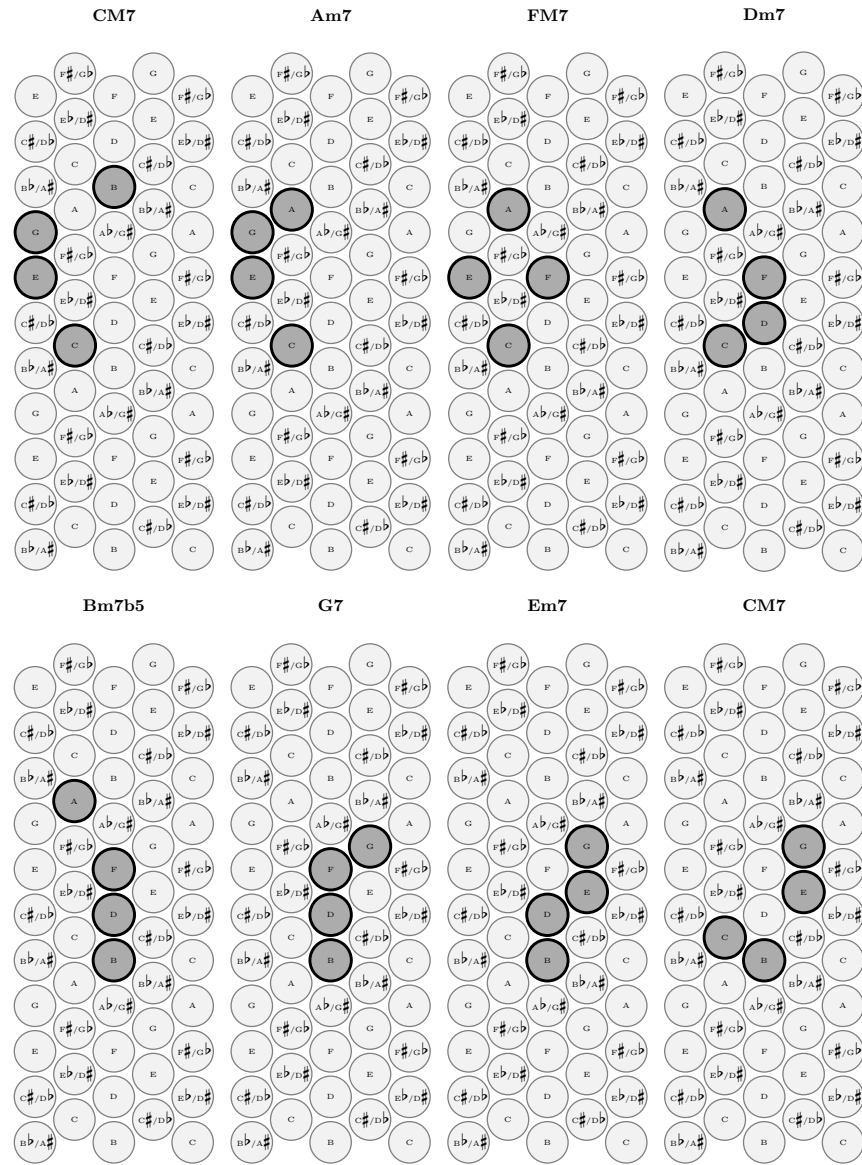


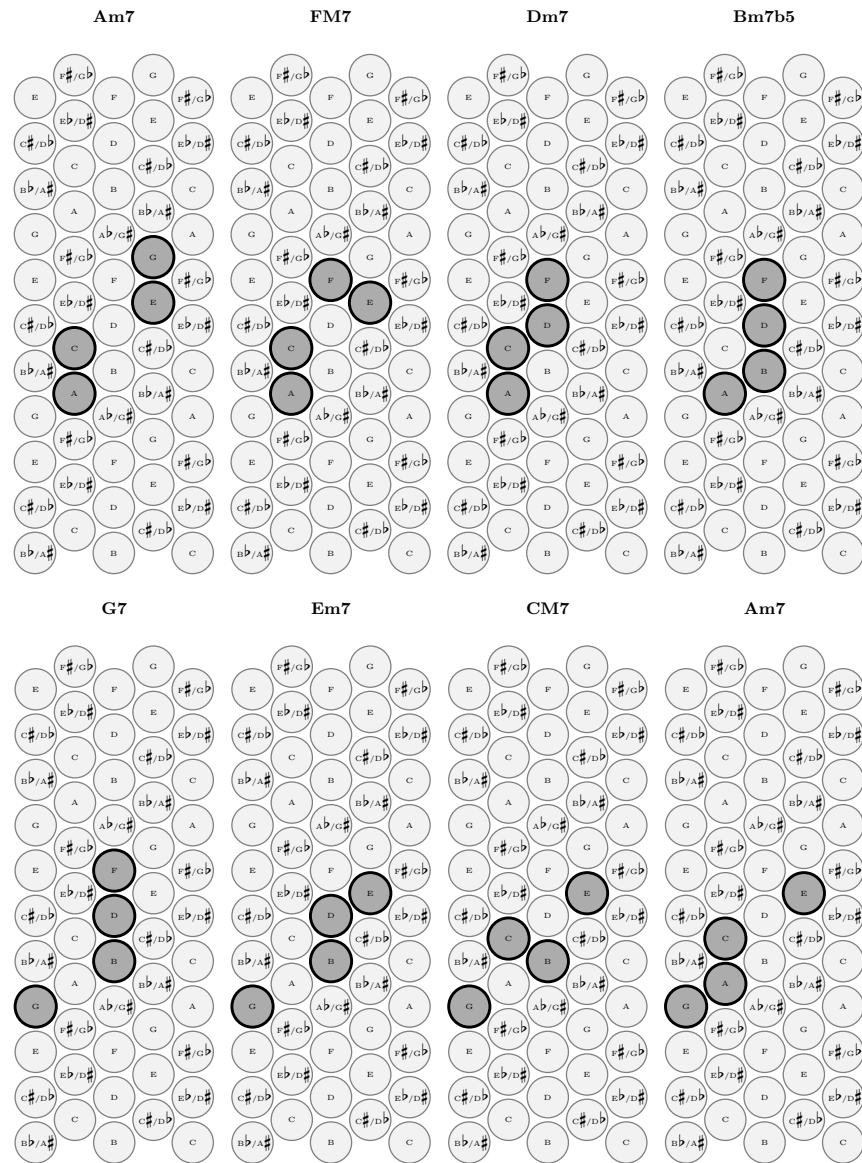


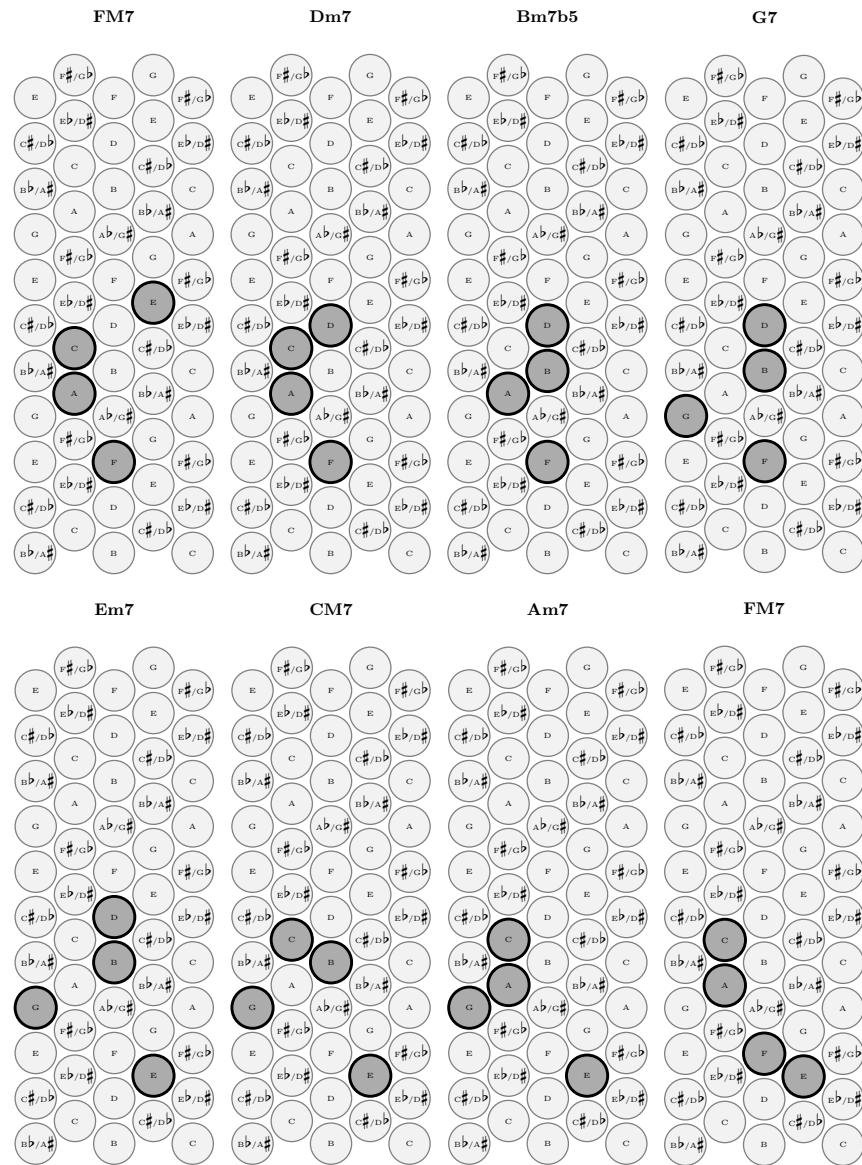


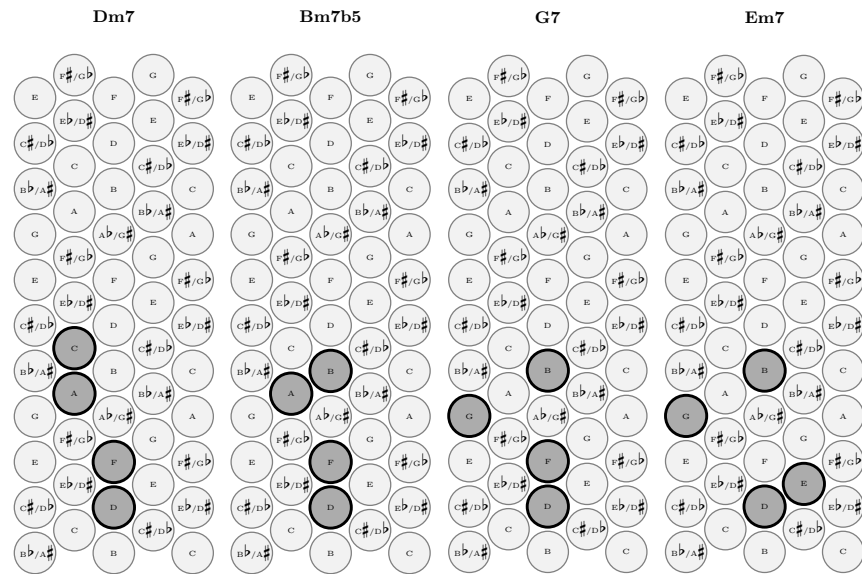


8.6 7th Chord Voicings from C Major Scale (Descending Voice Leading by Diatonic 6ths)

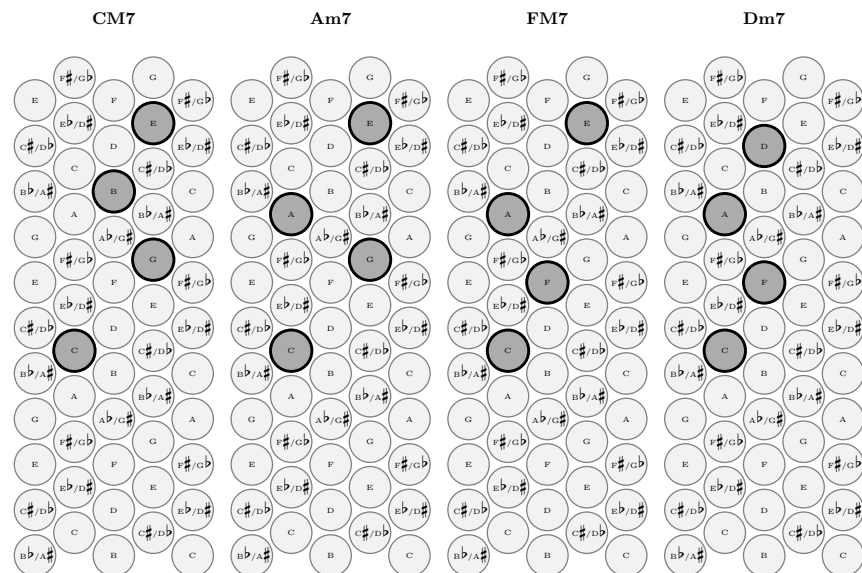


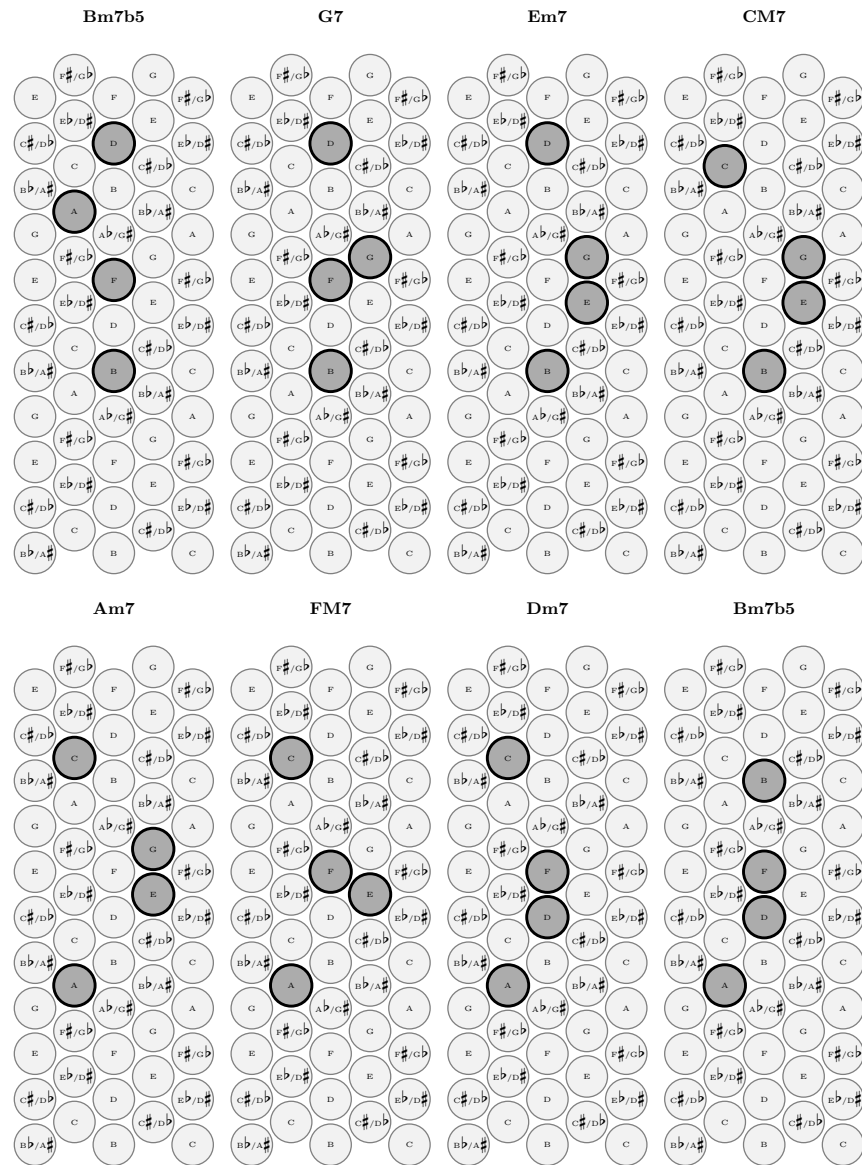


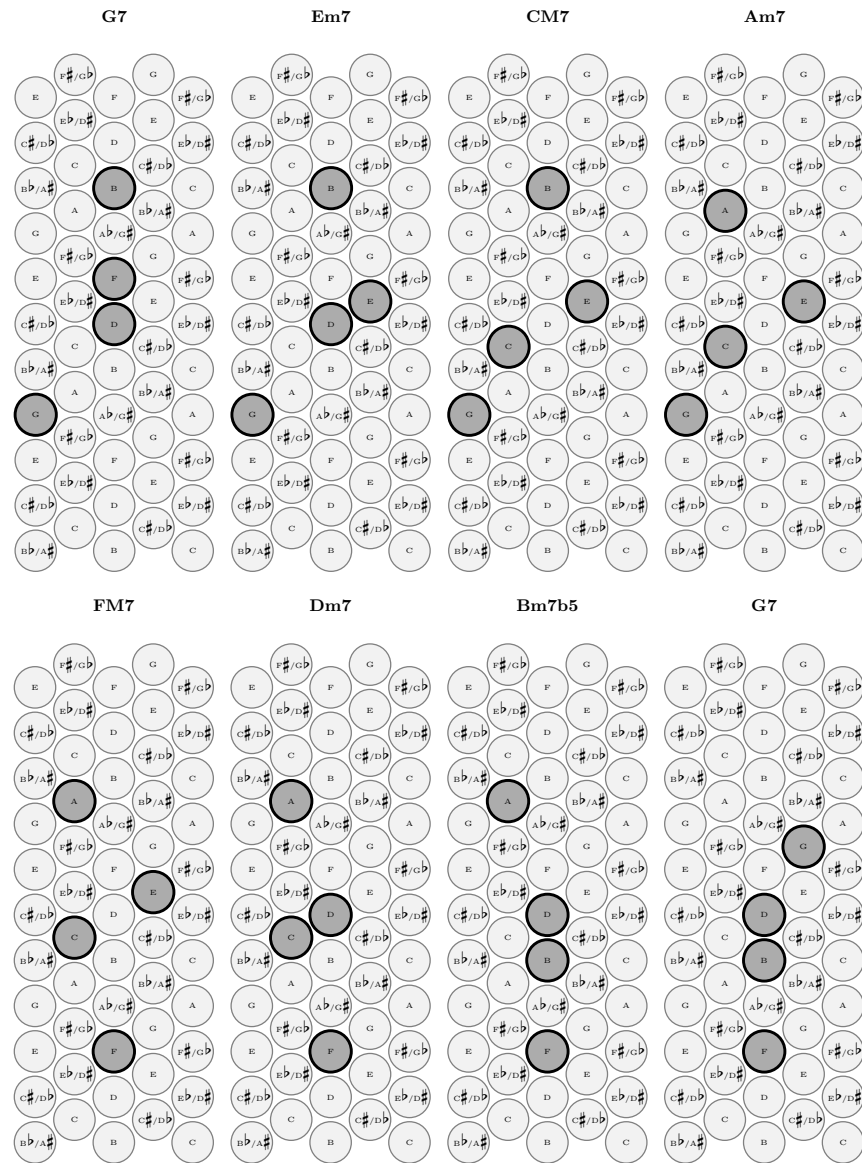


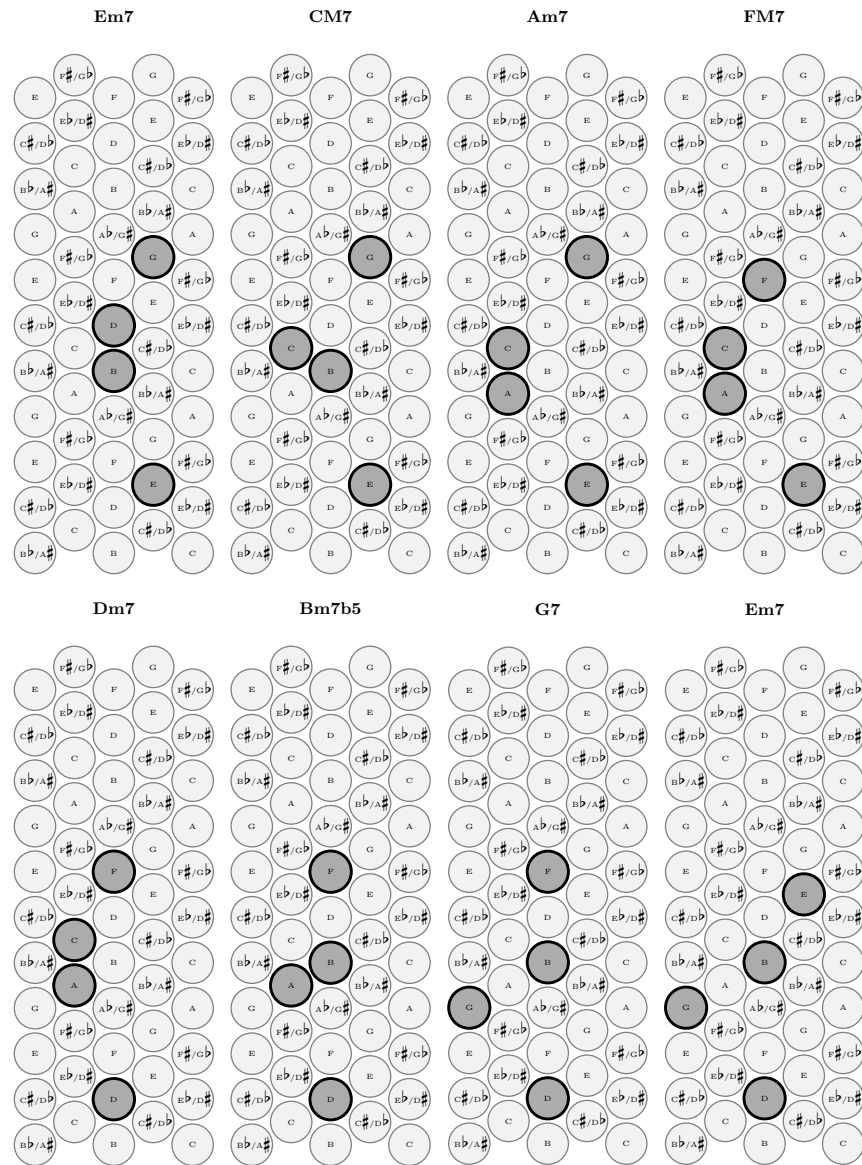


8.7 Drop 2 Diatonic 7th Chords from C Major Scale (Cycle 6)

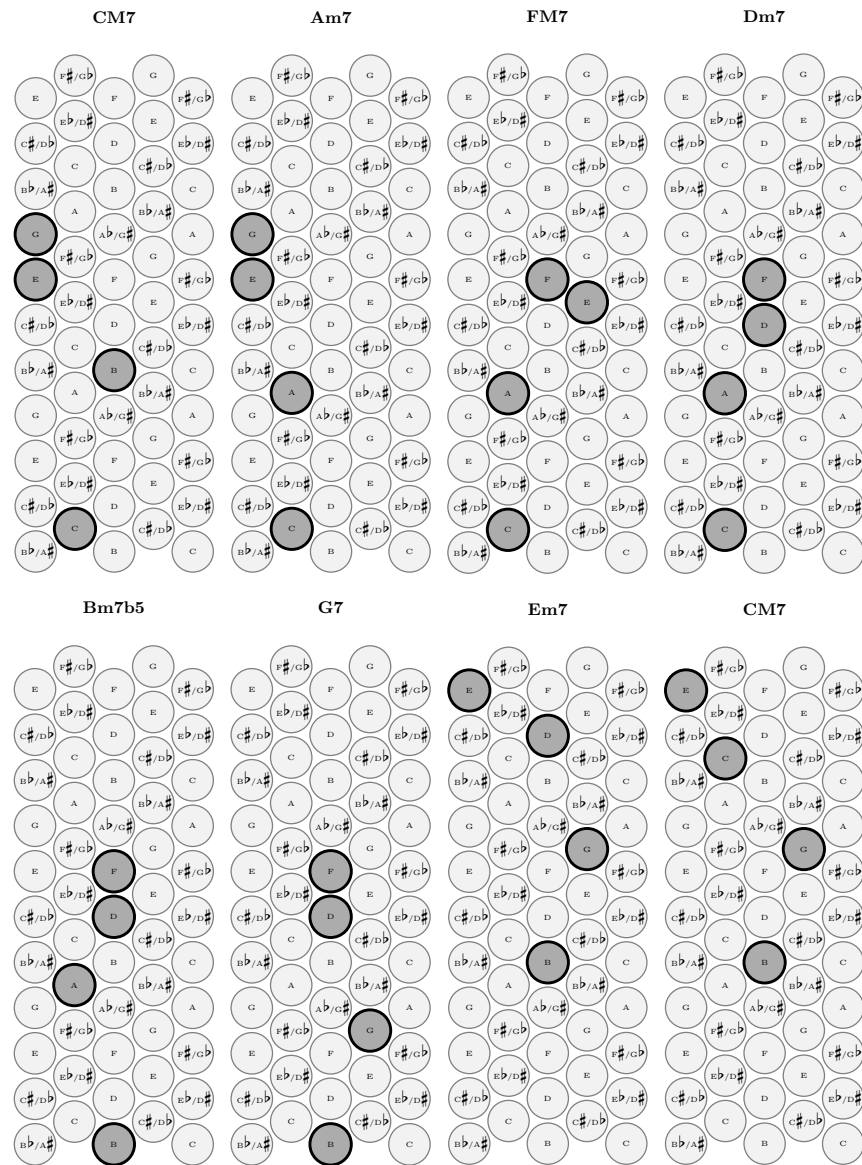


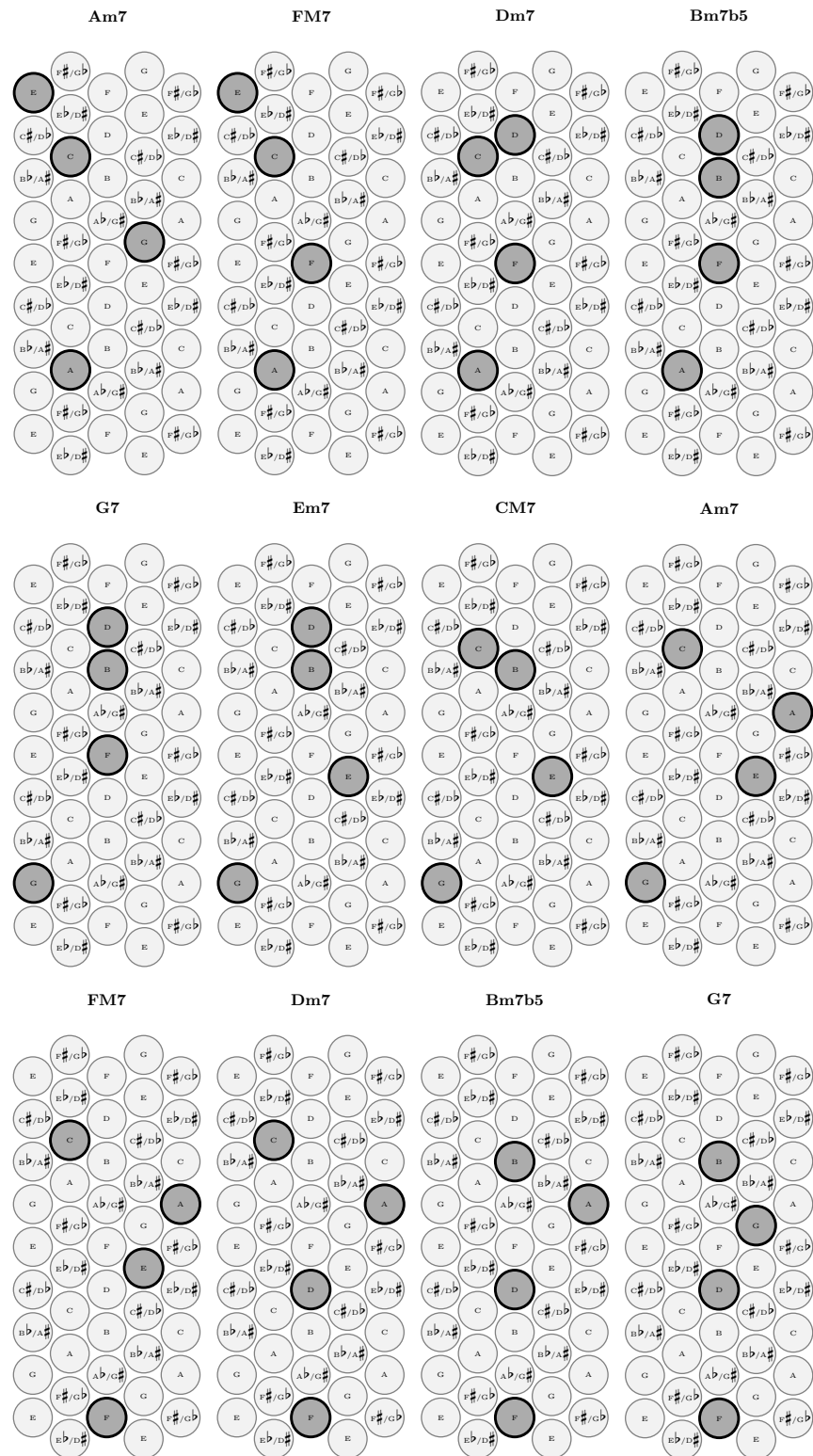


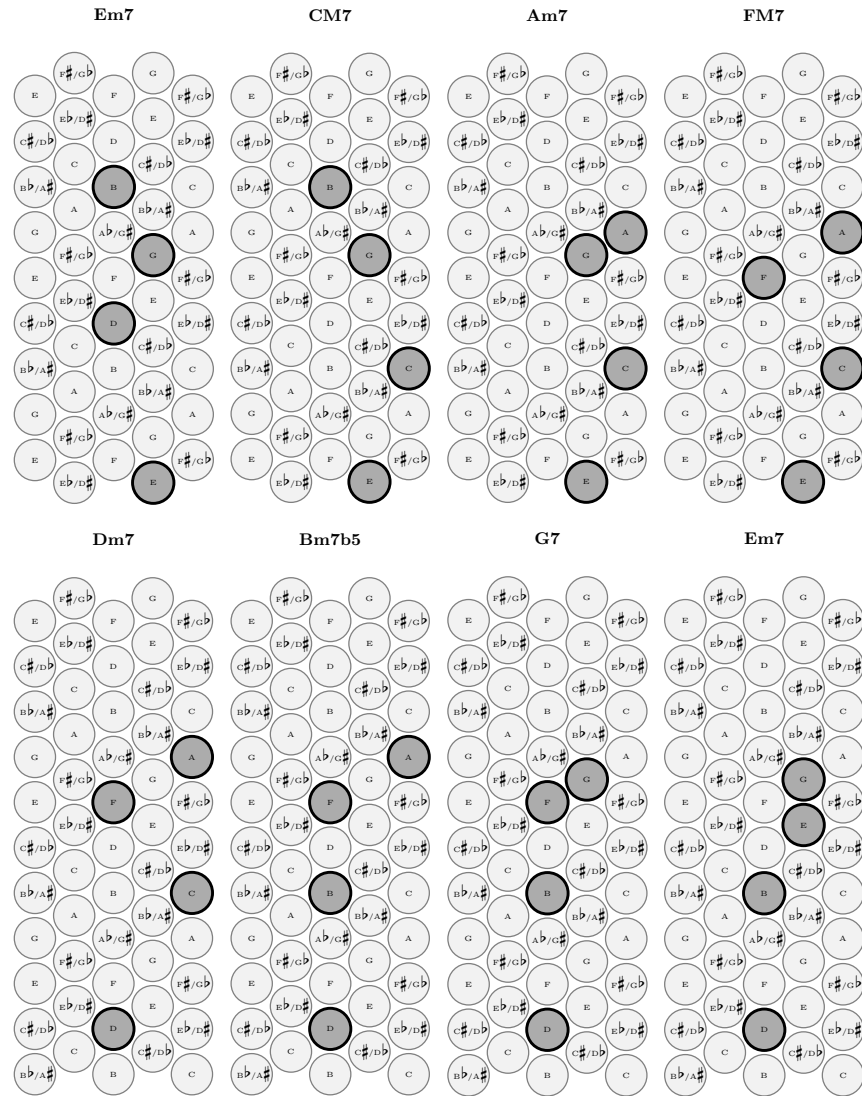




8.8 Drop 3 Diatonic 7th Chords from C Major Scale (Cycle 6)

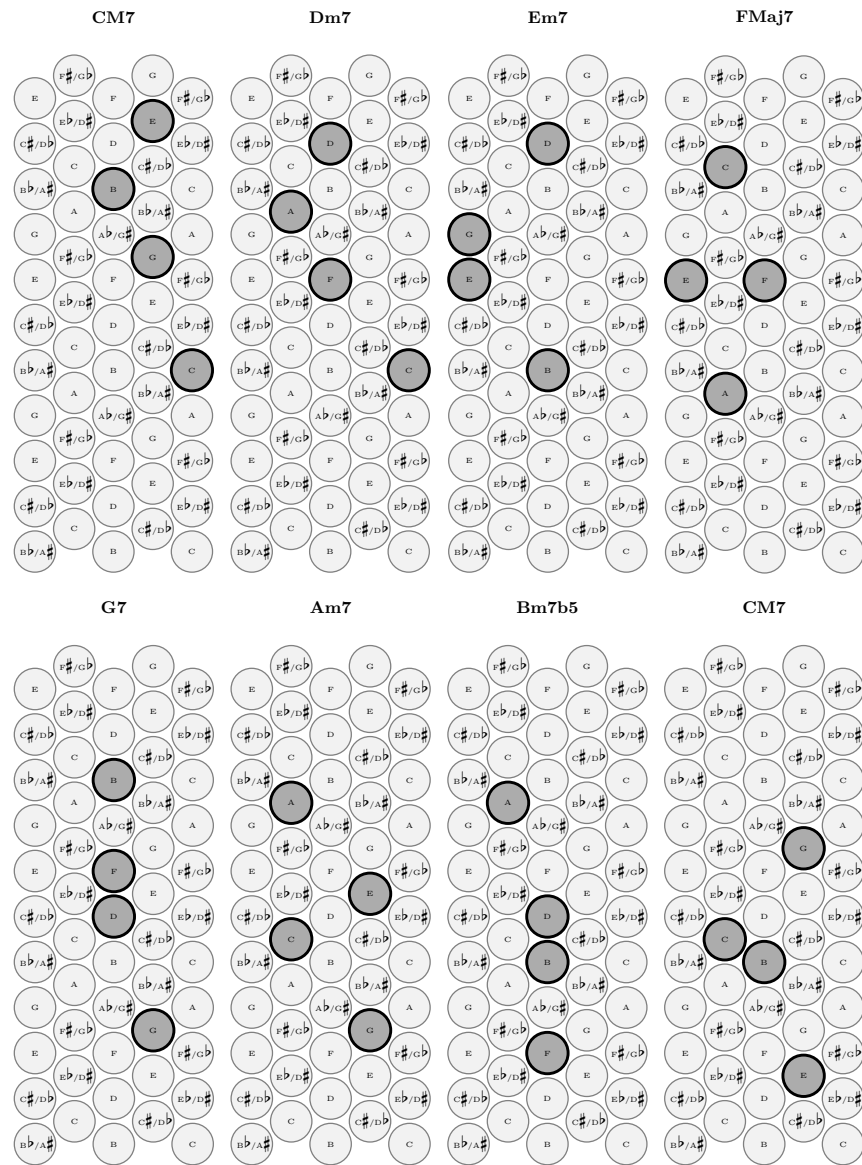


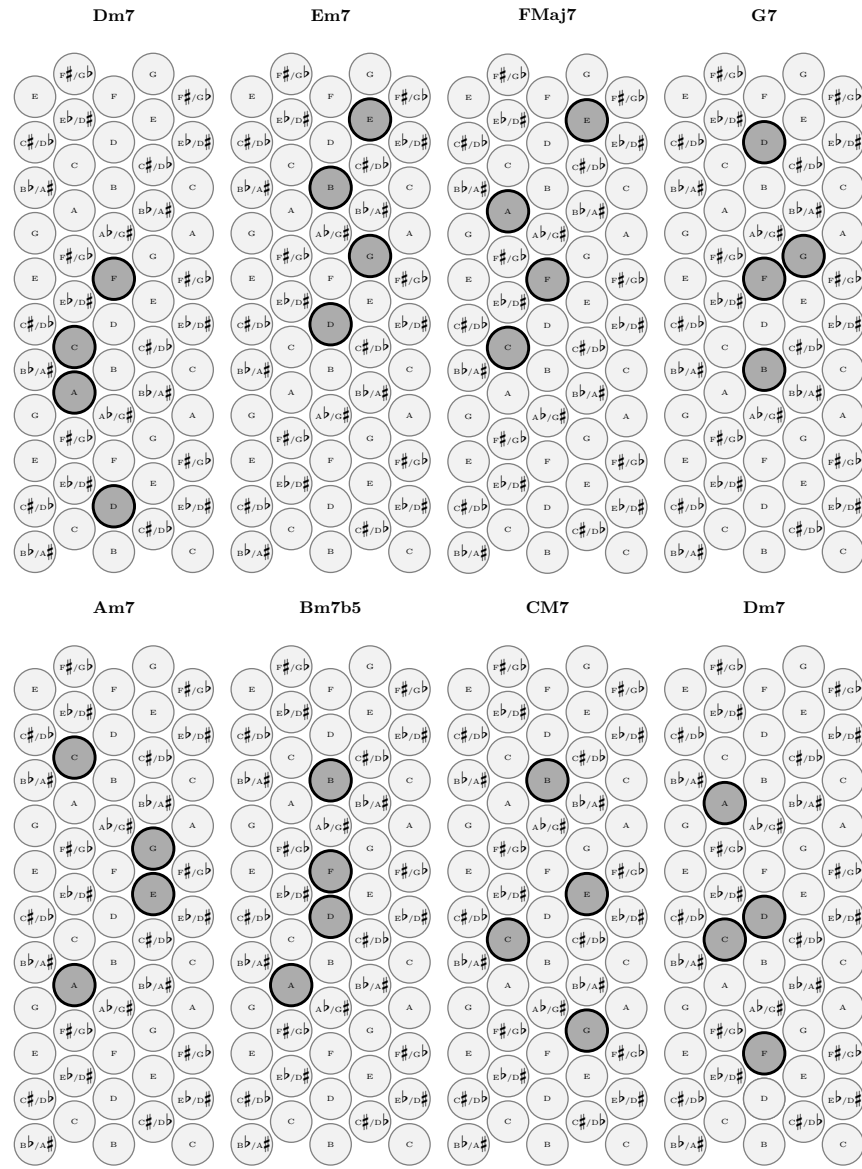


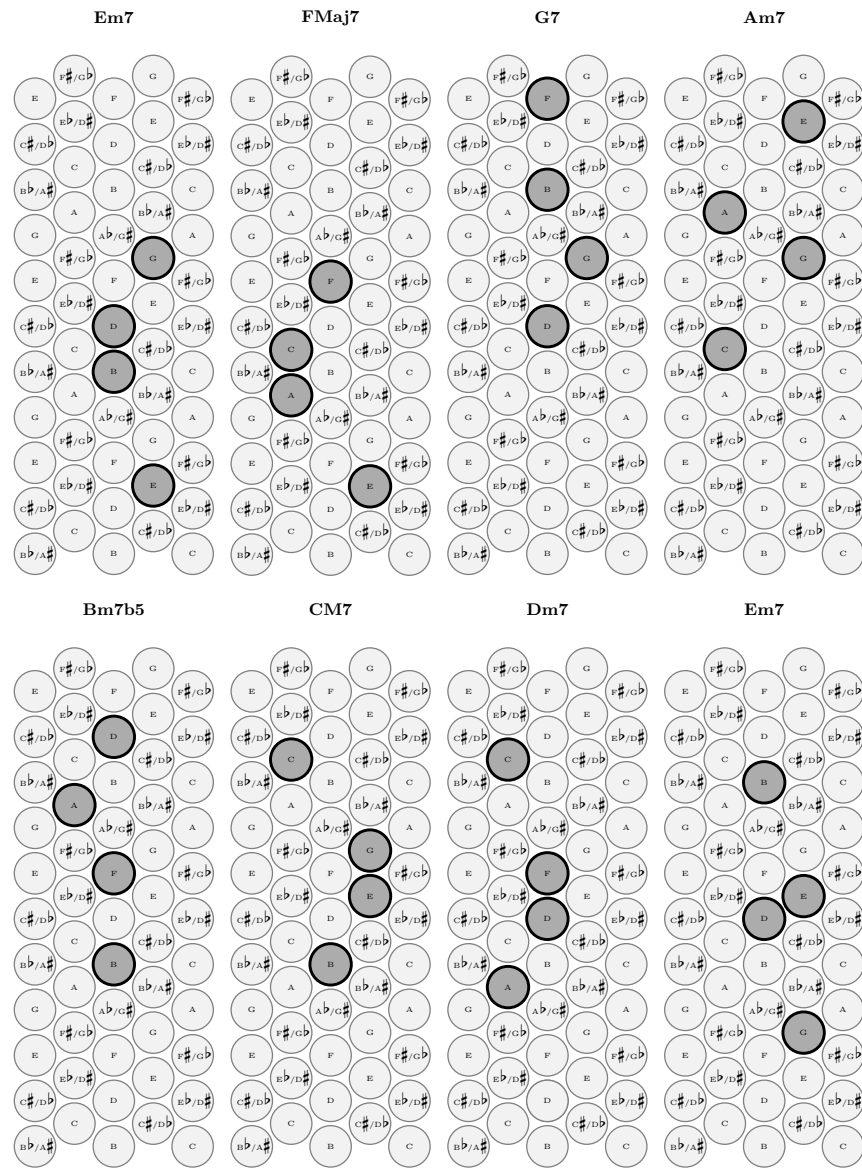


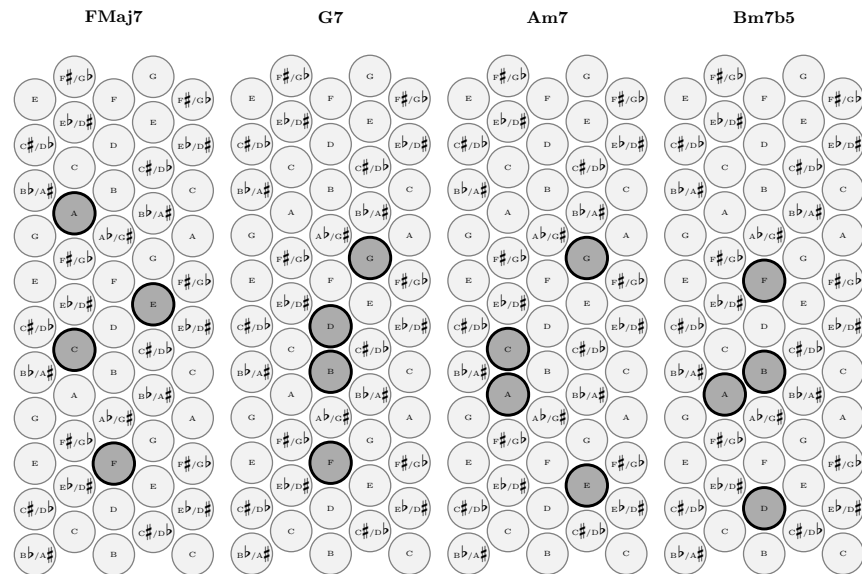
8.9 Drop 2 Diatonic 7th Chords — C Major, Roots Ascending in 2nds (Cycle 2)

Descending Drop 2 seventh chords from the C major scale with roots ascending by step (cycle of 2nds). Each voicing is arranged for minimal movement and best voice leading. Initial sequence: $CGBE \rightarrow CFAD \rightarrow BEGD \rightarrow AEFC \rightarrow GDFB \rightarrow GCEA \rightarrow FBDA \rightarrow EBCG$.

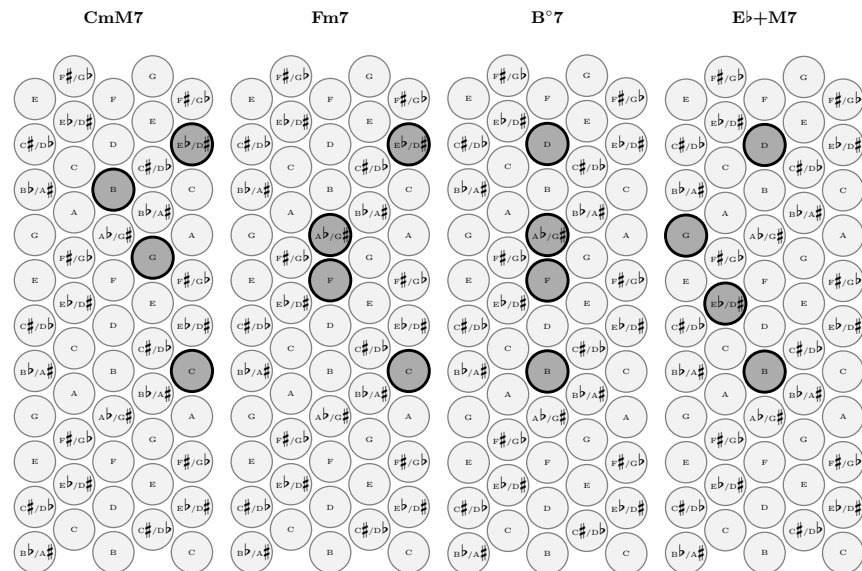


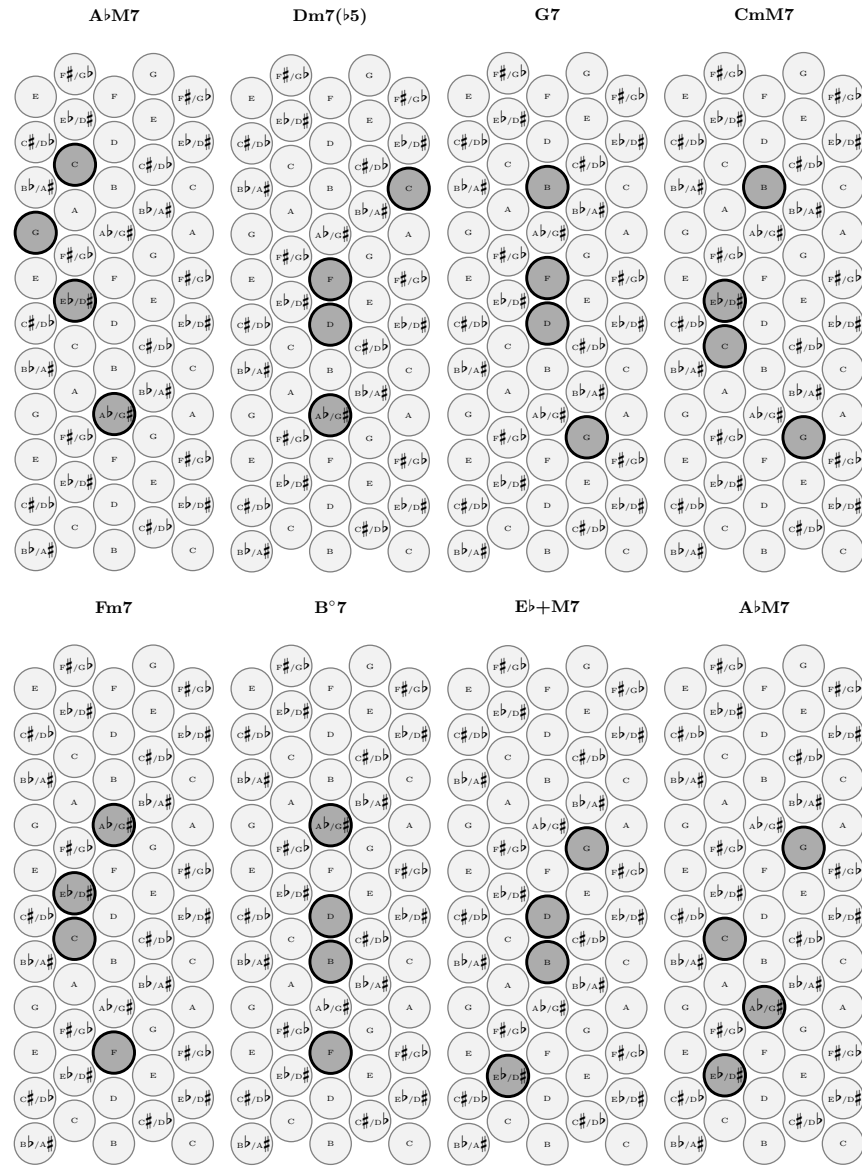


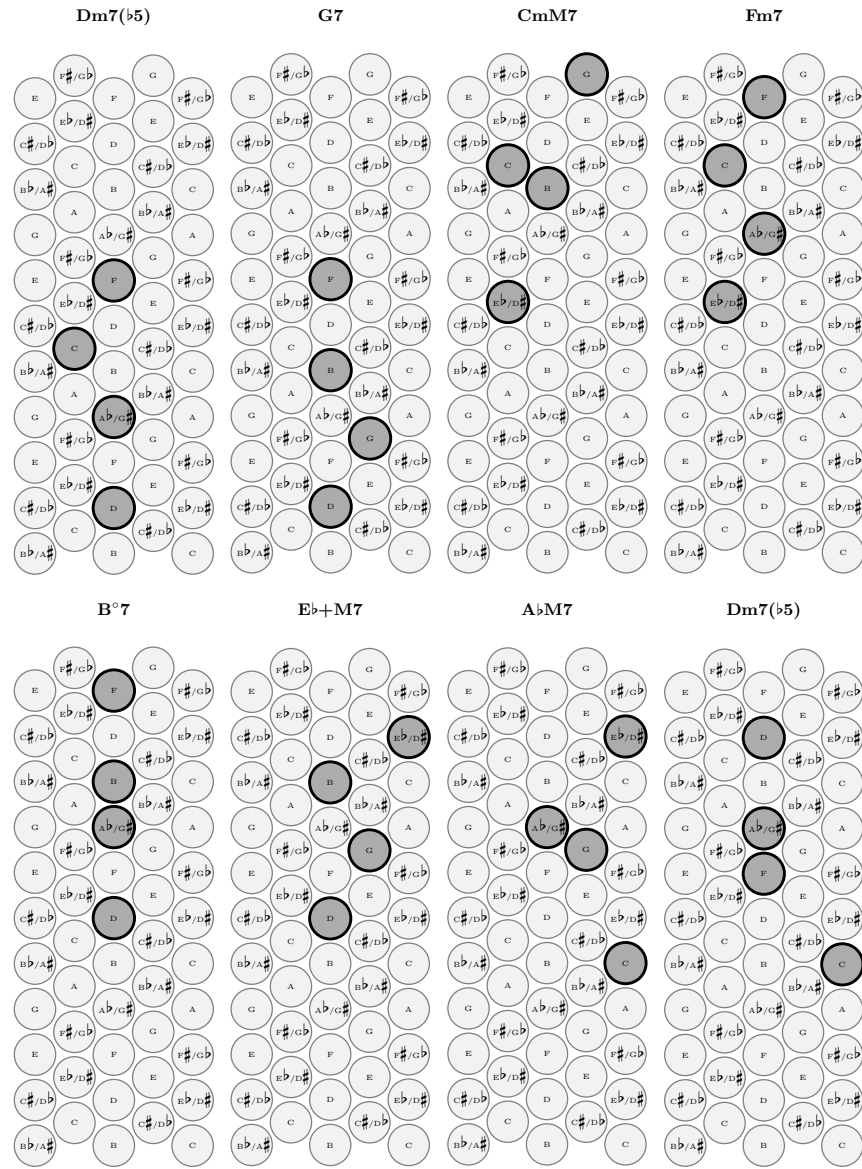


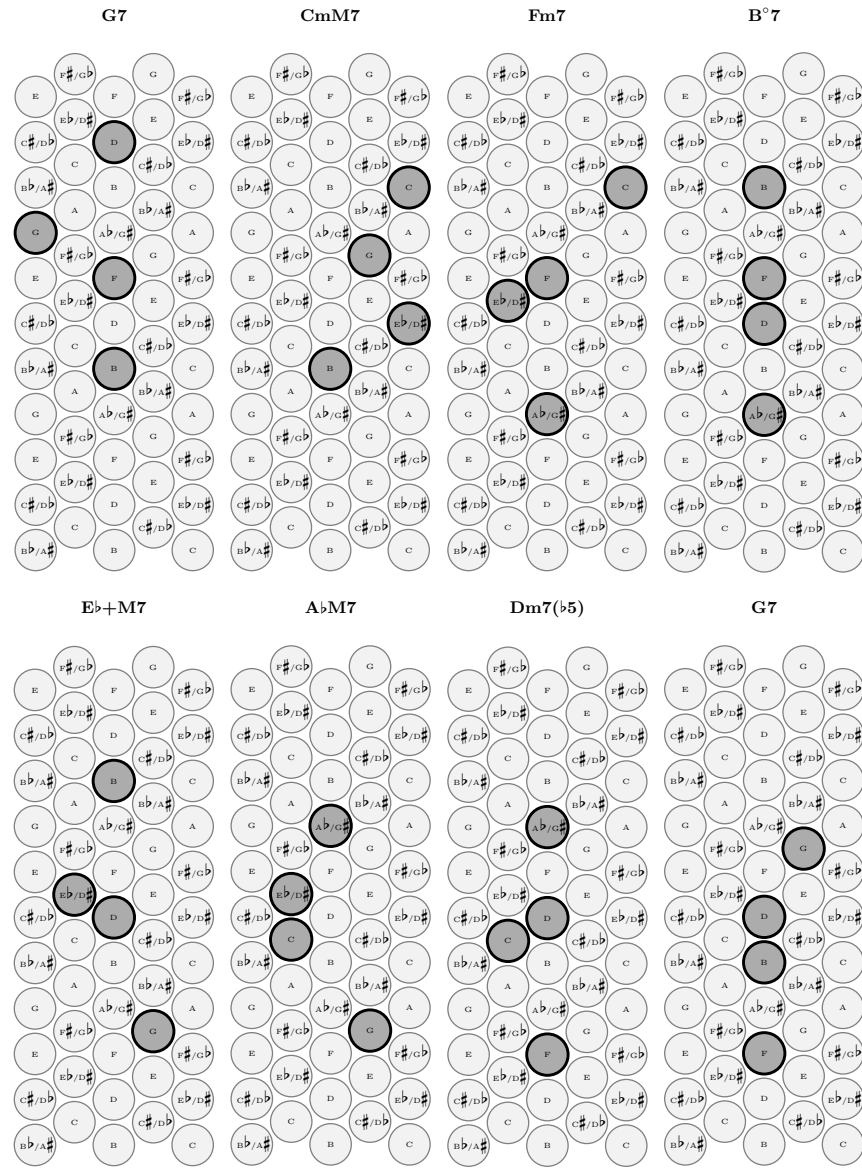


8.10 Progression of Drop 2 Diatonic Chords from C Harmonic Minor Scale

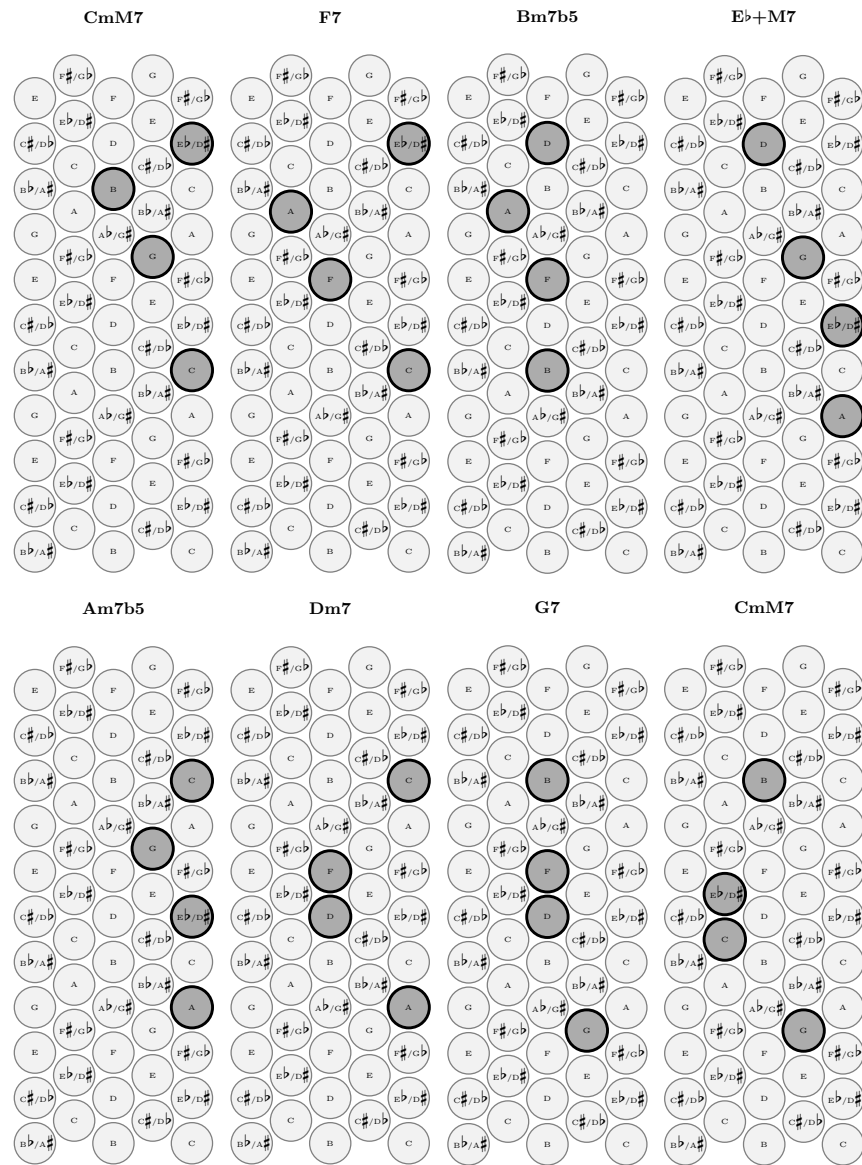


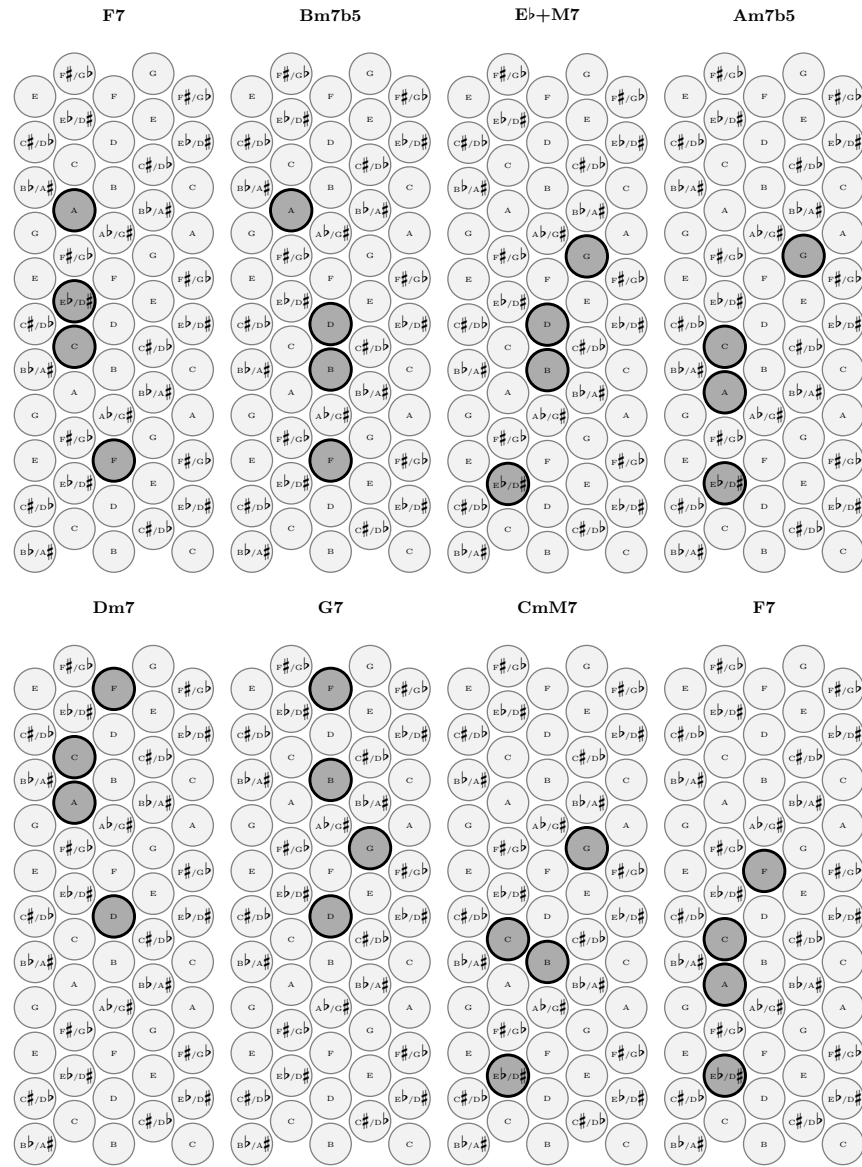


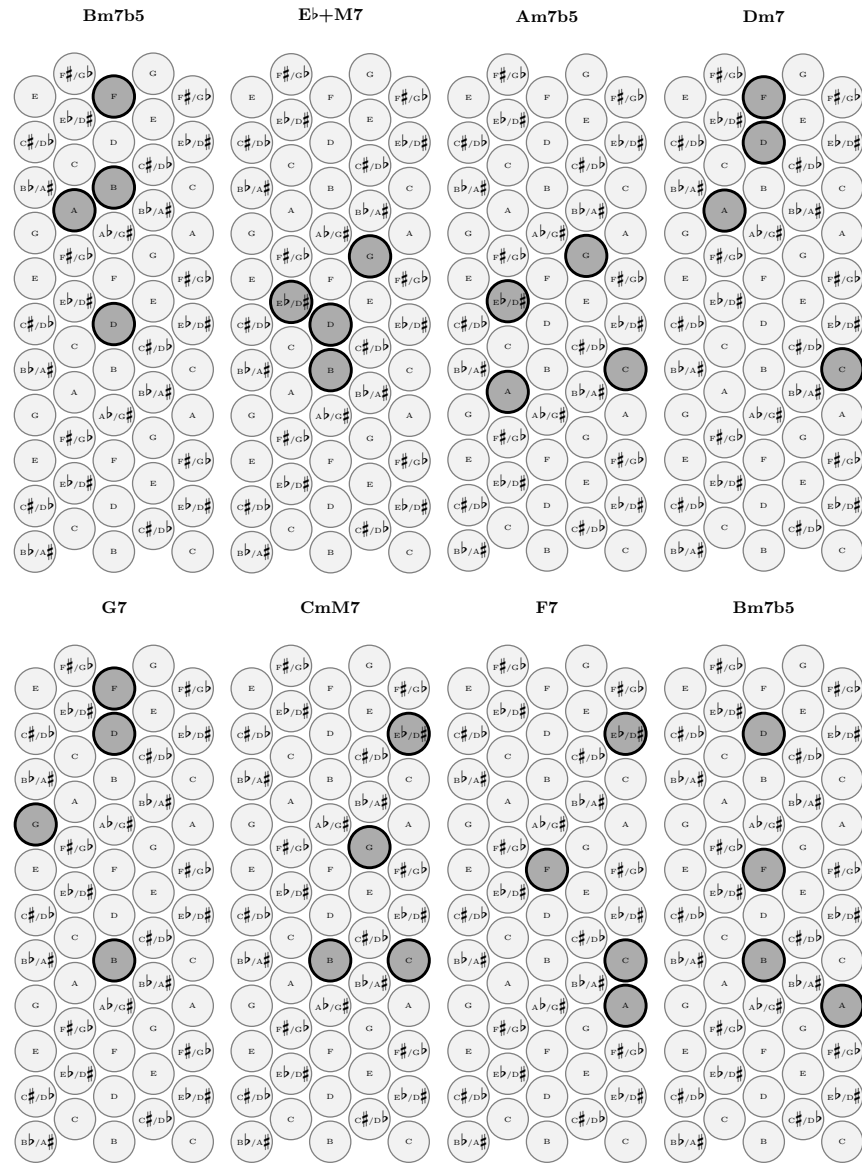


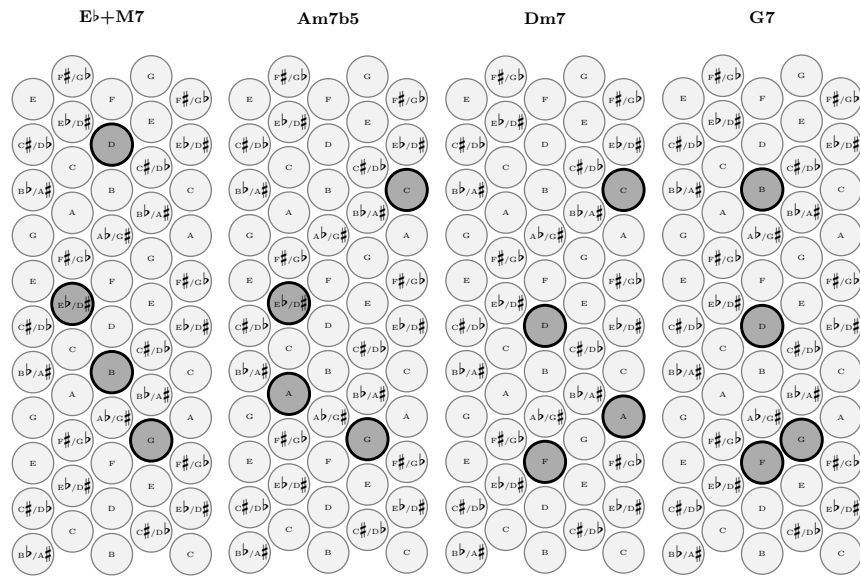


8.11 Progression of Drop 2 Diatonic Chords from C Melodic Minor Scale









Chapter 9

Chord Progressions

9.1 2-5-1 Progressions of Maj7 Chords with Voice Leading (naturally follows Circle of Fifths)

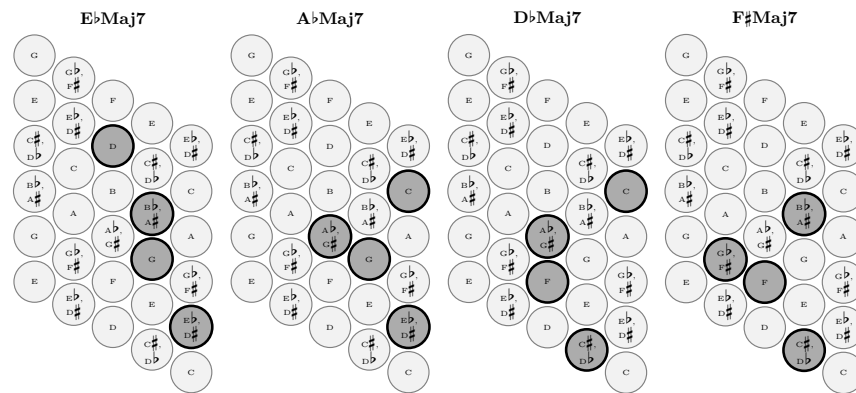


Figure 9.1.1

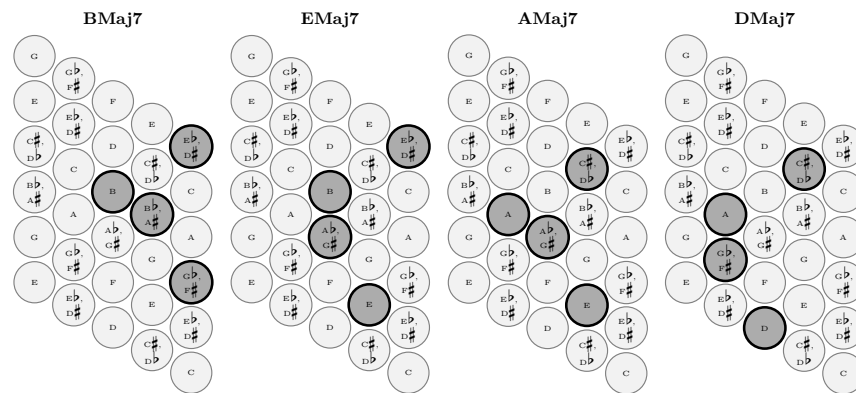


Figure 9.1.2

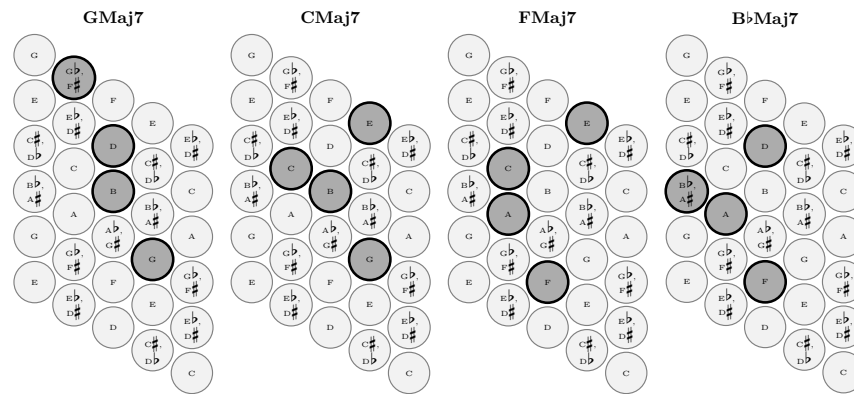


Figure 9.1.3

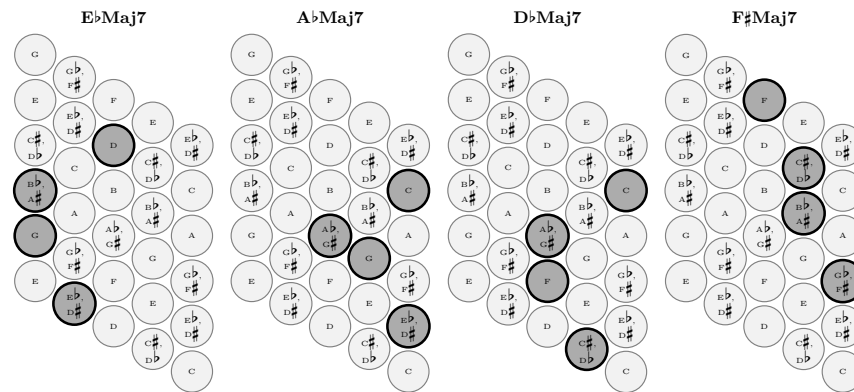
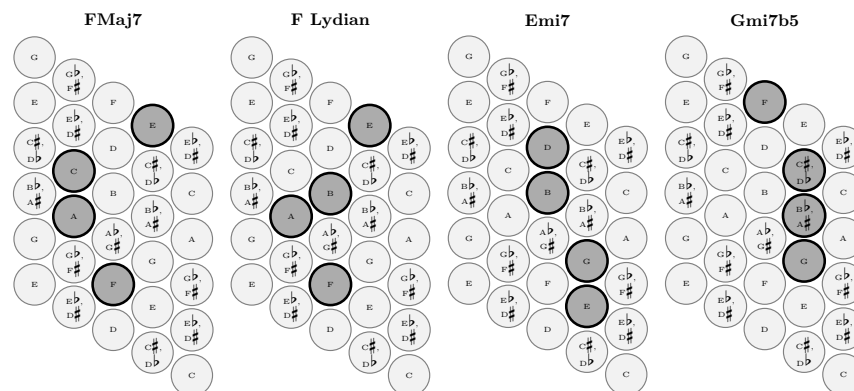


Figure 9.1.4

9.2 2-5-1 Closed in C Major

This progression is built from a 4 note grip (chord formed by content of the grip shown above the shape) and a bass note that is the Root of the Progression Chord (named in the notation).



external/notation/251-r1-notation.pdf

Figure 9.2.1

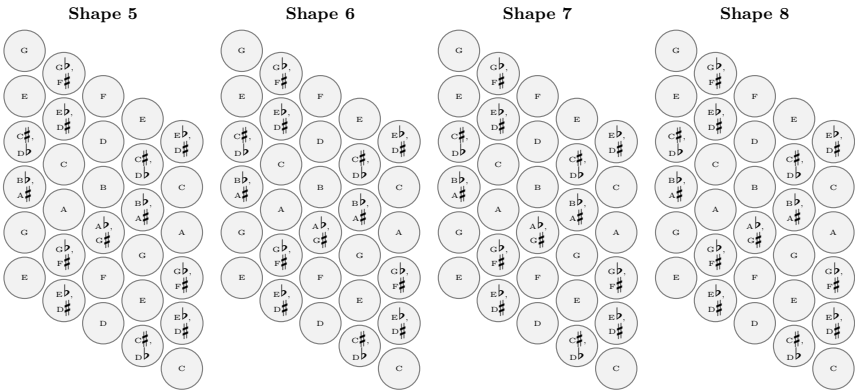


Figure 9.2.2

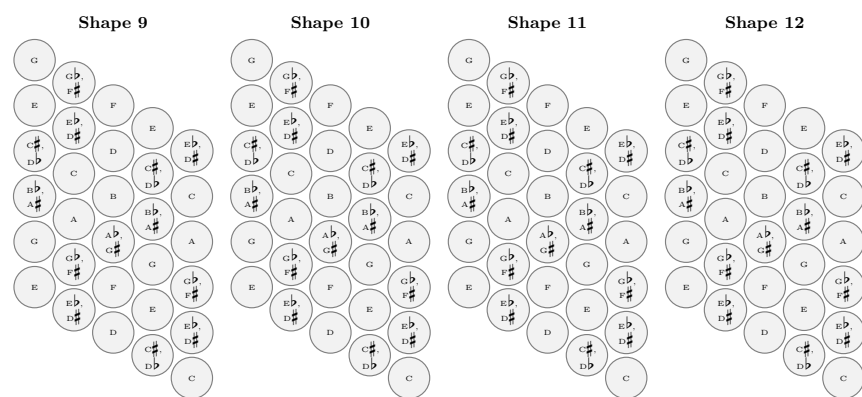


Figure 9.2.3

Chapter 10

The Circle of Fifths

The **Circle of Fifths** is useful in understanding musical concepts far beyond showing the different Key Signatures.

The **The Circle of Fifths** can be confusing. It is important to realise that you can progress round the circle clockwise or anticlockwise. It is also important to appreciate that the pitch may be ascending or descending as progress is made round the circle in either direction.

- It is a diagram showing the 12 musical Keys laid out in a circular format.
- The notes are written such that, if travelling round the circle in a clockwise direction, and they are ascending in pitch they are separated by an Interval of a **Perfect 5th**.
- If travelling in an anticlockwise direction and the notes are ascending in pitch, they are separated by an Interval of a **Perfect 4th**.
- If travelling in a clockwise direction and the notes are descending in pitch, they are separated by an Interval of a **Perfect 4th**.
- If travelling in an anticlockwise direction and the notes are descending in pitch, they are separated by an Interval of a **Perfect 5th**.
- You can see from the above, that the Circle could be legitimately called The Circle of Fourths and sometimes is. It all depends on direction of travel and ascending or falling pitch.
- With C at the 12 O'Clock position, C Major has no Accidentals in its' Key Signature, going clockwise G Major has 1#, D Major has 2#'s, A Major has 3#'s and so on.
- From C going anticlockwise F Major has 1b, BbMajor has 2 b's, EbMajor has 3 b's etc;
- The Relative Minor is shown on an inner circle opposite its' Relative Major, eg; A Minor is at the 12 O'clock position opposite to C Major.
- On a musical staff the position of the accidental symbol (# or b) is positioned where the # or b note would be printed. From this you can see that in D Major the # notes are F and C, similarly in Eb Major the b notes are E, B and A.

For the chromatic button accordion player, the Circle of Fifths is an especially powerful tool. Because the keyboard layout is isomorphic, the geometric relationships visible on the circle translate directly into physical patterns on the instrument. Understanding the circle helps with transposition, chord progressions, and improvisation.

10.1 Structure of the Circle

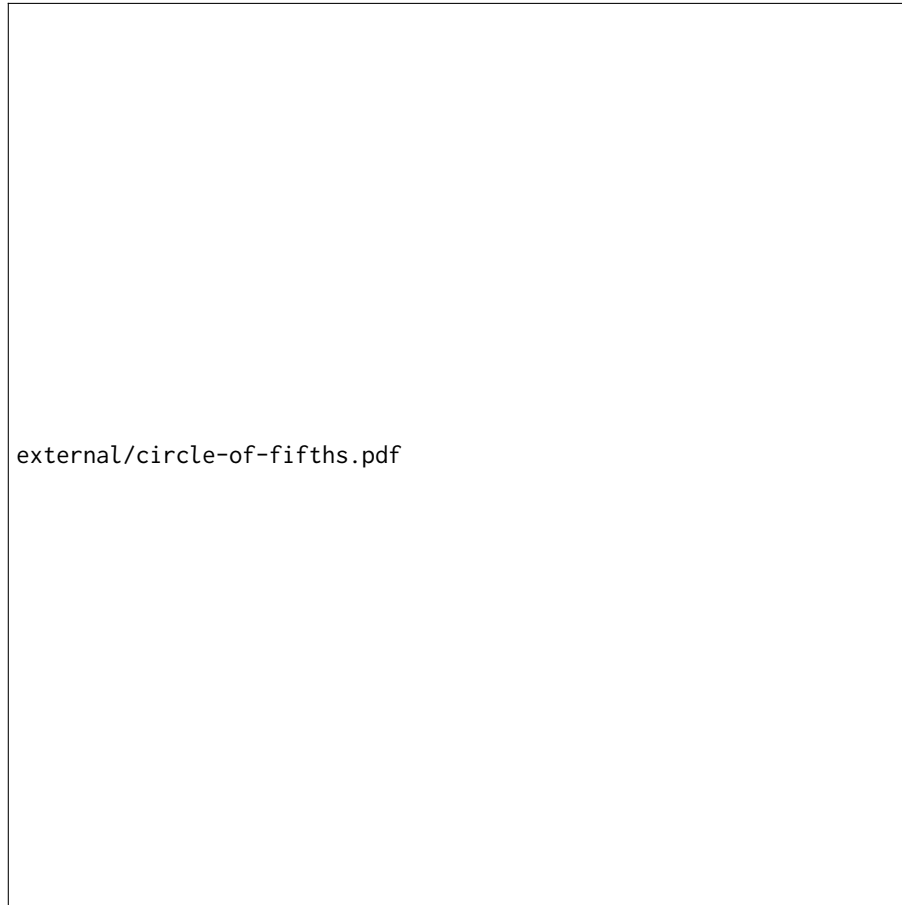


Figure 10.1.1 The Circle of Fifths. Outer circle contains Major Keys, the adjacent inner circle shows the Relative Minor.

Starting from C at the top of the circle and moving clockwise by perfect fifths, the twelve keys appear in the following order:

C — G — D — A — E — B — F $\sharp$ /G $\flat$ — D $\flat$ — A $\flat$ — E $\flat$ — B $\flat$ — F — back to C.

Each step clockwise adds one sharp to the key signature. Each step anticlockwise adds one flat. The key of C major, at the top, has no sharps or flats.

Table 10.1.2 Key Signatures Around the Circle of Fifths

Key	Sharps / Flats	Key Signature
C major	0	—
G major	1 $\sharp$	F $\sharp$
D major	2 $\sharp$	F $\sharp$, C $\sharp$
A major	3 $\sharp$	F $\sharp$, C $\sharp$, G $\sharp$
E major	4 $\sharp$	F $\sharp$, C $\sharp$, G $\sharp$, D $\sharp$
B major	5 $\sharp$	F $\sharp$, C $\sharp$, G $\sharp$, D $\sharp$, A $\sharp$
F $\sharp$ /G $\flat$ major	6 $\sharp$ / 6 $\flat$	Enharmonic equivalent
D $\flat$ major	5 $\flat$	B $\flat$, E $\flat$, A $\flat$, D $\flat$, G $\flat$
A $\flat$ major	4 $\flat$	B $\flat$, E $\flat$, A $\flat$, D $\flat$
E $\flat$ major	3 $\flat$	B $\flat$, E $\flat$, A $\flat$
B $\flat$ major	2 $\flat$	B $\flat$, E $\flat$
F major	1 $\flat$	B $\flat$

10.2 Relative Minor Keys

Every major key has a **relative minor** that shares the same key signature. The relative minor is found three semitones (a minor third) below the major key. On the Circle of Fifths, the relative minor keys form an inner circle:

Table 10.2.1 Major Keys and Their Relative Minors

Major Key	Relative Minor
C major	A minor
G major	E minor
D major	B minor
A major	F $\sharp$ minor
E major	C $\sharp$ minor
B major	G $\sharp$ minor
F $\sharp$ major	D $\sharp$ minor
D $\flat$ major	B $\flat$ minor
A $\flat$ major	F minor
E $\flat$ major	C minor
B $\flat$ major	G minor
F major	D minor

10.3 Chord Progressions and the Circle

An accordion player can access important Chord Progressions Using **The Circle of Fifths**:

- By adding a Chord Quality to each Pitch Identity on the Circle some of the most important Chord Progressions, used in all types of music, can be obtained by using the Circle Diagram.
- For this purpose all cycling through the Circle will be Counterclockwise.
- Direction of travel Chordwise will therefore be C to F to B $\flat$ to E $\flat$ and so on using the "pitch labels" in the Circle.
- Figure 8.3.1 illustrates all the possible ii-V-I progressions. Probably the most frequently occurring progression in all modern music.

- Figure 8.3.1 also shows one of the more important chord resolutions in music, The Dominant 7 (V7) to Major(Maj7)
- Figure 8.3.1 illustrates all the Dominants of Dominants, another frequently used chordal device. V7 to V7.
- In this discussion the information given in the Diagram remains the same. The "main-note-name" positions around the circle, remain fixed but the Chord Quality changes eg; from a Major Chord to a Minor etc; .
- All progressions cycle in an anticlockwise direction through the same Root note sequence, but with different Chord Qualities.

In Summary, the Circle of Fifths reveals the most common chord progressions in Western music. Chords that are adjacent on the circle sound naturally connected. The most important progressions follow the circle:

- **V — I (dominant to tonic):** Moving one step anticlockwise. For example, G to C. This is the strongest resolution in tonal music.
- **ii — V — I:** Moving two steps anticlockwise. For example, Dm — G — C. This is the most common jazz and popular music progression.
- **IV — I (subdominant to tonic):** Moving one step clockwise. For example, F to C. A plagal cadence, often heard as the “Amen” cadence.
- **I — IV — V — I:** The three primary chords of any key are always adjacent on the circle. In C major: C, F, and G sit next to each other.

Recognising these relationships on the Circle of Fifths helps the accordion player anticipate chord changes and navigate progressions fluently across the keyboard.



Figure 10.3.1 Example of ii-V-I progression using the circle diagram. .

The three inner rings show how the **ii–V–I** progression — the most common chord movement in tonal music — sits within each key around the circle. For any major key in the outer ring, the IMaj7, iim7 and V7 chords always

appear together in the same shaded section in the annular bands. Reading anticlockwise, the pattern repeats every three segments: the V7 chord resolves to the IMaj7, and the iim7 chord sets up the V7. Because the circle is arranged in fifths, every adjacent key shares two of these three chords, which is why modulation between neighbouring keys sounds so natural.

10.4 The Circle of Fifths on the Chromatic Button Accordion

On the C-system chromatic button accordion, the perfect fifth interval always corresponds to the same physical movement on the keyboard, regardless of the starting note. This means that moving around the Circle of Fifths is a repeating geometric pattern on the instrument.

On the Stradella bass system, the Circle of Fifths is built directly into the layout. The bass note rows are arranged in fifths, so adjacent bass buttons follow the circle exactly. This makes common chord progressions (such as ii—V—I) physically adjacent on the bass side of the instrument.

Practising scales and chord progressions around the Circle of Fifths is one of the most effective ways to build fluency on the chromatic button accordion. Start from any key, move through all twelve keys by fifths, and observe how the same finger patterns repeat across the entire keyboard.

Chapter 11

Scales, Modes and Chords

Chords, Scales and Modes are constructed from the same collection of notes. Scales are constructed by stacking 2nds on the Root of a Chromatic Scale (See Chapter 2.1) and Chords are constructed by stacking 3rds on the Root of Scale. (See Chapter 5). Whilst scales have notes which are played sequentially and arranged horizontally, chords notes are diatonic to a scale and are drawn from that scale. Chords notes are arranged vertically and the notes are played simultaneously. Modes are just a different way of thinking about scales and keys. A mode/scale can be assigned to every single chord in a progression.

A scale or mode is a subset of the 12 notes of a chromatic scale (See Chapter 2.1) of one octave; it has two components:

1. The notes they are comprised of;
2. A tonal centre (or root note).

Table 11.0.1 The Seven Modes of the Major Scale (C Major)

Mode	Degree	Scale Notation	Scale Formula
Ionian	I	external/notation/mode-ionian.pdf	1 2 3 4 5 6 7
Dorian	II	external/notation/mode-dorian.pdf	1 2 b3 4 5 6 b7
Phrygian	III	external/notation/mode-phrygian.pdf	1 2 b3 4 5 b6 b7
Lydian	IV	external/notation/mode-lydian.pdf	1 2 3 #4 5 6 7
Mixolydian	V	external/notation/mode-mixolydian.pdf	1 2 3 4 5 6 b7

NB The Natural 11th is the 4th degree of the scale and is a perfect 4th above the root, but the natural 11th, interferes with our ability to hear the root and clashes with the 3rd of a chord, therefore the 11th is generally ignored when thinking about chords and scales.

The following are examples of the relationship between Scales and Chords

- The C major scale or C Ionian mode (C D E F G A B) has the same notes as a CMaj13 chord (C E G B D F A). The natural 11th (F) is present in both the scale and the chord, but we ignore it when thinking about the chord and scale relationship. The clash with the 3rd is the problem. The 3rd and 7th are responsible for describing the quality of the chord and are therefore very important notes, but the 5th (if it's a perfect 5th) doesn't really add anything. So the 5th is also often omitted from a voicing, whether because it's clashing or because we don't have enough fingers.
- The C Mixolydian scale or mode (C D E F G A B) has the same notes as a C13 chord (C E G B D F A) (ignoring the 11).
- The C Lydian scale or mode (C D E F# G A B) has the same notes as a CMaj13#11 chord (C E G B D F# A).

These scales could be used to improvise over these chords.

Table 11.0.2 A table identifying all Major and Melodic Minor Modes and associated Chords

Mode	Chord	Chord Tone Notation (1 3 5 7 9 11 13)
Major Modes:		

A 13th chord uses all the notes in the scale to which it is diatonic (C Major). This means that chords not extended all the way to the 13th are slightly ambiguous, as they are missing some chord (and therefore scale) notes. The key they appear in depends on the surrounding chords in the progression. It is therefore possible to play multiple scales over the same chord (provided the chord is not a 13th chord). For example, because a CMaj7 chord (C E G B) does not contain an F, you can use either of the following scales to improvise over it:

- C Ionian (C D E F G A B); or
- C Lydian (C D E F# G A B).

Both scales contain all the notes that comprise the CMaj7 chord. Because the CMaj7 does not have a 9th, 11th or 13th, we do not know what key it is in, and therefore we can use both scales over this chord. When thinking in terms of keys (rather than modes) we can say that the CMaj7 chord can be derived from both the key of C Major and G Major (C Lydian).

Looking at the whole chord progression, the CMaj7 chord is preceded by a G7 chord, so it's reasonable to assume that the CMaj7 chord is in the key of C Major (rather than G Major). That doesn't mean that we cannot use the C Lydian mode over the CMaj7 chord. The Chord-Scale System looks at individual chords in isolation and allocates scales to each individual chord. We do NOT need to know what key a particular chord is in, in order to choose a scale to play over that chord. In the above example, it doesn't matter what key the CMaj7 chord is in, we can use both scales to improvise over it. It's often preferable to choose the scale outside the key of the chord progression in order to give a more harmonically complex and jazzy sound. If we had the chord progression: | G7 | CMaj7 || Many Jazz musicians would use the C Lydian mode over the CMaj7 chord precisely because the chord progression is NOT in the key of C Lydian (G Major).

Similarly, you could use the following scales over a Cm7 chord:

1. C Dorian (C D E F G A B);
2. C Aeolian (C D E F G A B).

You could use the following scales over a C7 chord:

1. C Mixolydian (C D E F G A B);
2. C Lydian Dominant (C D E F# G A B).

Chords and scales both consist of the same limited number of notes. All music theory can be broken down into notes and intervals. Scales and Chords are constructed from the 12 notes of the octave.

Using Different Scales Over the Same Chord. When improvising, you can use any scale over a particular chord, as long as that scale has all the notes found in that particular chord. This works even if the whole progression is in a particular key. For example, considering a II-V-I in C Major:

Table 11.0.3 II-V-I chord progressions using scales from different keys

Comments	II	V	I
Chord progression is IN the key of C Major. All chords are FROM the key of C Major.	Dm7	G7	CMaj7
Chord progression is IN the key of C Major. II-V are FROM the key of C Aeolian.	Dm7 $\flat$	Gm7	CMaj7

- Even though the top chord progression is IN the key of C Major, we can still use the C Lydian scale (G Major) to improvise over the CMaj7 chord.

- Even though a chord progression is in the key of C Major, we can still borrow chords from a different key. In the case of the bottom chord progression, we borrowed the II-V chords from the key of C Aeolian (E Major).

Omitting and Changing Notes. The most important notes of a chord are the Guide Tones (3rd & 7th) and to a lesser extent the root (which establishes tonality). The 5th isn't required which means you can play any scale over a chord, provided that scale contains the root, 3rd and 7th of the chord. For example, you can use any scale with a C, E and B in it over a CMaj7 chord, even if it doesn't contain a G.

The following also applies:

- You can omit notes; and
- You can treat notes enharmonically (i.e. as a different chord degree).

Where a scale has BOTH a minor 3rd and a Major 3rd, the Major 3rd is the True 3rd and the minor 3rd is a disguised #9. And it is possible for a scale to omit the 3rd or 7th of a chord and yet still work over that chord — the 3rd or 7th is implied by the harmony. For example, you can use a scale that omits the 3rd over a CMaj7 chord, and the 3rd is implied by the chord.

Considering the C7 chord, the following scales can be used to improvise over it.

Table 11.0.4 Scales that work over a C7 chord

Scale	Degrees	Comments
C Dominant Bebop scale	1 2 3 4 5 6 7 7	Maj 7 used as passing note
Whole-tone scale	1 2 3 5 6 7	1, 3 & 7
Altered scale	1 2 3 4 5 6 7	3 = #2 = #9; 4 = 3
Blues scale	1 3 4 #4 5 7	3 = #2 = #9; Omits 3

- The C Dominant Bebop scale fits perfectly over a C7 chord because it starts on the root (C) and has the 3 and 7. The 7 acts only as a passing note.
- The C Wholetone scale fits perfectly over a C7 chord because it starts on the root (C) and has the 3 and 7.
- The C Altered scale also fits over a C7 chord even though it has a 3 (E). This is because the 3 acts as a #9 (D#) and the True 3rd is the 4 (E). So the 3 is enharmonically the same as a #9 and the 4 is enharmonically the same as a 3. So this scale fits over a C7 chord.
- The C Blues scale also fits over a C7 chord. The 3 is treated as a #9 and the 3 is omitted. The 3 is implied by the chord C7. The scale then has an implied 3 and a 7, so fits over C7.

(Note #1: The 3 always overrides the minor 3 and turns it into a #9. This generally only applies to V7 chords and works because the #9 is an available tension over a C7 chord).

(Note #2: This idea does NOT apply to 7ths. You cannot turn a 7 into a #6 or #13. This is because a #13 is not an available tension over any chord).

Conclusion. Below is a selection of scales that can be used over particular chords, since they share the root, 3rd and 7th of the chord.

Table 11.0.5 Scales for common chord types, examples in C

Chord	Possible Scale	Degrees	From C
Major 7 (any scale with Maj 3 & Maj 7)	Ionian	1 2 3 4 5 6 7	C D E F G A B
Lydian	1 2 3 #4 5 6 7	C D E F# G A B	
Lydian Augmented	1 2 3 #4 #5 6 7	C D E F# G# A B	
Major Bebop Scale	1 2 3 4 5 6 6 7	C D E F G A A B	
Minor 7 (any scale with min 3 & min 7)	Aeolian	1 2 3 4 5 6 7	C D E F G A B
Dorian	1 2 3 4 5 6 7	C D E F G A B	
Phrygian	1 2 3 4 5 6 7	C D E F G A B	
Half Diminished (any scale with min 3, 5 & min 7)	Locrian	1 2 3 4 5 6 7	C D E F G A B
Half-diminished	1 2 3 4 5 6 7	C D E F G A B	
Dominant 7 (any scale with Maj 3 & min 7)	Mixolydian	1 2 3 4 5 6 7	C D E F G A B
Lydian Dominant	1 2 3 #4 5 6 7	C D E F# G A B	
Mixolydian 6 Scale	1 2 3 4 5 6 7	C D E F G A B	
Altered	1 2 3 4 5 6 7	C D E F G A B	
Dorian 2	1 2 3 4 5 6 7	C D E F G A B	
Major Pentatonic	1 2 3 5 6	C D E G A	
Blues	1 3 4 #4 5 7	C E F F# G B	
5th mode of Harmonic Minor	1 2 3 4 5 6 7	C D E F G A B	
5th mode of Harmonic Major	1 2 3 4 5 6 7	C D E F G A B	
Dominant Bebop	1 2 3 4 5 6 7 7	C D E F G A B B	
Dominant Diminished	1 2 #2 3 #4 5 6 7	C D D# E F# G A B	
Whole Tone	1 2 3 #4 #5 7	C D E G A B	

Chapter 12

Barry Harris and Jazz Piano Theory

This chapter presents the harmonic and theoretical system developed by the Detroit jazz pianist and educator Barry Harris (1929–2021). Harris spent more than six decades teaching jazz in workshops around the world, and his approach offers one of the most coherent and practical frameworks for understanding bebop harmony ever assembled. His system is centred on two complementary ideas: the **bebop scale** and the **6th chord / diminished 7th system**.

Although Barry Harris was primarily a pianist, his theories describe universal properties of tonal harmony and are equally applicable to any chromatic instrument. For the accordion player, the isomorphic layout of the chromatic button keyboard is a particular advantage: every pattern, every scale, every block-chord exercise has exactly the same fingering in all twelve keys. The daily exercises Harris recommended in *every* key can be mastered more systematically on the accordion than on most other instruments.

The chapter is organised as follows: the introduction presents Harris’s background and philosophy; this is followed by a detailed treatment of the three bebop scales with notation; the 6th chord / diminished 7th system is then explained both theoretically and in terms of four-voice keyboard voicings; and finally, a set of daily scale and chord exercises — drawn directly from Harris’s workshop teaching — is presented with practice guidance.

12.1 Introduction to Barry Harris and His Philosophy

Barry Harris (1929–2021) was an American jazz pianist, born in Detroit, Michigan, and widely regarded as one of the most important custodians of the bebop tradition. He performed and recorded with many of the giants of jazz, including Coleman Hawkins, Cannonball Adderley, Yusef Lateef, Lee Morgan, and Wes Montgomery, and was a trusted companion and collaborator of Charlie Parker and Thelonious Monk.

Beyond his playing, Barry Harris was a devoted and prolific educator. He founded the Barry Harris Jazz Workshop in New York City in 1982, which continued for nearly four decades, attracting students from across the world. He was known as the “keeper of the bebop flame,” insisting that the language of bebop — its vocabulary, its rhythmic feel, its harmonic logic — needed to be understood deeply, not imitated superficially.

12.1.1 Core Philosophy

Barry Harris’s approach to jazz theory is built on a small number of powerful ideas that unlock the entire harmonic language of bebop and post-bop jazz. His central claim is simple and radical:

Everything in jazz comes from the major scale.

In Harris’s system, the major scale is extended into an eight-note **bebop scale** by inserting a single chromatic passing tone. This eight-note scale is not an arbitrary addition — it is the key that allows chord tones to fall consistently on the strong beats of the bar when playing through a scale passage. This rhythmic alignment of chord tones with strong beats is what gives bebop melody its characteristic inside, swinging quality.

The second pillar of Harris’s system is the **6th chord / diminished 7th system**, sometimes called the “mother chord” concept. Every major key contains a 6th chord and a related diminished 7th chord whose notes, taken together, form the bebop scale. By learning to move between these two chords through all their inversions, a jazz musician develops smooth, stepwise voice leading that works across the entire range of the keyboard.

These ideas are highly practical for the accordion player. The chromatic button accordion’s isomorphic layout means that any pattern learned in one key can be transposed immediately to all twelve keys by shifting position. Barry Harris’s exercises, which Barry himself insisted be practised in all twelve keys, are particularly well-suited to this instrument.

12.2 The Bebop Scales

A **bebop scale** is an eight-note scale created by inserting a single chromatic passing tone into a standard seven-note scale. The purpose of this addition is rhythmic: when a bebop scale is played from the root descending (or ascending, starting on beat 1), chord tones fall on every strong beat (beats 1, 2, 3, 4) and passing tones fall on the off-beats (the “ands”). This produces naturally swinging melodic lines in which the harmonic structure is reinforced by the rhythm.

Barry Harris taught three primary bebop scales, each derived from a standard scale by adding a chromatic passing tone at a specific location. All three are taught and practised over a major 6th chord or dominant 7th chord in the corresponding key.

12.2.1 The Major Bebop Scale

The **major bebop scale** is built from the major scale by inserting a chromatic passing tone between the 5th and 6th scale degrees — that is, between G and A in C major. This passing tone is the augmented 5th ($\sharp 5$, or $G\sharp$ in the key of C).

Table 12.2.1 C Major Bebop Scale — Scale Degrees and Notes

Degree	1	2	3	4	5	$\sharp 5$	6	7	8
Note (C)	C	D	E	F	G	$G\sharp$	A	B	C

The $G\sharp$ is a passing tone only — it does not belong to the C major chord, but allows smooth connection between G and A. When descending, the $G\sharp$ appears between A and G as a chromatic neighbour.



Figure 12.2.2 C major bebop scale, ascending and descending. The #5 (G#) is the added chromatic passing tone between the 5th and 6th scale degrees.

12.2.2 The Dominant Bebop Scale

The **dominant bebop scale** is built from the Mixolydian scale (the natural mode used over a dominant 7th chord) by inserting the natural 7th as a chromatic passing tone between the b7 and the octave. In the key of C this means adding B natural between Bb and the upper C.

Table 12.2.3 C Dominant Bebop Scale — Scale Degrees and Notes

Degree	1	2	3	4	5	6	b7	7	8
Note (C)	C	D	E	F	G	A	Bb	B	C

The dominant bebop scale is the most widely used of Barry Harris’s scales. It is the scale associated with the dominant 7th chord (C7 in C) and is used extensively in ii–V–I progressions. The chord tones of the C7 chord (C, E, G, Bb) fall on strong beats when the scale is descended from the root.



Figure 12.2.4 C dominant bebop scale, ascending and descending. The natural 7 (B) is the chromatic passing tone inserted between the ♭7 and the octave.

12.2.3 The Minor Bebop Scale

Barry Harris’s **minor bebop scale** is derived from the harmonic minor scale by adding the natural 6th as a chromatic passing tone between the ♭6 and the raised 7th. This produces an eight-note scale that contains both the characteristic minor-key colours (the ♭3, ♭6) and a smooth leading tone (the natural 7th of the harmonic minor) with an additional passing tone (the natural 6th).

Table 12.2.5 C Minor Bebop Scale — Scale Degrees and Notes

Degree	1	2	♭3	4	5	♭6	6	7	8
Note (C)	C	D	E♭	F	G	A♭	A	B	C

Notice that the C minor bebop scale contains the notes of both the C minor 6th chord (C, E♭, G, A) and the B diminished 7th chord (B, D, F, A♭), the same pairing that underpins the 6th/diminished system described in the next section.

external/notation/barry-harris-minor-bebop-c.pdf

Figure 12.2.6 C minor bebop scale, ascending and descending. The natural 6 (A) is the chromatic passing tone inserted between $\flat 6$ ($A\flat$) and the raised 7th (B).

12.3 The 6th Chord / Diminished 7th System

The heart of Barry Harris’s harmonic teaching is the relationship between a major 6th chord and its companion diminished 7th chord. Harris called this pairing the **mother chord** because virtually all jazz harmony can be derived from it. Understanding and internalising this system is, in Harris’s view, the foundation upon which all jazz improvisation and accompaniment is built.

12.3.1 Constructing the 6th and Diminished Pair

In the key of C major, the two chords of the system are:

- The **C major 6th chord** ($C6$): C — E — G — A. This is a C major triad with an added major 6th (A). It is built on the tonic (scale degree 1) of C major. The notes C, E, G, A are scale degrees 1, 3, 5, and 6.
- The **B diminished 7th chord** ($B^\circ 7$): B — D — F — $A\flat$. This chord is built on scale degree 7 of C major, using a flattened 7th to complete the diminished 7th structure. The notes B, D, F, $A\flat$ fill in the four chromatic gaps between the notes of the $C6$ chord.

Together, these two chords contain exactly eight distinct pitches: C, D, E, F, G, A $\flat$, A, B. Arranged in scale order, these are the notes of the *C major bebop scale*. The 6th chord and its diminished companion are, therefore, two complementary halves of the same eight-note scale.

Table 12.3.1 C6 and B $^{\circ}$ 7 — Notes of the 6th / Diminished System

Chord	Voice 1	Voice 2	Voice 3	Voice 4
C6 (tonic)	C	E	G	A
B $^{\circ}$ 7 (passing)	B	D	F	A $\flat$
Combined (ascending)	C — D	E — F	G — A $\flat$	A — B

12.3.2 Voice Leading Between the Two Chords

The most remarkable property of this chord pair is the voice leading between them. Each note of the C6 chord moves by just a semitone or whole tone to reach the nearest note of the B $^{\circ}$ 7 chord. Specifically:

- C moves down a semitone to B (or up a semitone to C $\sharp$ /D $\flat$ in reverse direction)
- E moves down a whole tone to D
- G moves down a semitone to F $\sharp$ /G $\flat$, or up to A $\flat$
- A moves down a semitone to A $\flat$

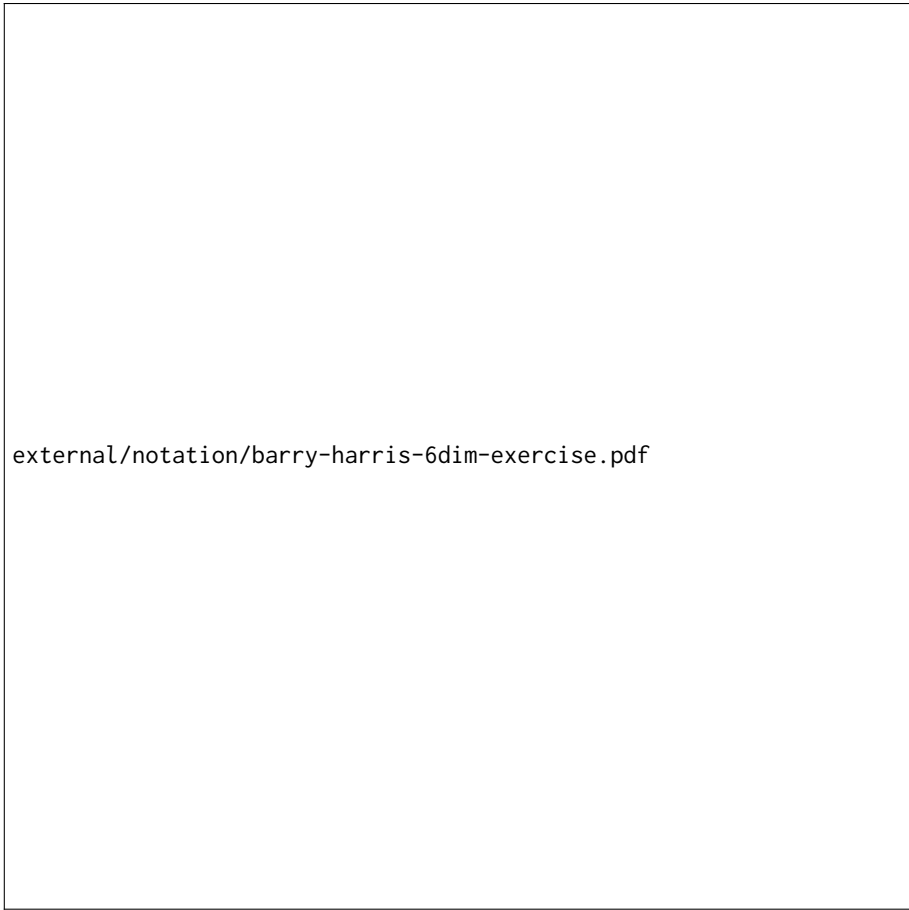
This smooth, stepwise motion is what creates the characteristic “bebop sound” when alternating between these two chords. Because the diminished 7th chord is symmetrical (it divides the octave into four equal minor thirds), the B $^{\circ}$ 7 chord has only three distinct inversions, and each inversion is enharmonically equivalent to a different diminished 7th chord root: B $^{\circ}$ 7 = D $^{\circ}$ 7 = F $^{\circ}$ 7 = A $\flat$ $^{\circ}$ 7.

12.3.3 The Four Inversions Through One Octave

Barry Harris’s fundamental keyboard exercise involves alternating between C6 and B $^{\circ}$ 7 through all four positions as you ascend (or descend) the keyboard. Because both chords are four-voice chords, and the combined scale is eight notes, one complete octave of the ascending exercise passes through exactly four C6 inversions and four B $^{\circ}$ 7 inversions.

Table 12.3.2 Ascending C6 / B $^{\circ}$ 7 Exercise — Four-Voice Close Voicings

Position	Chord	Voice 4 (top)	Voice 3	Voice 2	Voice 1 (bottom)
1	C6	A	G	E	C
2	B $^{\circ}$ 7/D	B	A $\flat$	F	D
3	C6/E	C	A	G	E
4	B $^{\circ}$ 7/F	D	B	A $\flat$	F
5	C6/G	E	C	A	G
6	C6/A	G	E	C	A
7	B $^{\circ}$ 7	A $\flat$	F	D	B
8	C6	A	G	E	C (octave up)



external/notation/barry-harris-6dim-exercise.pdf

Figure 12.3.3 The ascending C6 / B°7 chord exercise. Each four-note block chord moves upward by step, alternating between the tonic 6th chord (C6) and its passing diminished chord (B°7) through their inversions.

Practise this exercise hands separately and then together. The left hand can hold sustained bass notes (C — G — C, or the full chord root progression) while the right hand moves through the ascending inversions.

12.3.4 Using the Major 6th Diminished Scale

The following transcription, arranged by Robert Glover from a Barry Harris demonstration, shows how the major 6th diminished scale is applied in practice. The piece cycles through the alternating C6 and B°7 voicings, illustrating how the scale generates smooth voice-leading through its inversions. Study each phrase carefully, noting how every melodic movement connects chord tones of one voicing to the nearest chord tones of the next. The original video demonstration by Barry Harris can be viewed at youtu.be/OGA64aGWX98.

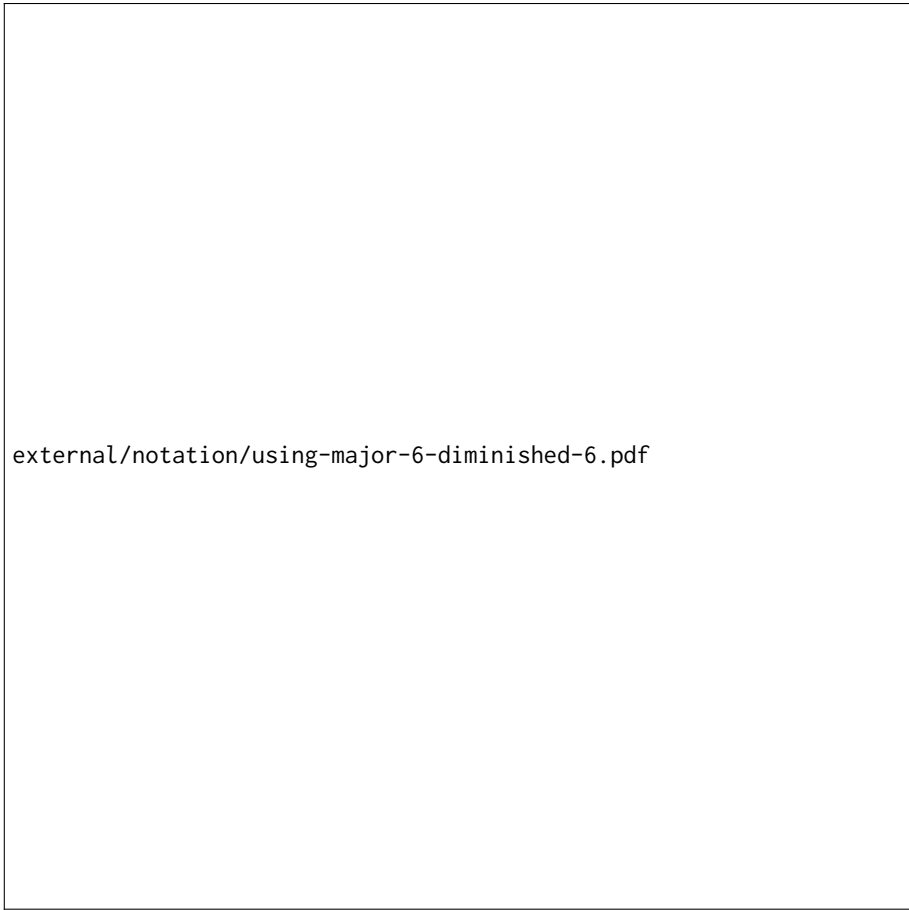
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external/notation/using-major-6-diminished-3.pdf

external/notation/using-major-6-diminished-4.pdf

external/notation/using-major-6-diminished-5.pdf



external/notation/using-major-6-diminished-6.pdf

Figure 12.3.4 Using the Major 6th Diminished Scale (Barry Harris, arr. Robert Glover). A demonstration of the C6 / B°7 system applied melodically and harmonically, showing practical voice-leading through the alternating voicings.

12.3.5 Barry Harris Diminished Chords Lesson

The following transcription, made by Robert Glover from a live Barry Harris workshop (Bilbao, Spain, 20 May 2016), captures both the musical examples and Harris’s spoken commentary as he demonstrated the borrow/surround technique for diminished chords. Harris shows how each note of the C diminished 7th chord is “surrounded” by the roots of its four related dominant chords, and how those borrowed tones create melodic motion back to the tonic. The original video was recorded by Montxo Garcia: youtu.be/JlQNLZiKTY0.

external/notation/barry-harris-dim-lesson-1.pdf

external/notation/barry-harris-dim-lesson-2.pdf

external/notation/barry-harris-dim-lesson-3.pdf

external/notation/barry-harris-dim-lesson-4.pdf

Figure 12.3.5 Barry Harris Diminished Chords Lesson (Bilbao, Spain, 2016; transcript R. Glover, 2019). The score illustrates the borrow/surround technique: dominant roots surrounding each note of the diminished 7th chord resolve stepwise back to the tonic chord tones. Green noteheads show the C diminished 7th chord tones; red noteheads show the borrowed dominant roots.

12.4 Basics of the Dominant 7th Scale

This section is based on a transcription by Robert Glover (26 August 2017) of a video lesson by Connor (a Barry Harris student) entitled “Barry Harris... “Basics” of Dom 7 Scale” (youtu.be/kCSB_iUW5Uk). The lesson demonstrates simple melodic patterns that can be derived from a dominant 7th scale, using B♭7 as the primary example. It is a practical companion to Barry Harris’s teaching on the 6th chord / diminished 7th system: once you understand the underlying harmony, these short patterns give you the raw material to build longer improvised phrases.

Connor opens by arguing that jazz improvisation is not a mysterious gift for a chosen few. Historically, great composers such as Bach, Mozart, Beethoven and Liszt were also accomplished improvisers. Bach is documented to have improvised full fugues for royalty on demand. What made this possible was not genius alone, but a thorough knowledge of scales and harmony. Connor draws the same conclusion for jazz: “This whole notion of improvisation can really be broken down to a science.” [1:04]

The lesson then demonstrates a series of increasingly sophisticated patterns

over B♭7, using the **B♭ dominant 7th scale** (also known as B♭ Mixolydian: B♭, C, D, E♭, F, G, A♭, B♭). For each pattern Connor first plays an ascending *and* descending version, then shows how adding a half-step approach note below the first note of each group enriches the sound.

12.4.1 Scale Outline (from the Previous Video)

Connor begins with a brief recap of the scale-outline technique from his previous lesson (the 12-bar blues form), demonstrating how a B♭7 scale outline moves to an E♭7 in the context of a simple chord progression.

external/notation/connor-dom7-basics-1.pdf

Figure 12.4.1 Mm. 1–5. Scale outline over B♭7 and E♭7, as introduced in the previous video. [0:00–3:32]

12.4.2 Scale Up and Down; Thirds; Triads; Full 7th Chords

“So, by now you should be able to just run, for instance, a B♭ dominant 7th scale by itself.” [3:32]

Connor then demonstrates the same scale in thirds (playing scale degree 1 with scale degree 3, degree 2 with degree 4, and so on), then in triads (three-note groups built from each scale degree), and finally as full seventh chords. Each of these represents a distinct set of “puzzle pieces” that can be recombined to make improvised phrases — just as short words are learned before full sentences.

When we practice small things that come from scales, what we can do is we can take the small things and we can put them together in

an infinite number of different ways. It's almost like speech... You learn about small words, the small phrases first. And then you put them together. [5:16]

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Figure 12.4.2 Mm. 6–37. B♭7 scale up and down (mm. 6–9); scale in thirds (mm. 10–12); scale in triads (mm. 13–18); scale outline / fast thirds (mm. 19–21); half-step approach added (mm. 22–37). [3:32–6:56]

12.4.3 Major Arpeggios, Augmented Arpeggio, and Diminished 7th Applications

Connor next moves to arpeggios. For B♭7, the three important arpeggios are built on scale degrees 1, 5, and ♭7:

- B♭ major arpeggio (scale degree 1): B♭ – D – F
- F minor arpeggio (scale degree 5): F – A♭ – C
- A♭ major arpeggio (scale degree ♭7): A♭ – C – E♭

All three arpeggios should be practised in all inversions. This is connected to Charlie Parker's bebop head “Chi Chi”, which opens with an inversion of the A♭ major triad: [8:10]

Connor also introduces the **augmented arpeggio** (a major arpeggio with a sharpened fifth: B♭ – D – F♯), which often appears in standard tunes. Like all other patterns, its inversions should also be practised.

Finally, Connor reminds the viewer of a fundamental Barry Harris principle: “A dominant seventh chord comes from a fully diminished seventh chord.” [10:20] This means that when improvising over a dominant quality chord, you can use the corresponding fully diminished 7th chord and its associated scale.

external/notation/connor-dom7-basics-3.pdf

Figure 12.4.3 Mm. 38–63. Major arpeggios on scale degrees 1, 5, and $\flat 7$ (mm. 38–40); inversions (mm. 41–50); Charlie Parker “Chi Chi” reference (mm. 51–52); descending versions (mm. 53–59); fully diminished 7th chord over $B\flat 7$ (mm. 60–63). [6:56–10:40]

12.5 Daily Scale and Chord Exercises

Barry Harris was insistent that theory without daily practice is meaningless. His workshop exercises are designed to be drilled slowly and methodically, at a tempo where every note is even, every chord change is smooth, and the sound of each chord type is committed to memory. He recommended beginning in C major and then transposing each exercise to all twelve keys before moving on.

The exercises below form the core daily practice routine that Harris taught at his workshops. They should be practised with a metronome and, once comfortable, with a rhythm section playalong.

12.5.1 Exercise 1: The Bebop Scale

Play each of the three bebop scales (major, dominant, and minor) ascending and descending for two octaves in every key. Pay close attention to the placement

of the chromatic passing tone — it should sound smooth and connected, not like an accidental. Barry Harris often said that the bebop scale should feel like a single phrase, not like a seven-note scale with a wrong note added.

Practice progression:

1. Right hand alone, one octave ascending and descending, slowly (quarter = 60).
2. Left hand alone, one octave ascending and descending.
3. Hands together in parallel motion (both hands playing the same scale in the same direction).
4. Hands together in contrary motion (right hand ascending while left hand descends).
5. Increase tempo gradually to quarter = 120 before moving to the next key.

Begin with the dominant bebop scale, as it is the most common in jazz. The major bebop scale follows naturally, and the minor bebop scale should be introduced once the first two are secure.

12.5.2 Exercise 2: The 6th Chord / Diminished Alternation

This is the central exercise of the Barry Harris method. It trains the ear, the fingers, and the musical imagination simultaneously. Play the ascending 6th chord and diminished chord exercise (shown in the previous section) using close four-voice block chords.

Practice routine:

1. Right hand alone, ascending: C6 — B°7/D — C6/E — B°7/F — C6/G — C6/A — B°7 — C6 (upper octave).
2. Right hand alone, descending: reverse the above, moving from the upper C6 back to the root position.
3. Left hand plays sustained root (C in the bass), right hand ascends and descends.
4. Left hand plays the bass note that corresponds to the inversion in the right hand (walking through the scale: C — D — E — F — G — A — B — C).
5. Play the full exercise in all twelve keys.

Listen carefully to the way each chord connects to the next. Barry Harris emphasised that the exercise should produce a *singing, melodic sound*, not a mechanical arpeggio. The top voice traces the major bebop scale — this is the melody you are harmonising.

12.5.3 Exercise 3: The Harmonized Major Scale

This exercise combines the melodic and harmonic aspects of the system. Play the C major scale in the right hand (single notes, one octave ascending) while assigning each note its correct chord symbol from the 6th / diminished system:

- Scale degrees 1, 3, 5, 6 (C, E, G, A) belong to the C6 chord.
- Scale degrees 2, 4, 7 (D, F, B) belong to the B°7 chord.

In practice, each note of the melody is played as the top voice of the corresponding four-note block chord. The figure below shows the chord symbol that applies to each scale degree as you ascend.

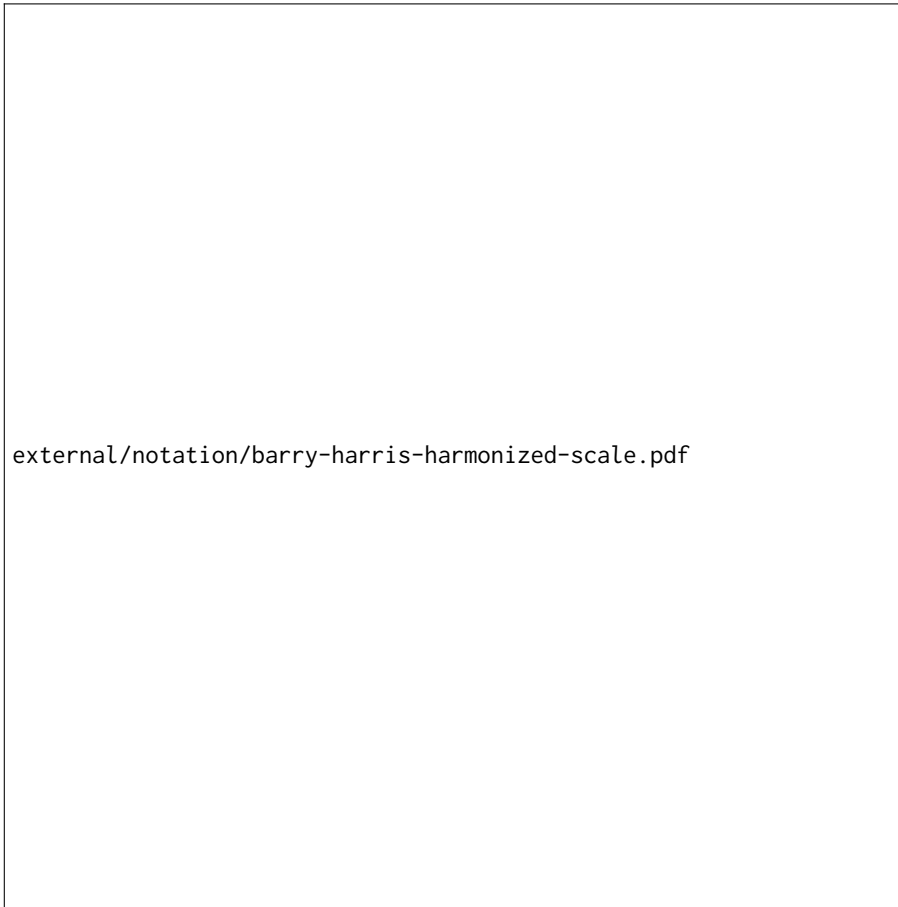


Figure 12.5.1 C major scale with Barry Harris 6th / diminished chord assignments. Each note of the ascending scale is labelled with its corresponding chord: C6 for chord tones, B°7 for passing tones.

Practice routine:

1. Sing or play the scale degree while naming the chord aloud (for example: “C — C6, D — B°7, E — C6, F — B°7 ...”).
2. Play the melody note as the top voice of a full four-voice block chord in the right hand. Use the inversions from the table in the previous section, moving upward smoothly.
3. Add a bass note in the left hand: the root of the chord for chord tones, or the bass note of the inversion for passing diminished chords.
4. Practise descending as well as ascending.
5. Transpose to all twelve keys.

12.5.4 Warm-Up Exercise

The following warm-up, transcribed by Adam Manness (Open Studio Jazz), distils the essential Barry Harris hand positions into a concise eleven-measure

exercise. It cycles through the alternating 6th chord and diminished 7th voicings in close position, ascending and descending over the C major *bebop* tonal centre. Practise it slowly until every voicing is even and balanced, then transpose to all twelve keys.

external/notation/barry-harris-warm-up.pdf

Figure 12.5.2 Barry Harris Warm-Up exercise (Adam Manness / Open Studio Jazz). Alternating C6 and B°7 voicings ascending and descending in close four-voice position.

Work through the exercise in the following stages:

1. Right hand alone, ascending through the full exercise.
2. Right hand alone, descending.
3. Add the bass note in the left hand, matching the inversion in the right hand.
4. Play the complete exercise hands together at a slow, steady tempo (quarter = 60).
5. Increase speed gradually; aim for quarter = 100 before transposing to the next key.

12.5.5 Practice Tips from the Barry Harris Workshop

Barry Harris offered the following guidance consistently throughout his teaching:

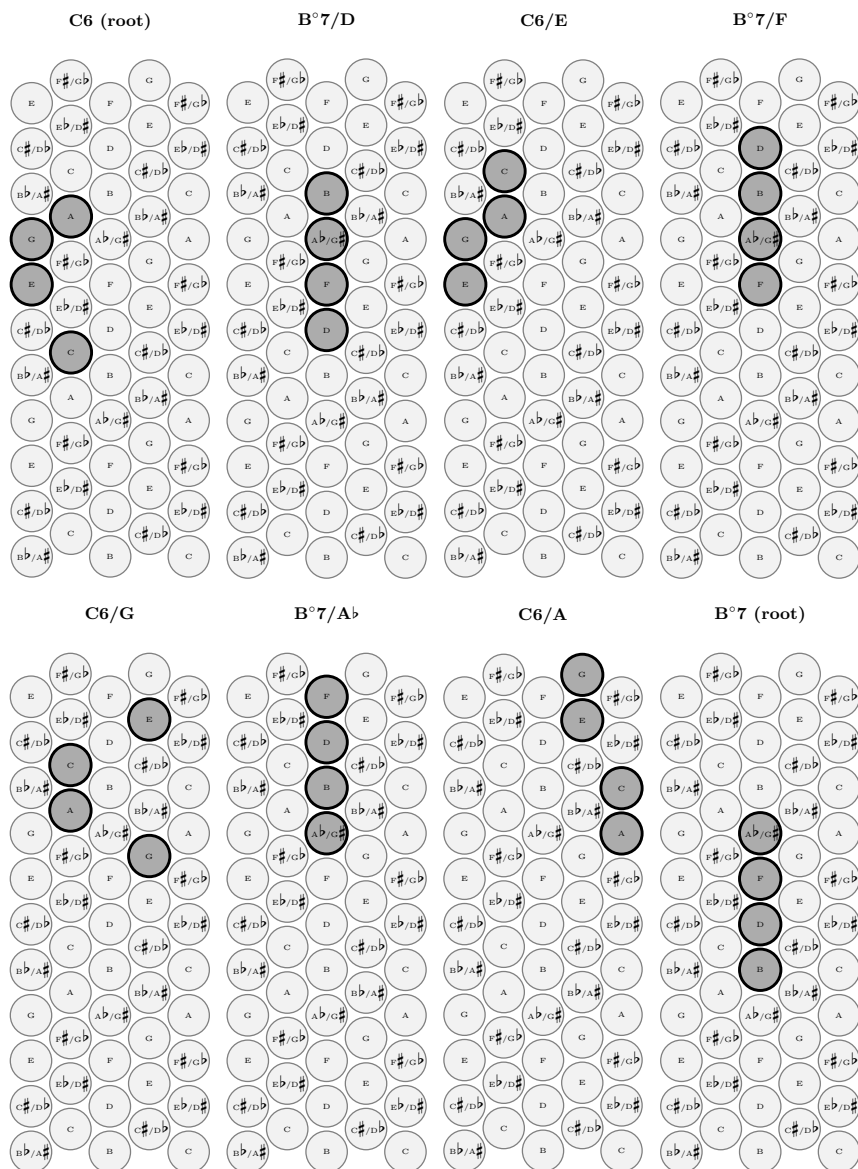
- *Slow is fast.* Always begin at a tempo where every note speaks clearly and every chord is balanced. Speed is a consequence of accuracy, not a goal in itself.
- *Twelve keys, every day.* A pattern learned in only one or two keys is not truly learned. The chromatic button accordion makes this particularly accessible, since the fingering pattern for any exercise is identical in every key.
- *Listen to the top voice.* When practising block chord exercises, the top voice is the melody. Train yourself to hear it singing above the harmony, and ensure it has a connected, vocal quality.
- *Hear the chord before you play it.* Harris insisted that students should be able to sing or audiate each voicing before placing their hands on the keys. The ear leads, the hands follow.
- *The diminished is not a chord, it is a passageway.* The B°7 is a connecting structure — a set of four passing tones that link one position of the C6 chord to the next. Understanding it this way keeps the ear focused on the tonic harmony and prevents the diminished chord from sounding like an unrelated intrusion.

12.6 C6 / B°7 Chord Shapes on the Accordion Keyboard

The eight diagrams below show all eight chord positions of the ascending C6 / B°7 exercise as they appear on the chromatic button accordion keyboard. Each diagram highlights the four notes of one chord voicing: four are C major 6th chord inversions (labelled C6) and four are B diminished 7th inversions (labelled B°7), alternating as you ascend through one octave.

The B°7 chord tones (B, D, F, A♭) all fall on the same vertical column (column 3) of the isomorphic keyboard — they form a single vertical chain. The C6 chord tones (C, E, G, A) spread naturally across columns 1, 2, and 5 in compact close-position clusters. Because both chords are symmetrical (each has only three distinct shapes across all twelve keys due to the diminished 7th symmetry), once the pattern for C major is learned the shapes for all other keys follow by uniform shifting.

Shape 8 shows the B°7 chord in its root position at the lower register, illustrating how the system completes a cycle: after ascending through all seven positions, the diminished chord returns to its root-position shape ready to re-enter the pattern one octave higher.



Notice that all B°7 inversions (shapes 2, 4, 6, and 8) highlight four buttons in a single vertical column of the keyboard — each one simply two rows higher than the previous. This is a direct consequence of the diminished 7th chord's symmetry: all four of its notes (B, D, F, A♭) lie on the same column in the isomorphic layout, separated by equal intervals of a minor 3rd.

The C6 inversions (shapes 1, 3, 5, and 7) each show a compact cluster spread across columns 1, 2, and 4–5, ascending by one scale step with each successive shape.

Chapter 13

Turning Exercises into real musical lines.

This exercise is designed to help you play solos that feel and sound like jazz, not just mechanically cycling through scales and arpeggios.

13.1 The Exercise

A practical way to learn a song and connect each chord to the right scale.

For illustration, we'll use the song *How High The Moon* See Section 13.5 for score. As you play through the song, you play the scale that matches each chord in the progression.

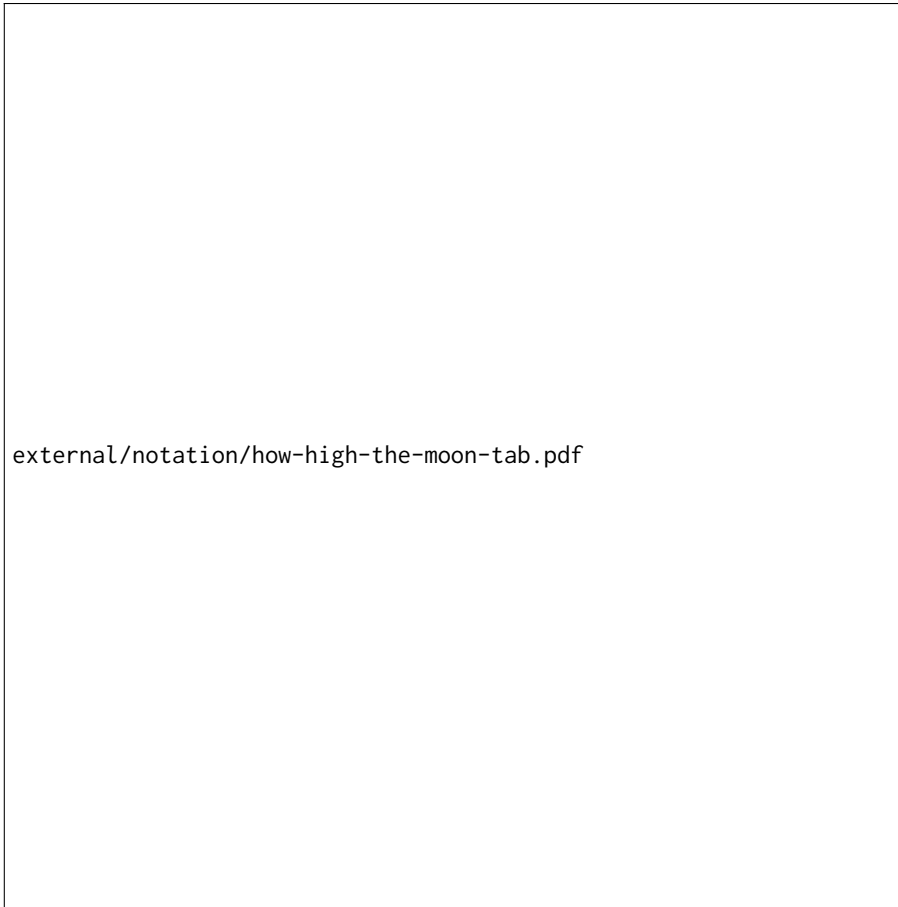


Figure 13.1.1 First four bars of *How High The Moon* — melody with guitar tablature, showing scale runs matching each chord in the progression.

This approach helps you understand the song's structure, the chord changes, and the scale that fits each chord:

- If a chord lasts one bar (like Gm7 to C7), play the chord's scale from the root to the 7th note of the scale.

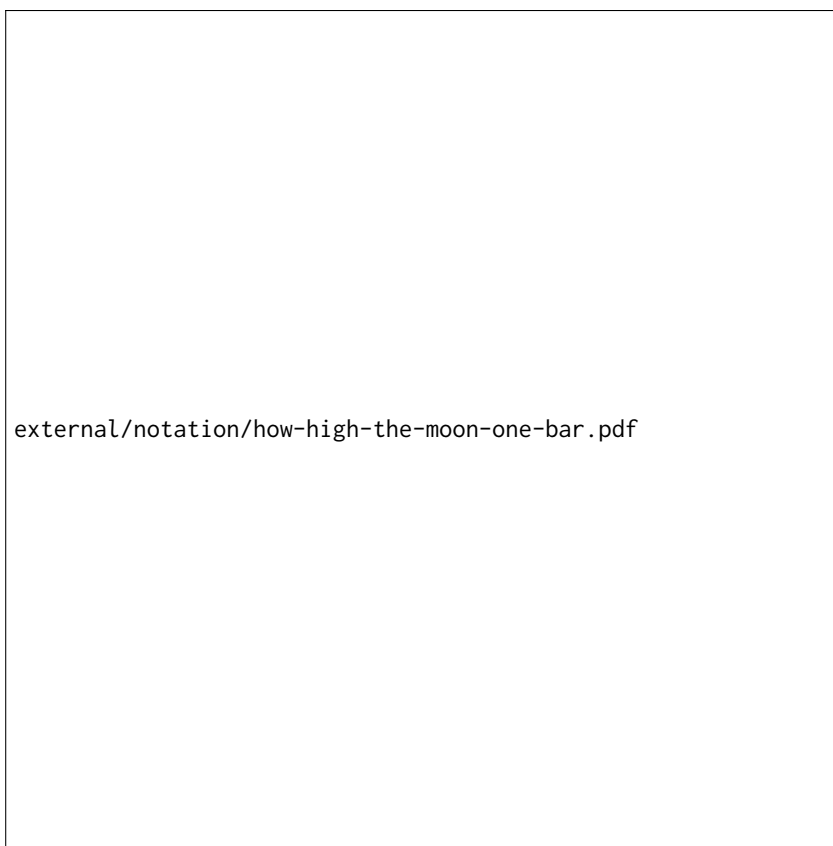
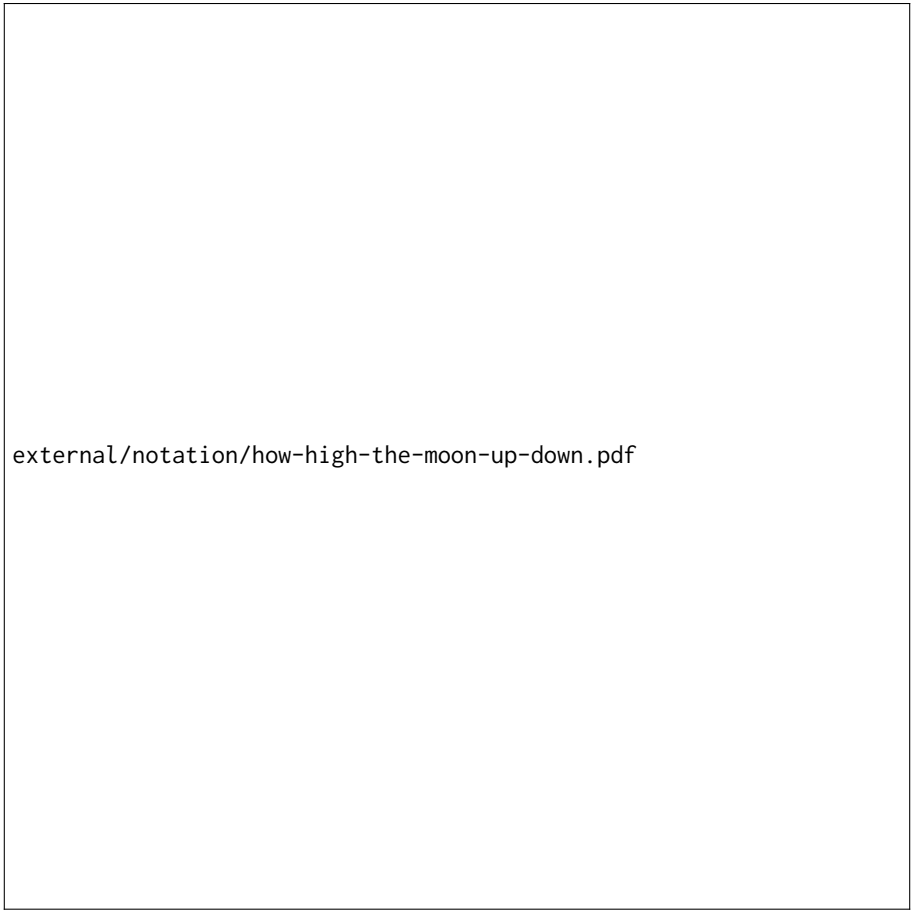


Figure 13.1.2 Gm7 and C7 each lasting one bar — scale played from root to 7th (*From The Root To The 7th*).

- If a chord lasts two bars (like the first two bars), play root up to 7th, then back down to the root again.



external/notation/how-high-the-moon-up-down.pdf

Figure 13.1.3 Gmaj7 over two bars — scale ascending from root to 7th, then descending back to root (*Up & Down From The Root*).

This gives you an overview of the song, allowing you to understand how the chords and scales relate to each other, which is essential for improvising. Having started with this, you can then build the solo by replacing scales for real solo phrases.

13.2 Building Your Jazz Vocabulary

First learn the scales for a song. Then in order to create melodic jazz lines, use arpeggios for building licks that start to sound like jazz.

Start by playing the basic arpeggios through the first four bars of *How High The Moon*.

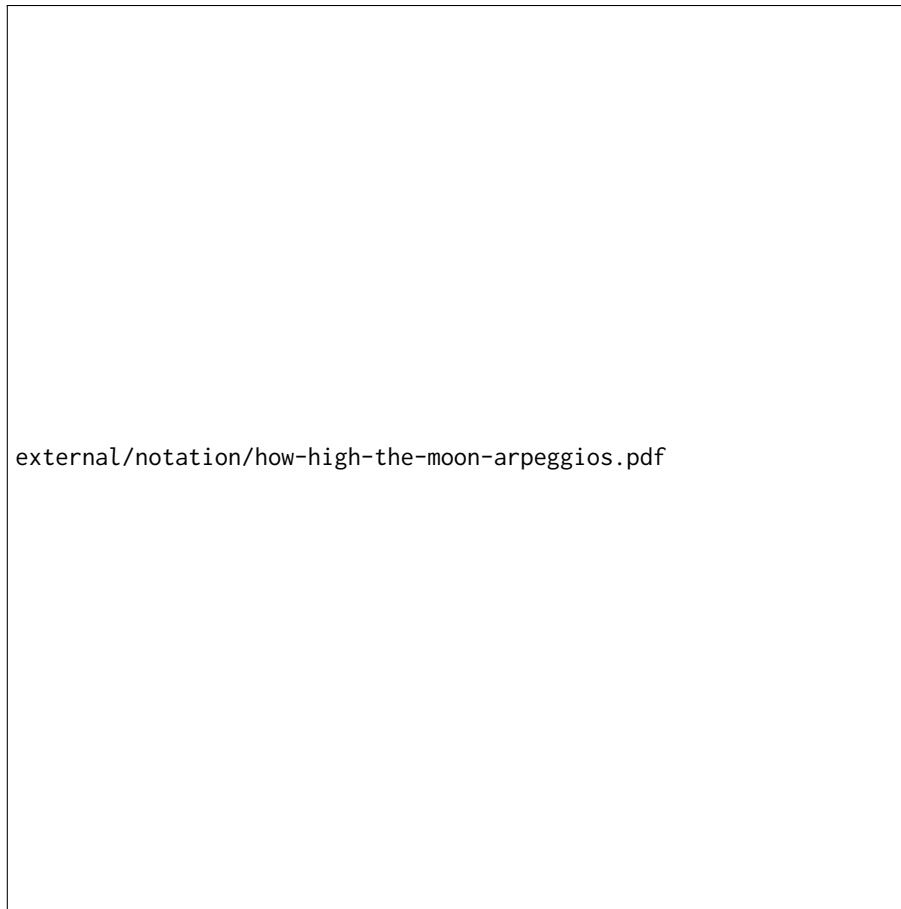


Figure 13.2.1 Basic arpeggios through the first four bars of *How High The Moon* — Gmaj7, Gm7, and C7 with guitar tablature.

Then level it up:

- Add a triplet rhythm to give it some bebop energy.



Figure 13.2.2 Triplet rhythm applied to the arpeggio — giving it that bebop energy.

- Throw in a leading note — a chromatic note just before the first arpeggio note — and finish with a scale run with another chromatic note.

external/how-high-the-moon-leading-note-1.pdf

Figure 13.2.3 Leading note — chromatic approach into the first arpeggio note.

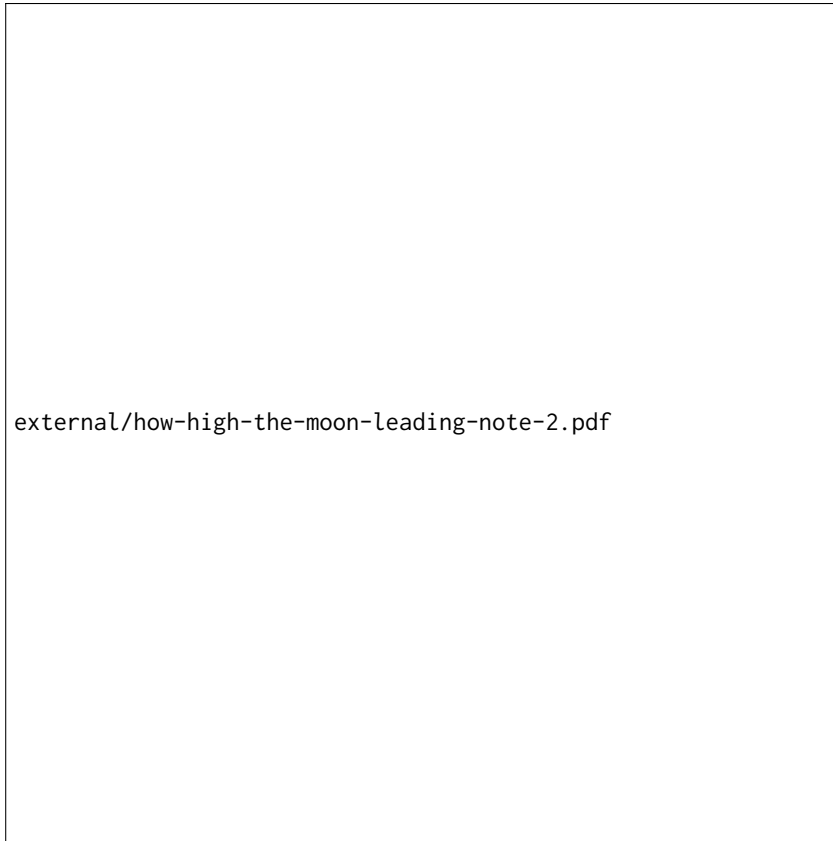


Figure 13.2.4 Scale run finishing with a chromatic leading note.

Hear how much more like jazz that sounds? Now try moving this lick around the song. All three versions are beautiful jazz lines you can use right away.

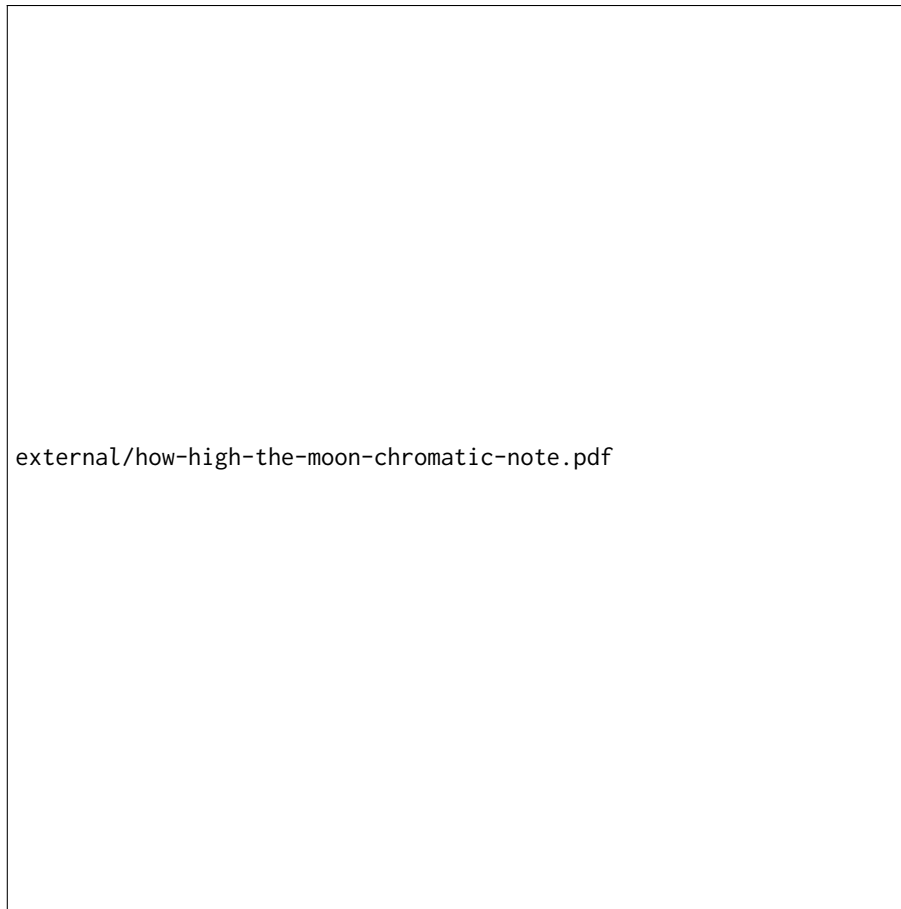


Figure 13.2.5 Chromatic approach note leading into the arpeggio.

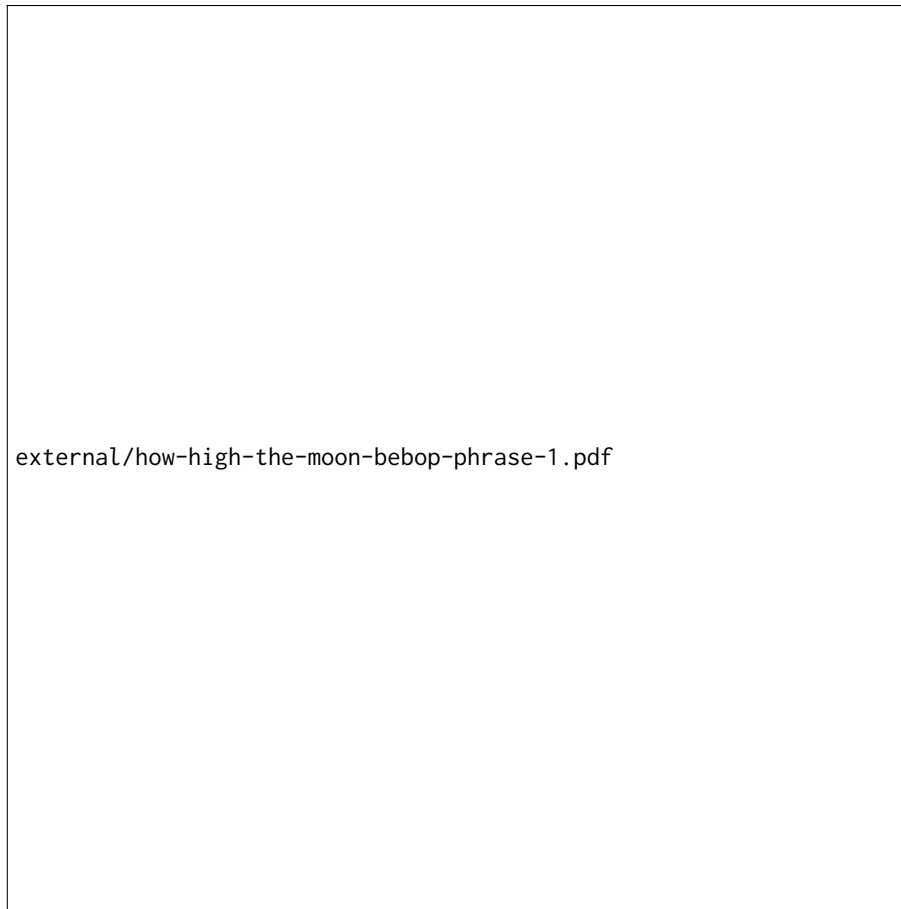
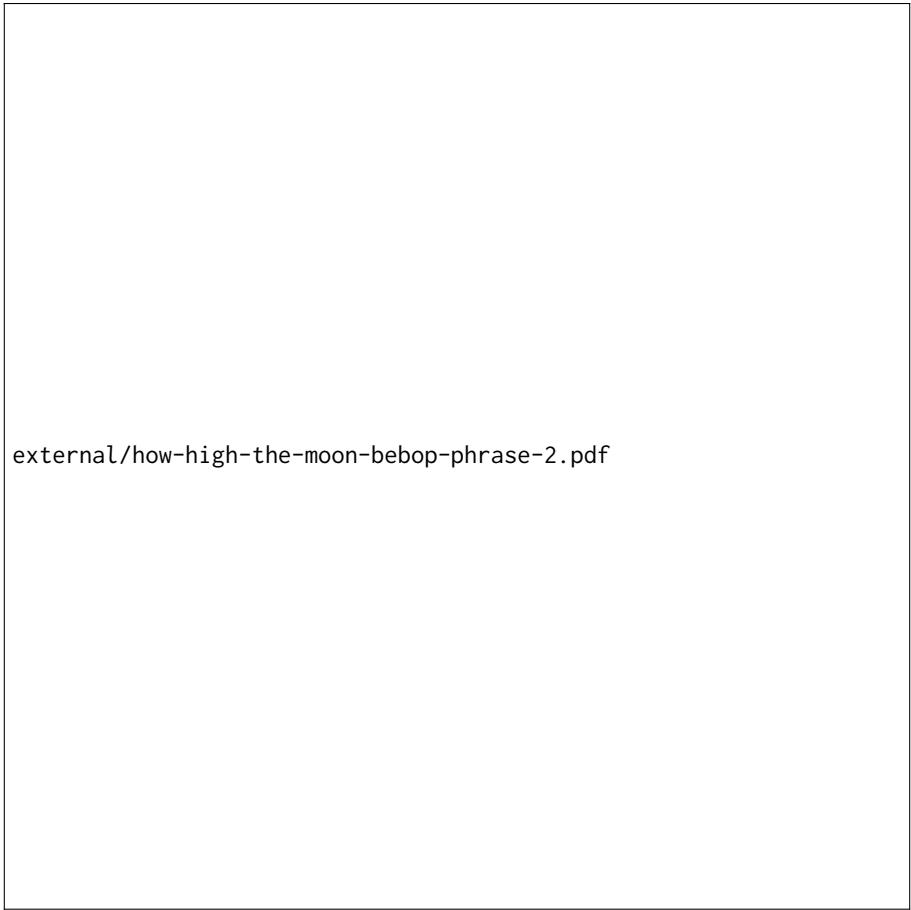


Figure 13.2.6 Bebop phrase — first variation with arpeggio as a triplet and descending scale run.



external/how-high-the-moon-bebop-phrase-2.pdf

Figure 13.2.7 Bebop phrase — second variation with leading note finishing the scale run.

The foundation for this exercise is to first take the phrase through the progression *rubato* and figure out how you can make it fit the chord. That helps you understand the building blocks: chromatic note, arpeggio as a triplet, and a descending scale run with a leading note.

This approach opens up a whole new world for your playing. Suddenly you're not just playing the right notes over a chord, but you're creating strong, melodic lines that sound like jazz. By connecting the lick to the song, you understand how it works and can easily move it around the progression. This is how you build your jazz vocabulary — by creating and internalising licks that are directly connected to the music you want to play, rather than just mechanically cycling through scales.

13.3 Don't Just Practice in All 12 Keys

You've probably heard this advice a thousand times: "take this lick through all 12 keys." Of course that is useful, but it's still "just" an exercise, and not the fastest way to learn to play jazz.

Instead, here's what's way more powerful: *take your lick through an actual song*. When you do this:

- You're not just moving your fingers.
- You're connecting it directly to the music you want to play.

- You're turning it into a flexible phrase that works on lots of chords.
- In the process, you also figure out *how* the lick works, not just practising how to play it.

That's how you get better at creating your own licks that actually sound like jazz, instead of just playing the right notes over a chord.

Here's a Gmaj7 lick to illustrate. First, find the basic chord tones: G, B, F# and D. Then look at what connects them: a leading note into the low 3rd (notice how great this interval skip sounds), and a scale note between the 7th and the 5th.

Once you understand the structure, moving the lick to a new chord becomes easy — for example onto Gm7 (G B F D) and then onto C7.

Next, take it through the song slowly. Don't worry about tempo; out of time is the best place to start. Just move it around the progression. Once that feels natural, start using it inside a solo. That's when it stops being an exercise, and becomes real jazz vocabulary.

13.4 Barry Harris' Arpeggio From the 3rd

You may have heard the advice to use the arpeggio from the 3rd of the chord — it's what Charlie Parker and Wes Montgomery do all the time. But knowing the concept and being able to use it in your solos are two very different things. This is where the approach above becomes incredibly useful.

13.4.1 Why Is the Arpeggio from the 3rd so Powerful?

For Gmaj7 the full arpeggio is G B D F#. The 3rd is B, and the diatonic arpeggio on B is Bm7: B D F# A. You are still playing three chord tones, but you have now added an A on top — a beautiful 9th that makes your solos more colourful.

Start by connecting the regular arpeggio to the arpeggio from the 3rd. First play the arpeggio from the root, then connect it to the version starting from the 3rd.

This helps you hear how the two relate, and also connects them on the neck so moving between them becomes natural and the arpeggio from the 3rd is much easier to find.

Now take the arpeggio from the 3rd through the song. Once that feels comfortable, start building short lines from it and move those lines through the progression, just like before. It will make the arpeggio from the 3rd a part of your playing in no time!

13.5 Transcription: How High the Moon

The following is a lead-sheet transcription of *How High the Moon* (Morgan Lewis / Nancy Hamilton, 1940), the song used throughout this chapter as a vehicle for the exercises above. Study the melody and chord changes before applying the scale and arpeggio exercises described in the preceding sections.

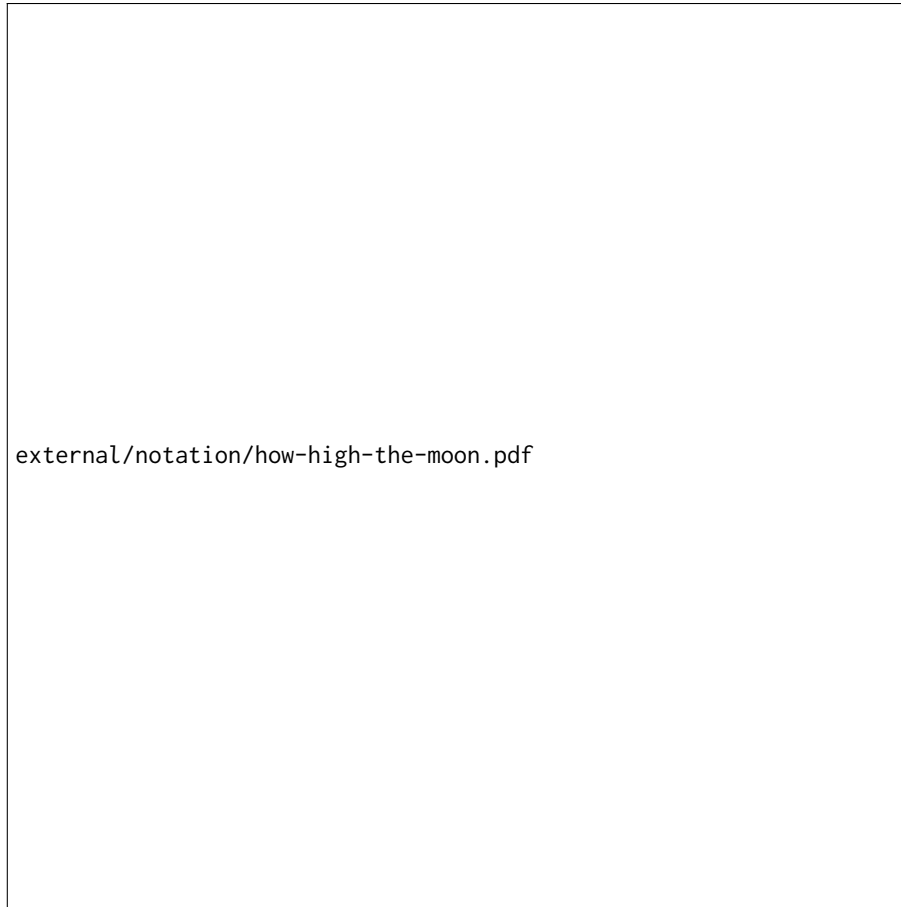


Figure 13.5.1 *How High the Moon* — lead sheet with melody and chord symbols.

Chapter 14

Templates and References

14.1 Blank Chord Shape Template

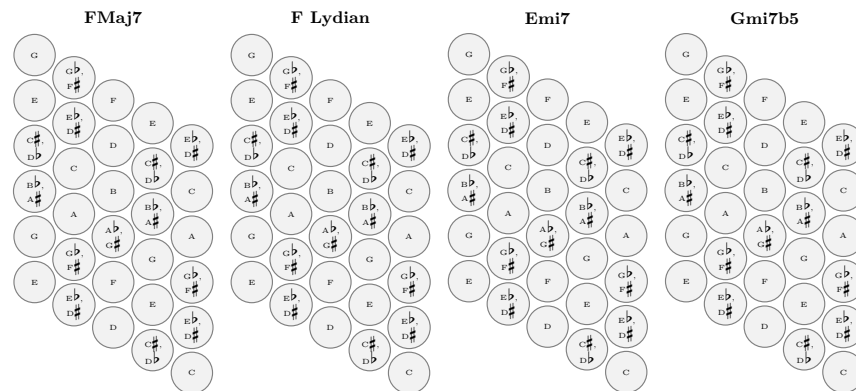


Figure 14.1.1

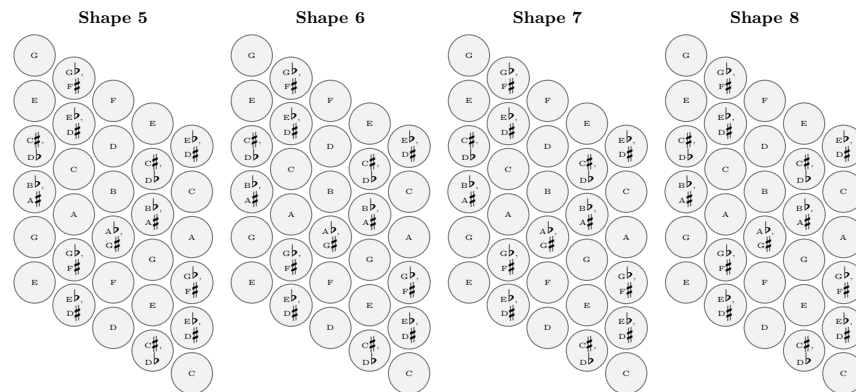


Figure 14.1.2

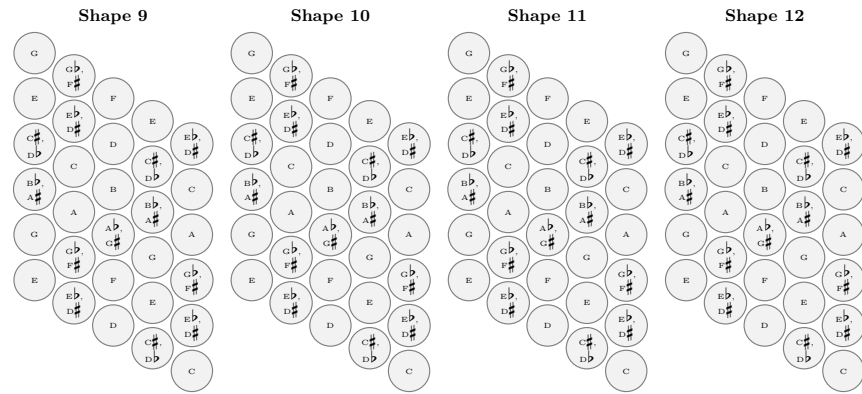


Figure 14.1.3

14.2 Extended Chord Progression Template

This template uses the shared 52-node extended 11-row grid layout, including the added lower and upper extension nodes.

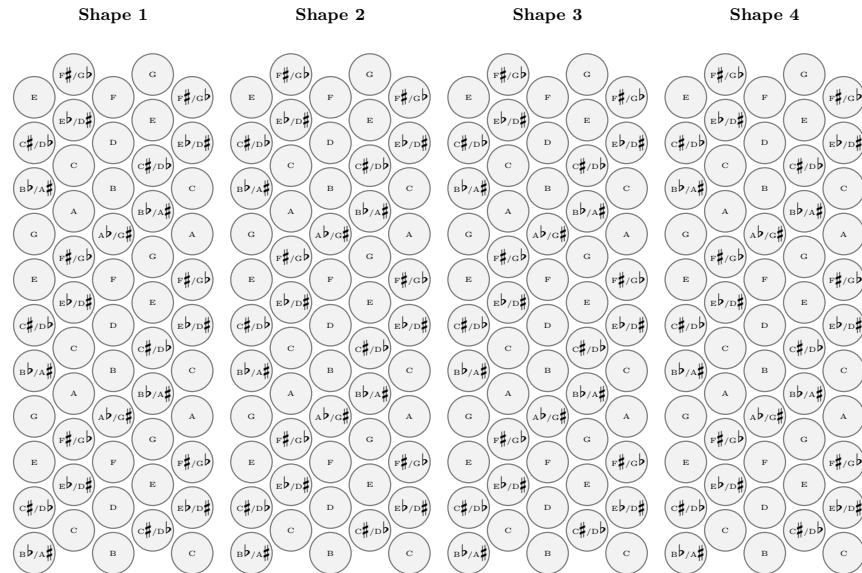


Figure 14.2.1

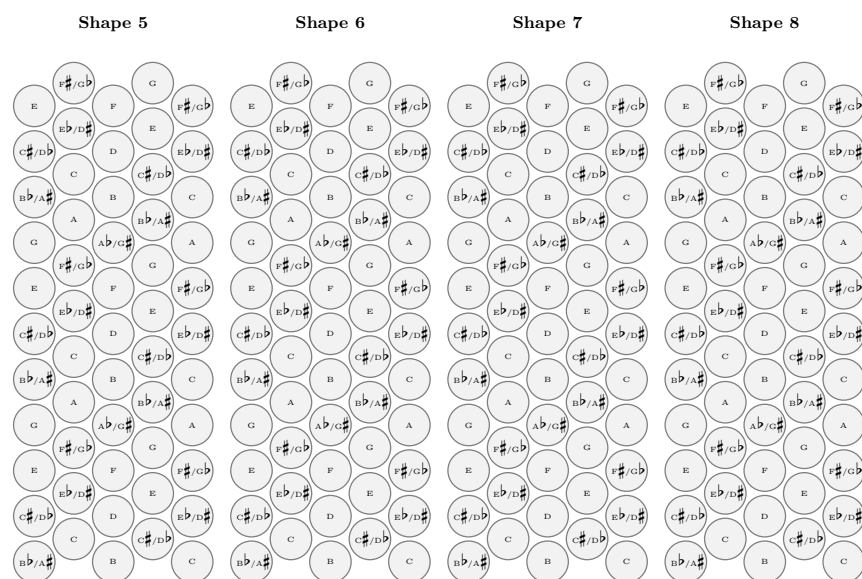


Figure 14.2.2

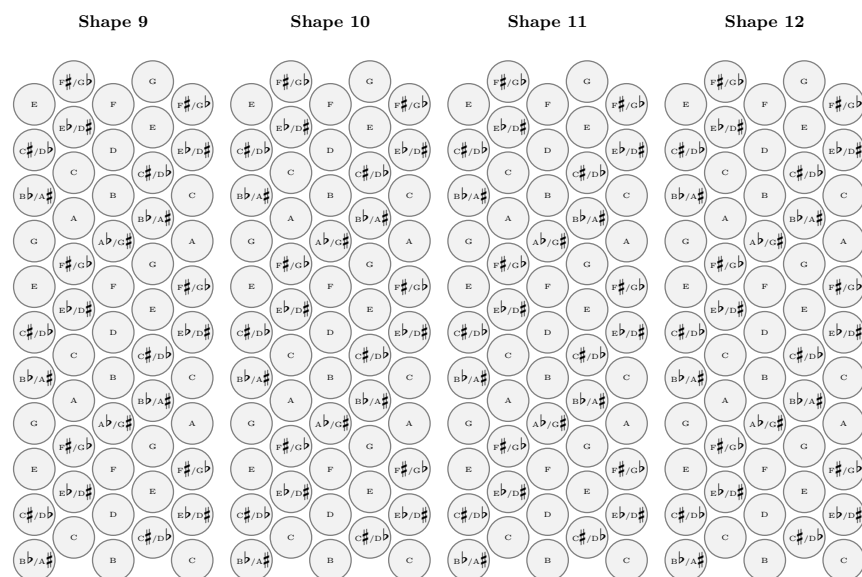


Figure 14.2.3

14.3 Node Reference Grid (1–30)

The grid below numbers every node from 1 (lowest C, bottom-right) to 30 (highest G, top-left). Each diagonal band of five nodes rises from col 5 (right) to col 1 (left). To request a chord shape, simply name the node numbers to highlight — for example “nodes 1, 7, 12” gives the C major triad (C, E, G).

Node Reference

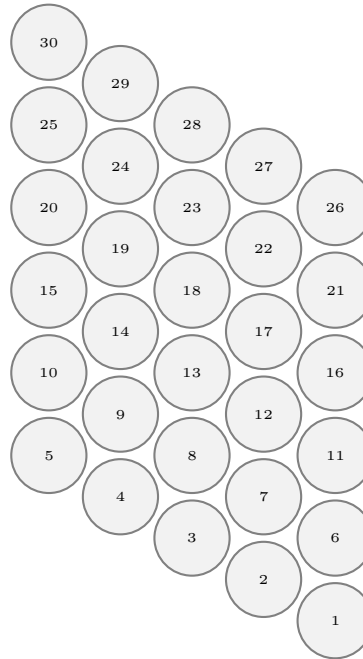


Figure 14.3.1 Node numbering reference — node 1 is the lowest C (bottom-right); node 30 is the highest G (top-left).

Full lookup table — node number, note name, and \gn address (col, row):

Node	Note	col	row	Node	Note	col	row
1	C	5	1.5	16	A	5	4.5
2	C/D	4	2	17	B/A	4	5
3	D	3	2.5	18	B	3	5.5
4	E/D	2	3	19	C	2	6
5	E	1	3.5	20	C/D	1	6.5
6	E/D	5	2.5	21	C	5	5.5
7	E	4	3	22	C/D	4	6
8	F	3	3.5	23	D	3	6.5
9	F/G	2	4	24	E/D	2	7
10	G	1	4.5	25	E	1	7.5
11	F/G	5	3.5	26	E/D	5	6.5
12	G	4	4	27	E	4	7
13	A/G	3	4.5	28	F	3	7.5
14	A	2	5	29	F/G	2	8
15	B/A	1	5.5	30	G	1	8.5

14.4 Grid 1 — Nodes 8, 14, 19, 27 (FMaj7)

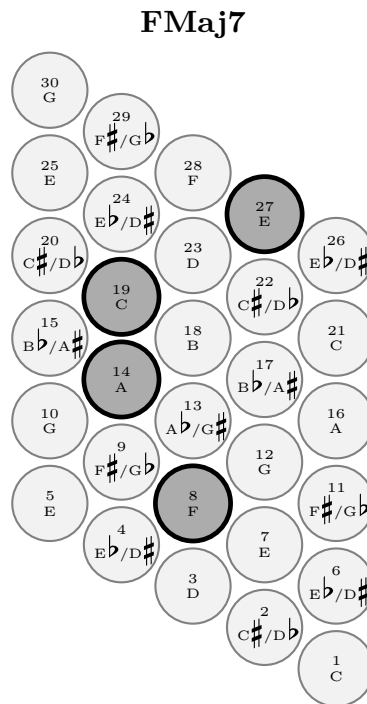


Figure 14.4.1 Nodes 8 (F), 14 (A), 19 (C), 27 (E) highlighted — FMaj7 shape.

14.5 Extended Grid 1 — 55-Node Reference

This reference shows all 55 positions in the extended 11-row grid, with each node retaining its note label and carrying an identification number, including the added lower and upper extension nodes used by the drop-2 and chord-progression template figures.

Extended Grid 1

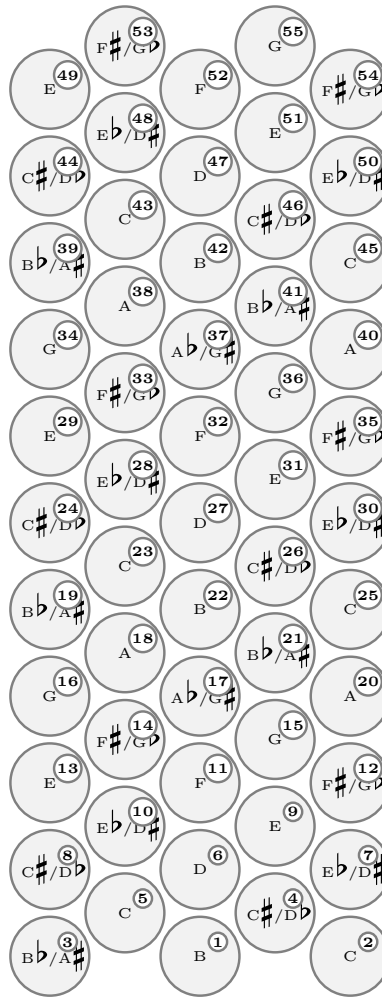


Figure 14.5.1 Extended 11-row reference showing note labels and identification numbers for all 55 positions in the shared extended grid layout.

Colophon

This book was authored in PreTeXt.